

Theory of Mathematics

Volume III: Measure, Fourier Analysis, Distributions and PDE

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Part I

Measure and Integration

Chapter 1

Sigma-Algebras and Measures

Measure theory begins by specifying which sets are stable under countable operations and which size functions are countably additive. The resulting continuity properties are the set-theoretic engine behind integration.

Learning goals

After this chapter, the reader should be able to:

- verify that a proposed family of sets is a sigma-algebra and identify which closure axiom fails when it is not;
- construct the sigma-algebra generated by elementary families and relate it to Borel sets in concrete spaces;
- derive monotonicity and countable subadditivity from countable additivity;
- apply continuity from below and continuity from above with the correct finiteness hypothesis;
- distinguish finite, sigma-finite, complete, and incomplete measure spaces by examples;
- construct null-set and completion examples showing why measurable structure matters beyond pointwise set inclusion;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

1.1 Sigma-algebras

A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences.

1.2 Generated sigma-algebras

The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators.

Example 1.1 (Generating a finite sigma-algebra). Let $X = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$. Any sigma-algebra containing A must contain $A^c = \{3, 4\}$, $\emptyset$, and X . These four sets are already closed under complements and countable unions, so

$$\sigma(A) = \{\emptyset, A, A^c, X\}.$$

This proves both closure and minimality rather than merely guessing the list.

1.3 Borel sets

The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets.

1.4 Measures

A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity.

1.5 Finite and sigma-finite measures

Finite measures have finite total mass; sigma-finite spaces are countable unions of finite-measure pieces and are the natural setting for product theory.

1.6 Continuity from below

If measurable sets increase to a union, their measures increase to the measure of that union.

1.7 Continuity from above

For decreasing measurable sets with finite measure at the first stage, measures decrease to the measure of the intersection.

Example 1.2 (Why continuity from above needs finite mass). For Lebesgue measure on $\mathbb{R}$, set $E_n = [n, \infty)$. Then $E_n \downarrow \emptyset$, but $m(E_n) = \infty$ for every n . Thus the measures do not decrease to $m(\emptyset) = 0$. The standard continuity-from-above theorem needs $\mu(E_1) < \infty$; this example isolates the exact hypothesis used when the proof subtracts from $\mu(E_1)$.

1.8 Completion and null sets

A complete measure space contains every subset of every measurable null set, so null-set modifications remain measurable.

Example 1.3 (Completing a measure space). If N is measurable with $\mu(N) = 0$ and $A \subset N$, the completion declares A measurable with measure zero. More generally, completed sets are obtained by adjoining subsets of measurable null sets to original measurable sets. This is why changing a function on an arbitrary null subset is safe only after completion of the measure space.

1.9 Core structural results

Theorem 1.4 (Continuity from below). *If $E_n \uparrow E$, then $\mu(E_n) \uparrow \mu(E)$.*

Proof. Disjointify the increase by writing E as E_1 together with the successive differences $E_n \setminus E_{n-1}$, then use countable additivity. $\square$

Theorem 1.5 (Continuity from above). *If $E_n \downarrow E$ and $\mu(E_1) < \infty$, then $\mu(E_n) \downarrow \mu(E)$.*

Proof. Apply continuity from below to $E_1 \setminus E_n \uparrow E_1 \setminus E$ and subtract from the finite quantity $\mu(E_1)$. $\square$

Theorem 1.6 (Countable subadditivity). *For measurable E_n , $\mu(\cup_n E_n) \leq \sum_n \mu(E_n)$.*

Proof. Replace the family by disjoint pieces $F_n = E_n \setminus \cup_{k < n} E_k$, use countable additivity, and compare $\mu(F_n) \leq \mu(E_n)$. $\square$

1.10 Worked examples

Example 1.7 (Sigma-algebras diagnostic). Give a rigorous worked analysis of Sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 1.8 (Generated sigma-algebras diagnostic). Give a rigorous worked analysis of Generated sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 1.9 (Borel sets diagnostic). Give a rigorous worked analysis of Borel sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 1.10 (Measures diagnostic). Give a rigorous worked analysis of Measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

1.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 1.11 (Sigma-algebras diagnostic). Give a rigorous worked analysis of Sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.12 (Generated sigma-algebras diagnostic). Give a rigorous worked analysis of Generated sigma-algebras in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.13 (Borel sets diagnostic). Give a rigorous worked analysis of Borel sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.14 (Measures diagnostic). Give a rigorous worked analysis of Measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.15 (Finite and sigma-finite measures diagnostic). Give a rigorous worked analysis of Finite and sigma-finite measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Finite measures have finite total mass; sigma-finite spaces are countable unions of finite-measure pieces and are the natural setting for product theory. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.16 (Continuity from below diagnostic). Give a rigorous worked analysis of Continuity from below in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If measurable sets increase to a union, their measures increase to the measure of that union. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.17 (Continuity from above diagnostic). Give a rigorous worked analysis of Continuity from above in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For decreasing measurable sets with finite measure at the first stage, measures decrease to the measure of the intersection. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.18 (Completion and null sets diagnostic). Give a rigorous worked analysis of Completion and null sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A complete measure space contains every subset of every measurable null set, so null-set modifications remain measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 1.19 (Continuity from below proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $E_n \uparrow E$, then $\mu(E_n) \uparrow \mu(E)$.

Solution. Disjointify the increase by writing E as E_1 together with the successive differences $E_n \setminus E_{n-1}$, then use countable additivity. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 1.20 (Continuity from above proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $E_n \downarrow E$ and $\mu(E_1) < \infty$, then $\mu(E_n) \downarrow \mu(E)$.

Solution. Apply continuity from below to $E_1 \setminus E_n \uparrow E_1 \setminus E$ and subtract from the finite quantity $\mu(E_1)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 1.21 (Countable subadditivity proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For measurable E_n , $\mu(\cup_n E_n) \leq \sum_n \mu(E_n)$.

Solution. Replace the family by disjoint pieces $F_n = E_n \setminus \cup_{k < n} E_k$, use countable additivity, and compare $\mu(F_n) \leq \mu(E_n)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 1.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Measure theory begins by specifying which sets are stable under countable operations and which size functions are countably additive. The resulting continuity properties are the set-theoretic engine behind integration. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

1.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 1.23. State and apply the central principle behind Sigma-algebras. Give one immediate consequence in the setting of this chapter.

Hint. Check closure under complements and countable unions first; the remaining sigma-algebra operations follow from De Morgan's laws. $\square$

Solution. A sigma-algebra contains the empty set and is closed under complements and countable unions; De Morgan's laws give closure under countable intersections and differences. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.24. State and apply the central principle behind Generated sigma-algebras. Give one immediate consequence in the setting of this chapter.

Hint. Use the generated-sigma-algebra definition as an intersection of all sigma-algebras containing the generators. $\square$

Solution. The sigma-algebra generated by a family is the intersection of all sigma-algebras containing it, giving the smallest measurable structure compatible with the generators. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.25. State and apply the central principle behind Borel sets. Give one immediate consequence in the setting of this chapter.

Hint. For Borel measurability on $\mathbb{R}$, reduce to intervals with rational endpoints rather than arbitrary open sets. $\square$

Solution. The Borel sigma-algebra is generated by open sets. On the real line, open intervals with rational endpoints already generate all Borel sets. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.26. State and apply the central principle behind Measures. Give one immediate consequence in the setting of this chapter.

Hint. Derive monotonicity by writing the larger set as a disjoint union of the smaller set and its difference. $\square$

Solution. A measure is a nonnegative countably additive set function with value zero at the empty set. Countable additivity forces monotonicity and subadditivity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.27. State and apply the central principle behind Finite and sigma-finite measures. Give one immediate consequence in the setting of this chapter.

Hint. For sigma-finiteness, exhibit an explicit countable cover by measurable sets of finite measure. $\square$

Solution. Finite measures have finite total mass; sigma-finite spaces are countable unions of finite-measure pieces and are the natural setting for product theory. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.28. State and apply the central principle behind Continuity from below. Give one immediate consequence in the setting of this chapter.

Hint. Disjointify the increasing sequence and apply countable additivity term by term. $\square$

Solution. If measurable sets increase to a union, their measures increase to the measure of that union. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.29. State and apply the central principle behind Continuity from above. Give one immediate consequence in the setting of this chapter.

Hint. For continuity from above, subtract from the finite set E_1 ; that is exactly where $\mu(E_1) < \infty$ is used. $\square$

Solution. For decreasing measurable sets with finite measure at the first stage, measures decrease to the measure of the intersection. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 1.30. State and apply the central principle behind Completion and null sets. Give one immediate consequence in the setting of this chapter.

Hint. To check completeness, start with a measurable null set and ask whether every one of its subsets is measurable. $\square$

Solution. A complete measure space contains every subset of every measurable null set, so null-set modifications remain measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

1.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 1.31 (Generated family). On $X = \{1, 2, 3\}$, find the sigma-algebra generated by $\mathcal{A}$.

Hint. Close under complements and unions. $\square$

Solution. It is empty, $\{1\}$, $\{2, 3\}$, X . $\square$

Exercise 1.32 (Continuity below). If E_n increases to E and $\mu(E_n) = 1 - 2^{-n}$, find $\mu(E)$.

Hint. Take the monotone limit. $\square$

Solution. $\mu(E) = 1$. $\square$

Exercise 1.33 (Subadditivity). If $\mu(E) = 2$ and $\mu(F) = 4$, give a universal upper bound for $\mu(E \cup F)$.

Hint. Use subadditivity. $\square$

Solution. The bound is 6. $\square$

Exercise 1.34 (Sigma-finite line). Explain why Lebesgue measure on the real line is sigma-finite.

Hint. Cover by bounded intervals. □

Solution. The real line is the union of $[-n, n]$, each having finite measure. □

Exercise 1.35 (Null completion). If A is a subset of a measurable null set in a complete space, what is $\mu(A)$?

Hint. Use completeness and monotonicity. □

Solution. A is measurable and has measure 0. □

Proofs

Exercise 1.36 (Monotonicity). Prove $A \subset B$ implies $\mu(A) \leq \mu(B)$.

Hint. Write B as a disjoint union. □

Solution. $B = A \cup (B \setminus A)$, so nonnegativity gives the inequality. □

Exercise 1.37 (Continuity below). Prove continuity from below.

Hint. Disjointify the increasing sequence. □

Solution. Write the union as E_1 plus successive differences and use countable additivity. □

Exercise 1.38 (Continuity above). Derive continuity from above from continuity below under finite first measure.

Hint. Apply continuity below to E_1 minus E_n . □

Solution. Subtract the increasing limit from the finite number $\mu(E_1)$. □

Exercise 1.39 (Subadditivity proof). Prove countable subadditivity.

Hint. Replace sets by disjoint new pieces. □

Solution. Disjointify and compare each new piece with the original set. □

Counterexamples and hypothesis testing

Exercise 1.40 (Finite-mass necessity). Give a decreasing sequence showing continuity from above can fail without finite first measure.

Hint. Use tails of the real line. □

Solution. $[n, \infty)$ decreases to empty but every term has infinite measure. □

Exercise 1.41 (Algebra not sigma-algebra). Give a family closed under finite operations but not countable unions.

Hint. Use finite/cofinite subsets of the natural numbers. □

Solution. The union of even singleton sets is neither finite nor cofinite. □

Exercise 1.42 (Incomplete measure). Describe what failure of completeness means.

Hint. Look inside measurable null sets. □

Solution. A subset of a measurable null set can fail to be measurable. □

Applications and investigations

Exercise 1.43 (Probability events). Why does countable additivity matter for mutually exclusive probabilistic outcomes?

Hint. Union their events. □

Solution. It makes probability of a countable disjoint union equal the sum of probabilities. □

Exercise 1.44 (Null modifications). Why does completion matter for almost-everywhere arguments?

Hint. Exceptional sets can be arbitrary subsets of null sets. □

Solution. Completion keeps all such exceptional subsets measurable and null. □

Challenge problems

Exercise 1.45 (Rational generators). Explain why rational-endpoint intervals generate all Borel subsets of the real line.

Hint. Approximate open sets by countably many rational intervals. □

Solution. Every open set is a countable union of such intervals, so all Borel sets are generated. □

Exercise 1.46 (Borel versus Lebesgue). Why is the completed Lebesgue sigma-algebra larger than the Borel sigma-algebra?

Hint. Add subsets of Borel null sets. □

Solution. Completion contains all such subsets, including some that are not Borel. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 2

Measurable Functions

Measurable functions are those whose inverse images respect the measurable structure. Their closure under algebra, limits, and monotone simple approximation makes them the natural domain of Lebesgue integration.

Learning goals

After this chapter, the reader should be able to:

- verify measurability of a function from inverse images of Borel sets or from level-set criteria;
- prove that algebraic combinations and pointwise limits of measurable functions are measurable under the stated hypotheses;
- construct simple-function approximations to nonnegative measurable functions;
- compute measurable level sets for explicit functions and use them to analyze distribution functions;
- distinguish almost-everywhere equality from pointwise equality in measure-theoretic arguments;
- produce examples of nonmeasurable behavior by identifying where the measurable-structure hypotheses fail;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

2.1 Inverse-image definition

A real-valued function is measurable when inverse images of Borel sets are measurable.

2.2 Ray criteria

It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra.

Example 2.1 (Measurability from level sets). For $f(x) = x^2$, the sublevel set $\{f < a\}$ is empty when $a \leq 0$ and equals $(-\sqrt{a}, \sqrt{a})$ when $a > 0$. Every such set is Borel, so the ray criterion proves f measurable. Testing a generating family of Borel sets is enough; one need not compute the inverse image of every Borel set.

2.3 Algebraic operations

Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable.

2.4 Pointwise limits

Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable.

2.5 Simple functions

A simple measurable function has finite range and measurable level sets; it is the finite-data model for integration.

2.6 Approximation from below

Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions obtained by dyadic quantization and truncation.

Example 2.2 (Dyadic simple approximation). For $f \geq 0$, truncate at height n and replace values by lower dyadic endpoints of mesh 2^{-n} . The resulting ϕ_n is simple and measurable, satisfies $0 \leq \phi_n \leq f$, and increases pointwise to f . On the region where $f \leq n$, the quantization error is below 2^{-n} . This is the constructive bridge from measurable functions to Lebesgue integration.

2.7 Almost-everywhere equivalence

On complete measure spaces, changing a measurable function on a null set preserves measurability.

Example 2.3 (Almost-everywhere modification). Assume the measure space is complete and $f = g$ outside a null set N . For every Borel set B , the symmetric difference between $f^{-1}(B)$ and $g^{-1}(B)$ is contained in N . Completeness makes that difference measurable, hence $g^{-1}(B)$ is measurable whenever $f^{-1}(B)$ is. Thus measurability survives null-set modification.

2.8 Composition

Composing a measurable map with a continuous or Borel measurable scalar map preserves measurability.

2.9 Core structural results

Theorem 2.4 (Pointwise limit theorem). *A pointwise limit of measurable real-valued functions is measurable.*

Proof. Represent the limit using limsup and liminf, each built from countable suprema and infima of measurable functions. $\square$

Theorem 2.5 (Algebra closure). *If f and g are measurable, then $f + g$ and fg are measurable.*

Proof. The map $x \mapsto (f(x), g(x))$ is measurable into $\mathbb{R}^2$, and addition and multiplication are continuous Borel maps. $\square$

Theorem 2.6 (Simple approximation). *If $f \geq 0$ is measurable, there exist simple $\phi_n \uparrow f$.*

Proof. Truncate at height n and replace values by the lower endpoints of dyadic intervals of width 2^{-n} ; measurability of the level sets gives simple functions and the mesh tends to zero. $\square$

2.10 Worked examples

Example 2.7 (Inverse-image definition diagnostic). Give a rigorous worked analysis of Inverse-image definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A real-valued function is measurable when inverse images of Borel sets are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 2.8 (Ray criteria diagnostic). Give a rigorous worked analysis of Ray criteria in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 2.9 (Algebraic operations diagnostic). Give a rigorous worked analysis of Algebraic operations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 2.10 (Pointwise limits diagnostic). Give a rigorous worked analysis of Pointwise limits in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

2.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 2.11 (Inverse-image definition diagnostic). Give a rigorous worked analysis of Inverse-image definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A real-valued function is measurable when inverse images of Borel sets are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.12 (Ray criteria diagnostic). Give a rigorous worked analysis of Ray criteria in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.13 (Algebraic operations diagnostic). Give a rigorous worked analysis of Algebraic operations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.14 (Pointwise limits diagnostic). Give a rigorous worked analysis of Pointwise limits in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.15 (Simple functions diagnostic). Give a rigorous worked analysis of Simple functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A simple measurable function has finite range and measurable level sets; it is the finite-data model for integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.16 (Approximation from below diagnostic). Give a rigorous worked analysis of Approximation from below in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions obtained by dyadic quantization and truncation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.17 (Almost-everywhere equivalence diagnostic). Give a rigorous worked analysis of Almost-everywhere equivalence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On complete measure spaces, changing a measurable function on a null set preserves measurability. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.18 (Composition diagnostic). Give a rigorous worked analysis of Composition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Composing a measurable map with a continuous or Borel measurable scalar map preserves measurability. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 2.19 (Pointwise limit theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: A pointwise limit of measurable real-valued functions is measurable.

Solution. Represent the limit using limsup and liminf, each built from countable suprema and infima of measurable functions. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 2.20 (Algebra closure proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If f and g are measurable, then $f + g$ and fg are measurable.

Solution. The map $x \mapsto (f(x), g(x))$ is measurable into $\mathbb{R}^2$, and addition and multiplication are continuous Borel maps. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 2.21 (Simple approximation proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \geq 0$ is measurable, there exist simple $\phi_n \uparrow f$.

Solution. Truncate at height n and replace values by the lower endpoints of dyadic intervals of width 2^{-n} ; measurability of the level sets gives simple functions and the mesh tends to zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 2.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Measurable functions are those whose inverse images respect the measurable structure. Their closure under algebra, limits, and monotone simple approximation makes them the natural domain of Lebesgue integration. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

2.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 2.23. State and apply the central principle behind Inverse-image definition. Give one immediate consequence in the setting of this chapter.

Hint. Use the inverse-image definition: test $f^{-1}((a, \infty))$ or another generating family of Borel sets. $\square$

Solution. A real-valued function is measurable when inverse images of Borel sets are measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.24. State and apply the central principle behind Ray criteria. Give one immediate consequence in the setting of this chapter.

Hint. Reduce measurability of algebraic combinations to measurability of the coordinate map on $\mathbb{R}^2$. $\square$

Solution. It is enough to test sets of the form $\{f < a\}$ or $\{f > a\}$ for all real a because such rays generate the Borel sigma-algebra. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.25. State and apply the central principle behind Algebraic operations. Give one immediate consequence in the setting of this chapter.

Hint. For pointwise limits, express the relevant level set using countable unions and intersections of level sets of the approximating functions. $\square$

Solution. Sums, products, absolute values, maxima, and minima of measurable real-valued functions remain measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.26. State and apply the central principle behind Pointwise limits. Give one immediate consequence in the setting of this chapter.

Hint. Approximate a nonnegative measurable function from below by dyadic simple functions. $\square$

Solution. Countable suprema, infima, limsup, liminf, and pointwise limits of measurable functions are measurable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.27. State and apply the central principle behind Simple functions. Give one immediate consequence in the setting of this chapter.

Hint. Write the distribution function as $\mu(\{|f| > t\})$ and compute the level set before its measure. $\square$

Solution. A simple measurable function has finite range and measurable level sets; it is the finite-data model for integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.28. State and apply the central principle behind Approximation from below. Give one immediate consequence in the setting of this chapter.

Hint. Almost-everywhere equality means the exceptional set has measure zero; identify that set explicitly. $\square$

Solution. Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions obtained by dyadic quantization and truncation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.29. State and apply the central principle behind Almost-everywhere equivalence. Give one immediate consequence in the setting of this chapter.

Hint. To prove measurability of a supremum or infimum, rewrite the level set in terms of countably many measurable level sets. $\square$

Solution. On complete measure spaces, changing a measurable function on a null set preserves measurability. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 2.30. State and apply the central principle behind Composition. Give one immediate consequence in the setting of this chapter.

Hint. For a composition, verify that the outer map is Borel measurable so inverse images remain in the sigma-algebra. $\square$

Solution. Composing a measurable map with a continuous or Borel measurable scalar map preserves measurability. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

2.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 2.31 (Indicator). Why is the indicator of a measurable set measurable?

Hint. Inspect threshold sets. $\square$

Solution. Every threshold set is empty, the set, its complement, or the whole space. $\square$

Exercise 2.32 (Absolute value). If f is measurable, what about $|f|$?

Hint. Compose with a continuous map. $\square$

Solution. It is measurable. $\square$

Exercise 2.33 (Maximum). If f and g are measurable, what about $\max(f, g)$?

Hint. Use algebra and absolute value. $\square$

Solution. $\max(f, g) = (f + g + |f - g|)/2$, so it is measurable. $\square$

Exercise 2.34 (Pointwise limit). What can be said about a pointwise limit of measurable real functions?

Hint. Use the pointwise limit theorem. $\square$

Solution. The limit is measurable. □

Exercise 2.35 (Simple range). What distinguishes a simple function by its range?

Hint. Recall the definition. □

Solution. Its range is finite. □

Proofs

Exercise 2.36 (Ray criterion). Why does checking all sublevel rays suffice?

Hint. Rays generate the Borel sigma-algebra. □

Solution. Inverse images preserve complements and countable unions, extending measurability from generators. □

Exercise 2.37 (Supremum). Prove the supremum of countably many measurable functions is measurable.

Hint. Write a superlevel set as a union. □

Solution. $\sup f_n > a$ is the union of $f_n > a$. □

Exercise 2.38 (Limsup). Prove limsup of measurable functions is measurable.

Hint. Use tail suprema and countable infimum. □

Solution. $\limsup$ is $\inf_N \sup_{n \geq N} f_n$, built from measurable operations. □

Exercise 2.39 (Simple approximation). Explain why dyadic lower approximants converge upward.

Hint. Refine mesh and raise truncation level. □

Solution. The approximants never exceed f , become finer, and their pointwise error tends to zero. □

Counterexamples and hypothesis testing

Exercise 2.40 (Nonmeasurable indicator). If A is nonmeasurable, why is its indicator nonmeasurable?

Hint. Use a threshold such as $1/2$. □

Solution. The superlevel set is exactly A . □

Exercise 2.41 (Incomplete null change). Why can null-set modification fail in an incomplete space?

Hint. A subset of a null set can be nonmeasurable. □

Solution. The modified function can acquire a nonmeasurable level set. □

Exercise 2.42 (Pointwise intuition). Why does pointwise regularity alone not imply measurability?

Hint. Measurability concerns inverse images and the chosen sigma-algebra. □

Solution. A function can be defined using a nonmeasurable set, producing nonmeasurable level sets. □

Applications and investigations

Exercise 2.43 (Random variables). Why must a random variable be measurable?

Hint. Events such as $X < a$ need probabilities. □

Solution. Measurability ensures their inverse images are measurable events. □

Exercise 2.44 (Quantization). Why do simple functions matter computationally?

Hint. They use finitely many values and measurable regions. □

Solution. They approximate measurable functions while making integrals finite sums. □

Challenge problems

Exercise 2.45 (Composition). Prove continuous h composed with measurable f is measurable.

Hint. Take inverse images. □

Solution. $(h \circ f)^{-1}(B) = f^{-1}(h^{-1}(B))$, and $h^{-1}(B)$ is Borel. □

Exercise 2.46 (Tail envelope). Show tail suprema decrease to $\limsup$.

Hint. Removing early terms cannot raise a supremum. □

Solution. The decreasing sequence $\sup_{k \geq n} f_k$ converges by definition to $\limsup f_n$. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 3

The Lebesgue Integral

The Lebesgue integral is constructed by integrating simple functions and taking monotone suprema. It is insensitive to null-set changes, linear on integrable functions, and controlled by the integral of the absolute value.

Learning goals

After this chapter, the reader should be able to:

- compute Lebesgue integrals first for simple functions and then for nonnegative or integrable functions;
- use monotone approximation by simple functions to justify integral identities;
- separate positive and negative parts and determine when a signed function is integrable;
- apply basic comparison and linearity properties of the integral with all finiteness assumptions checked;
- compare Lebesgue and Riemann integrals on functions covered by both theories;
- construct examples of measurable functions with finite, infinite, or undefined signed integrals;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

3.1 Simple-function integral

For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$.

Example 3.1 (A simple-function integral). On $[0, 3]$, let

$$\phi = 2 1_{[0,1]} + 5 1_{(1,3]}.$$

The level sets are disjoint, so

$$\int \phi = 2 \cdot 1 + 5 \cdot 2 = 12.$$

This is the defining finite-sum calculation: value times measure, summed over measurable levels.

3.2 Nonnegative functions

For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$.

3.3 Positive and negative parts

Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$.

Example 3.2 (Positive and negative parts). Write $f = f^+ - f^-$ with $f^\pm \geq 0$ and $|f| = f^+ + f^-$. For $f(x) = 1/x$ on $(-1, 1)$, both $\int f^+$ and $\int f^-$ are infinite. Therefore the signed Lebesgue integral is not defined by symmetric cancellation: the expression would be the forbidden $\infty - \infty$.

3.4 Integrable functions

A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$.

3.5 Monotonicity

If $0 \leq f \leq g$, then $\int f \leq \int g$ because every simple lower approximation to f is also one for g .

3.6 Linearity

Integrable functions form a vector space and the integral is linear.

3.7 Null-set invariance

If $f = g$ almost everywhere and one is integrable, then both represent the same integral.

Example 3.3 (Null-set invariance). Let $f = 0$ on $[0, 1]$, and let g equal 1000 at $x = 1/2$ and zero elsewhere. Then $f = g$ almost everywhere and both integrals are zero. Lebesgue integration naturally ignores modifications on null sets, which is why L^p spaces use almost-everywhere equivalence classes.

3.8 Integrals over sets

Define $\int_E f = \int f 1_E$, making restriction to measurable subsets part of the same formalism.

3.9 Core structural results

Theorem 3.4 (Absolute integral inequality). *For integrable f , $|\int f| \leq \int |f|$.*

Proof. Use $-|f| \leq f \leq |f|$ and monotonicity of the integral. $\square$

Theorem 3.5 (Zero integral criterion). *If $f \geq 0$ and $\int f = 0$, then $f = 0$ almost everywhere.*

Proof. For $E_n = \{f \geq 1/n\}$, the inequality $\int f \geq (1/n)\mu(E_n)$ forces every E_n null; their union is $\{f > 0\}$. $\square$

Theorem 3.6 (Linearity). *If f, g are integrable and $a, b \in \mathbb{R}$, then $\int (af + bg) = a \int f + b \int g$.*

Proof. Prove first for nonnegative simple functions on a common refinement, pass to nonnegative measurable functions by monotone approximation, then use positive/negative parts. $\square$

3.10 Worked examples

Example 3.7 (Simple-function integral diagnostic). Give a rigorous worked analysis of Simple-function integral in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 3.8 (Nonnegative functions diagnostic). Give a rigorous worked analysis of Nonnegative functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 3.9 (Positive and negative parts diagnostic). Give a rigorous worked analysis of Positive and negative parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 3.10 (Integrable functions diagnostic). Give a rigorous worked analysis of Integrable functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

3.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 3.11 (Simple-function integral diagnostic). Give a rigorous worked analysis of Simple-function integral in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.12 (Nonnegative functions diagnostic). Give a rigorous worked analysis of Nonnegative functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.13 (Positive and negative parts diagnostic). Give a rigorous worked analysis of Positive and negative parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.14 (Integrable functions diagnostic). Give a rigorous worked analysis of Integrable functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.15 (Monotonicity diagnostic). Give a rigorous worked analysis of Monotonicity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $0 \leq f \leq g$, then $\int f \leq \int g$ because every simple lower approximation to f is also one for g . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.16 (Linearity diagnostic). Give a rigorous worked analysis of Linearity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Integrable functions form a vector space and the integral is linear. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.17 (Null-set invariance diagnostic). Give a rigorous worked analysis of Null-set invariance in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $f = g$ almost everywhere and one is integrable, then both represent the same integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.18 (Integrals over sets diagnostic). Give a rigorous worked analysis of Integrals over sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Define $\int_E f = \int f 1_E$, making restriction to measurable subsets part of the same formalism. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 3.19 (Absolute integral inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For integrable f , $|\int f| \leq \int |f|$.

Solution. Use $-|f| \leq f \leq |f|$ and monotonicity of the integral. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 3.20 (Zero integral criterion proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \geq 0$ and $\int f = 0$, then $f = 0$ almost everywhere.

Solution. For $E_n = \{f \geq 1/n\}$, the inequality $\int f \geq (1/n)\mu(E_n)$ forces every E_n null; their union is $\{f > 0\}$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 3.21 (Linearity proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If f, g are integrable and $a, b \in \mathbb{R}$, then $\int (af + bg) = a \int f + b \int g$.

Solution. Prove first for nonnegative simple functions on a common refinement, pass to nonnegative measurable functions by monotone approximation, then use positive/negative parts. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 3.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Lebesgue integral is constructed by integrating simple functions and taking monotone suprema. It is insensitive to null-set changes, linear on integrable functions, and controlled by the integral of the absolute value. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

3.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 3.23. State and apply the central principle behind Simple-function integral. Give one immediate consequence in the setting of this chapter.

Hint. Start with the simple-function formula $\int \sum a_k 1_{E_k} = \sum a_k \mu(E_k)$ on disjoint pieces. $\square$

Solution. For a nonnegative simple function $\phi = \sum a_k 1_{E_k}$ on disjoint measurable level sets, define $\int \phi = \sum a_k \mu(E_k)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.24. State and apply the central principle behind Nonnegative functions. Give one immediate consequence in the setting of this chapter.

Hint. For a nonnegative function, choose an increasing sequence of simple functions and use the defining supremum/monotone limit. $\square$

Solution. For measurable $f \geq 0$, define the integral as the supremum of simple-function integrals over $0 \leq \phi \leq f$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.25. State and apply the central principle behind Positive and negative parts. Give one immediate consequence in the setting of this chapter.

Hint. Split f into $f^+ - f^-$ and check that both positive-part integrals are finite before calling f integrable. $\square$

Solution. Write $f = f^+ - f^-$ with $f^+, f^- \geq 0$ and $|f| = f^+ + f^-$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.26. State and apply the central principle behind Integrable functions. Give one immediate consequence in the setting of this chapter.

Hint. Use monotonicity of the integral directly when one function is bounded above by another almost everywhere. $\square$

Solution. A measurable real-valued function is integrable when $\int |f| < \infty$, which rules out the indeterminate expression $\infty - \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.27. State and apply the central principle behind Monotonicity. Give one immediate consequence in the setting of this chapter.

Hint. Linearity for signed functions is safest after verifying absolute integrability of all terms involved. $\square$

Solution. If $0 \leq f \leq g$, then $\int f \leq \int g$ because every simple lower approximation to f is also one for g . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.28. State and apply the central principle behind Linearity. Give one immediate consequence in the setting of this chapter.

Hint. To compare with the Riemann integral, use the common simple/step approximations rather than treating the two definitions as identical. $\square$

Solution. Integrable functions form a vector space and the integral is linear. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.29. State and apply the central principle behind Null-set invariance. Give one immediate consequence in the setting of this chapter.

Hint. An infinite integral is allowed for nonnegative functions; distinguish this from an undefined expression $\infty - \infty$. $\square$

Solution. If $f = g$ almost everywhere and one is integrable, then both represent the same integral. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 3.30. State and apply the central principle behind Integrals over sets. Give one immediate consequence in the setting of this chapter.

Hint. For truncation, replace f by $\min(f, n)$ or by bounded-support cutoffs and then pass to the limit. □

Solution. Define $\int_E f = \int f 1_E$, making restriction to measurable subsets part of the same formalism. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

3.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 3.31 (Simple integral). Integrate a simple function equal to 3 on a set of measure 4 and 2 on a disjoint set of measure 5.

Hint. Multiply values by measures. □

Solution. The integral is 22. □

Exercise 3.32 (Indicator). What is the integral of the indicator of E ?

Hint. Use the simple-function definition. □

Solution. It equals $\mu(E)$. □

Exercise 3.33 (Positive part). If f is -2 on A and 3 on B , identify positive and negative parts.

Hint. Keep positive magnitudes separately. □

Solution. f_+ is 3 on B and f_- is 2 on A . □

Exercise 3.34 (Absolute inequality). State the basic absolute integral inequality.

Hint. Compare f with $|f|$. □

Solution. Absolute value of integral f is at most integral $|f|$. □

Exercise 3.35 (Set integral). How is integral over E defined?

Hint. Use an indicator. □

Solution. Integrate f times the indicator of E . □

Proofs

Exercise 3.36 (Zero criterion). Prove nonnegative f with zero integral is zero almost everywhere.

Hint. Use level sets where $f \geq 1/n$. □

Solution. Each such set has measure zero; their union is where $f > 0$. □

Exercise 3.37 (Monotonicity). Prove $0 \leq f \leq g$ implies $\int f \leq \int g$.

Hint. Compare simple lower approximations. □

Solution. Every simple lower approximation to f is also one for g . □

Exercise 3.38 (Absolute inequality). Prove the absolute integral inequality.

Hint. Use $-|f| \leq f \leq |f|$. □

Solution. Integrate the two inequalities and combine. □

Exercise 3.39 (Null invariance). Prove almost-everywhere equal integrable functions have equal integrals.

Hint. Apply the zero criterion to $|f-g|$. □

Solution. Its integral is zero, so the difference of integrals is zero. □

Counterexamples and hypothesis testing

Exercise 3.40 (Undefined signed integral). Give a measurable signed function whose positive and negative integrals are both infinite.

Hint. Use $1/x$ around zero. □

Solution. On $(-1,1)$, each side contributes infinite mass. □

Exercise 3.41 (Finite values not enough). Can a pointwise finite function fail to be integrable?

Hint. Use a slowly decaying tail. □

Solution. Yes; $1/x$ on $[1, \infty)$ is finite pointwise but has infinite integral. □

Exercise 3.42 (Lebesgue not Riemann). Give a bounded Lebesgue-integrable function on $[0,1]$ that is not Riemann integrable.

Hint. Use the rational indicator. □

Solution. It is zero almost everywhere, so Lebesgue integral zero, but oscillates on every interval. □

Applications and investigations

Exercise 3.43 (Expectation). Why is expectation a Lebesgue integral?

Hint. Probability is a measure and a random variable is measurable. ☐

Solution. Expectation is integral of the random variable against probability. ☐

Exercise 3.44 (Sparse corruption). Why can changing isolated sample values leave an integral unchanged?

Hint. A finite set has measure zero. ☐

Solution. Null-set changes do not affect the integral. ☐

Challenge problems

Exercise 3.45 (Layer-cake). Explain the level-set representation of a nonnegative integral.

Hint. Represent $f(x)$ as the integral over t of the indicator $f(x) > t$. ☐

Solution. Tonelli yields $\text{integral } f = \text{integral over } t \text{ of the measure of } f > t$. ☐

Exercise 3.46 (Power singularity). When is $x^{-\alpha}$ integrable on $(0,1)$?

Hint. Integrate the power near zero. ☐

Solution. Exactly when $\alpha < 1$. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 4

Convergence Theorems

Monotone convergence, Fatou's lemma, and dominated convergence govern the passage of limits through integrals. Together they separate the roles of positivity, one-sided control, and integrable domination.

Learning goals

After this chapter, the reader should be able to:

- apply monotone, Fatou, and dominated convergence theorems after verifying their distinct hypotheses;
- construct dominating functions suitable for dominated-convergence arguments;
- use Fatou's lemma to obtain one-sided integral inequalities from pointwise limits;
- distinguish pointwise, almost-everywhere, and norm convergence in examples where the conclusions differ;
- build counterexamples showing why domination or monotonicity cannot simply be omitted;
- choose the weakest convergence theorem that actually applies to a given sequence of measurable functions;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

4.1 Monotone convergence

Increasing nonnegative sequences allow direct interchange of pointwise limit and integral.

Example 4.1 (Monotone convergence for a nonnegative series). Let

$$f_n = \sum_{k=1}^n 2^{-k} 1_{[0,k]}.$$

Then $0 \leq f_n \uparrow f$. By MCT,

$$\int f = \lim_n \sum_{k=1}^n k 2^{-k} = 2.$$

No separate dominator is needed: positivity and monotonicity are the decisive hypotheses.

4.2 Fatou's lemma

The integral of a nonnegative liminf cannot exceed the liminf of the integrals.

4.3 Dominated convergence

Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions.

Example 4.2 (Dominated convergence with an explicit dominator). On $[0, 1]$, take $f_n(x) = x/(1 + nx)$. Then $f_n(x) \rightarrow 0$ for every x , and $0 \leq f_n(x) \leq x$. Since x is integrable on $[0, 1]$, DCT gives $\int f_n \rightarrow 0$. The proof is complete only after the dominating function has been named and checked to be integrable.

4.4 Almost-everywhere convergence

Null exceptional sets do not affect the integral but still require measurable representatives.

4.5 Convergence in measure

Error sets $\{|f_n - f| > \varepsilon\}$ shrinking in measure define a weaker convergence mode.

4.6 L^1 convergence

Norm convergence controls the integral of absolute error and implies convergence in measure.

4.7 Series applications

MCT handles nonnegative series through increasing partial sums; DCT handles dominated signed series.

4.8 Counterexamples

Moving spikes show that pointwise convergence alone does not control integrals.

Example 4.3 (A moving spike defeats pointwise convergence). Set $f_n = n1_{(0,1/n)}$ on $[0, 1]$. Then $f_n \rightarrow 0$ almost everywhere, but every integral equals 1. The mass concentrates into a shrinking region, so there is no single integrable dominator. This is the canonical warning that pointwise or almost-everywhere convergence alone does not control integrals.

4.9 Core structural results

Theorem 4.4 (Monotone convergence theorem). *If $0 \leq f_n \uparrow f$, then $\int f_n \uparrow \int f$.*

Proof. The integrals increase and are bounded above by $\int f$; every simple function below f is eventually captured up to an arbitrarily small factor on most of its level sets, forcing equality of the limit. $\square$

Theorem 4.5 (Fatou's lemma). *For nonnegative measurable f_n , $\int \liminf f_n \leq \liminf \int f_n$.*

Proof. Set $g_N = \inf_{n \geq N} f_n$. Then $g_N \uparrow \liminf f_n$ and $\int g_N \leq \inf_{n \geq N} \int f_n$; apply MCT and let $N \rightarrow \infty$. $\square$

Theorem 4.6 (Dominated convergence theorem). *If $f_n \rightarrow f$ a.e. and $|f_n| \leq g \in L^1$, then $\int f_n \rightarrow \int f$.*

Proof. Apply Fatou to the nonnegative sequences $g \pm f_n$ to obtain matching upper and lower bounds; also $|f| \leq g$ a.e., so f is integrable. $\square$

4.10 Worked examples

Example 4.7 (Monotone convergence diagnostic). Give a rigorous worked analysis of Monotone convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Increasing nonnegative sequences allow direct interchange of pointwise limit and integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 4.8 (Fatou's lemma diagnostic). Give a rigorous worked analysis of Fatou's lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The integral of a nonnegative liminf cannot exceed the liminf of the integrals. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 4.9 (Dominated convergence diagnostic). Give a rigorous worked analysis of Dominated convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 4.10 (Almost-everywhere convergence diagnostic). Give a rigorous worked analysis of Almost-everywhere convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Null exceptional sets do not affect the integral but still require measurable representatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

4.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 4.11 (Monotone convergence diagnostic). Give a rigorous worked analysis of Monotone convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Increasing nonnegative sequences allow direct interchange of pointwise limit and integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.12 (Fatou's lemma diagnostic). Give a rigorous worked analysis of Fatou's lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The integral of a nonnegative liminf cannot exceed the liminf of the integrals. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.13 (Dominated convergence diagnostic). Give a rigorous worked analysis of Dominated convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.14 (Almost-everywhere convergence diagnostic). Give a rigorous worked analysis of Almost-everywhere convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Null exceptional sets do not affect the integral but still require measurable representatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.15 (Convergence in measure diagnostic). Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Error sets $\{|f_n - f| > \varepsilon\}$ shrinking in measure define a weaker convergence mode. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.16 (L^1 convergence diagnostic). Give a rigorous worked analysis of L^1 convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Norm convergence controls the integral of absolute error and implies convergence in measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.17 (Series applications diagnostic). Give a rigorous worked analysis of Series applications in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. MCT handles nonnegative series through increasing partial sums; DCT handles dominated signed series. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.18 (Counterexamples diagnostic). Give a rigorous worked analysis of Counterexamples in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Moving spikes show that pointwise convergence alone does not control integrals. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 4.19 (Monotone convergence theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $0 \leq f_n \uparrow f$, then $\int f_n \uparrow \int f$.

Solution. The integrals increase and are bounded above by $\int f$; every simple function below f is eventually captured up to an arbitrarily small factor on most of its level sets, forcing equality of the limit. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 4.20 (Fatou's lemma proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For nonnegative measurable f_n , $\int \liminf f_n \leq \liminf \int f_n$.

Solution. Set $g_N = \inf_{n \geq N} f_n$. Then $g_N \uparrow \liminf f_n$ and $\int g_N \leq \inf_{n \geq N} \int f_n$; apply MCT and let $N \rightarrow \infty$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 4.21 (Dominated convergence theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f_n \rightarrow f$ a.e. and $|f_n| \leq g \in L^1$, then $\int f_n \rightarrow \int f$.

Solution. Apply Fatou to the nonnegative sequences $g \pm f_n$ to obtain matching upper and lower bounds; also $|f| \leq g$ a.e., so f is integrable. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 4.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Monotone convergence, Fatou's lemma, and dominated convergence govern the passage of limits through integrals. Together they separate the roles of positivity, one-sided control, and integrable domination. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

4.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 4.23. State and apply the central principle behind Monotone convergence. Give one immediate consequence in the setting of this chapter.

Hint. For monotone convergence, check $0 \leq f_n \uparrow f$ almost everywhere before passing the limit through the integral. $\square$

Solution. Increasing nonnegative sequences allow direct interchange of pointwise limit and integral. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.24. State and apply the central principle behind Fatou's lemma. Give one immediate consequence in the setting of this chapter.

Hint. Fatou gives a lower bound for $\int \liminf f_n$; write the $\liminf$ explicitly before applying it. $\square$

Solution. The integral of a nonnegative liminf cannot exceed the liminf of the integrals. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.25. State and apply the central principle behind Dominated convergence. Give one immediate consequence in the setting of this chapter.

Hint. For dominated convergence, find one integrable g with $|f_n| \leq g$ for every n . $\square$

Solution. Almost-everywhere convergence under a single integrable absolute bound permits limit-integral interchange for signed functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.26. State and apply the central principle behind Almost-everywhere convergence. Give one immediate consequence in the setting of this chapter.

Hint. Do not replace domination by pointwise boundedness; the dominating function must itself be integrable. $\square$

Solution. Null exceptional sets do not affect the integral but still require measurable representatives. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.27. State and apply the central principle behind Convergence in measure. Give one immediate consequence in the setting of this chapter.

Hint. To build a counterexample, use moving spikes whose heights and supports defeat uniform integral control. $\square$

Solution. Error sets $\{|f_n - f| > \varepsilon\}$ shrinking in measure define a weaker convergence mode. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.28. State and apply the central principle behind L^1 convergence. Give one immediate consequence in the setting of this chapter.

Hint. If the functions change sign, apply Fatou to a nonnegative modification or use dominated convergence instead. $\square$

Solution. Norm convergence controls the integral of absolute error and implies convergence in measure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.29. State and apply the central principle behind Series applications. Give one immediate consequence in the setting of this chapter.

Hint. Separate almost-everywhere convergence from convergence in measure or norm; the theorem only needs the stated mode. $\square$

Solution. MCT handles nonnegative series through increasing partial sums; DCT handles dominated signed series. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 4.30. State and apply the central principle behind Counterexamples. Give one immediate consequence in the setting of this chapter.

Hint. Choose the theorem with hypotheses you can actually verify rather than the strongest theorem you remember. $\square$

Solution. Moving spikes show that pointwise convergence alone does not control integrals. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

4.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 4.31 (MCT limit). If nonnegative f_n increases to f and integral f_n tends to 7, what is integral f ?

Hint. Apply MCT. $\square$

Solution. It is 7. $\square$

Exercise 4.32 (Fatou). If nonnegative f_n have $\liminf$ of integrals equal to 5, what upper bound follows for integral $\liminf f_n$?

Hint. Use Fatou. $\square$

Solution. At most 5. $\square$

Exercise 4.33 (DCT). If f_n tends to f almost everywhere and $|f_n|$ is bounded by an L^1 function, what happens to integrals?

Hint. Apply DCT. $\square$

Solution. The integrals converge to integral f . $\square$

Exercise 4.34 (L^1 to measure). What does L^1 convergence imply?

Hint. Use Chebyshev. $\square$

Solution. Convergence in measure. $\square$

Exercise 4.35 (Series MCT). Why can a nonnegative series be integrated term by term?

Hint. Partial sums increase. $\square$

Solution. MCT applies to the increasing partial sums. $\square$

Proofs

Exercise 4.36 (Fatou proof). Prove Fatou using tail infima.

Hint. Define g_N as the infimum of the tail. □

Solution. g_N increases to $\liminf f_n$; apply MCT and compare integrals. □

Exercise 4.37 (DCT integrability). Show the DCT limit is integrable.

Hint. The pointwise limit is bounded by the dominator almost everywhere. □

Solution. Integral $|f|$ is at most integral g . □

Exercise 4.38 (L1 implies measure). Prove L1 convergence implies convergence in measure.

Hint. Apply Chebyshev to the difference. □

Solution. Bad-set measure is at most L1 error divided by epsilon. □

Exercise 4.39 (MCT series). Explain why MCT is the natural theorem for nonnegative sums.

Hint. Partial sums form an increasing sequence. □

Solution. Their pointwise limit is the full series, so MCT interchanges limit and integral. □

Counterexamples and hypothesis testing

Exercise 4.40 (Moving spike). Give almost-everywhere convergence without integral convergence.

Hint. Use n on $(0, 1/n)$. □

Solution. It tends to zero almost everywhere while every integral is 1. □

Exercise 4.41 (No dominator). Why can the moving spike not have one L1 dominator?

Hint. Any dominator must be large arbitrarily close to zero. □

Solution. Its lower envelope behaves like a nonintegrable $1/x$ scale. □

Exercise 4.42 (Signed monotone caveat). Why is nonnegativity structural in standard MCT?

Hint. Signed infinite masses can create undefined subtraction. □

Solution. Additional lower-integrability hypotheses are needed for safe signed variants. □

Applications and investigations

Exercise 4.43 (Expectation limits). Why is DCT useful for limits of expectations?

Hint. Expectations are integrals. □

Solution. An integrable dominating random variable lets the limit pass through expectation. □

Exercise 4.44 (Domain truncation). How can MCT justify expanding-domain approximations of a nonnegative integral?

Hint. Use f times indicators of increasing sets. □

Solution. The truncated functions increase pointwise to f . □

Challenge problems

Exercise 4.45 (Scheffe insight). If nonnegative f_n tend to f almost everywhere and their integrals converge to integral f , what stronger convergence is plausible?

Hint. Compare $\min(f_n, f)$. □

Solution. Under these hypotheses one obtains L1 convergence. □

Exercise 4.46 (Uniform integrability). Why does uniform integrability address the moving-spike failure?

Hint. It prevents substantial integral mass on tiny sets. □

Solution. Combined with convergence in measure, it can force L1 convergence. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 5

Product Measures and Fubini Theory

Product measures combine measurable coordinate spaces, while Tonelli and Fubini convert multidimensional integration into iterated integration. Absolute integrability is the key condition that permits changing order safely.

Learning goals

After this chapter, the reader should be able to:

- construct product sigma-algebras and product measures in standard sigma-finite settings;
- apply Tonelli's theorem to nonnegative functions and Fubini's theorem to integrable functions with the correct hypotheses;
- justify changing the order of integration by checking absolute integrability or nonnegativity;
- compute iterated integrals on rectangles and simple product domains;
- identify measurable sections and use them to interpret product-measure statements;
- construct examples where changing integration order fails because the Fubini hypotheses are not satisfied;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

5.1 Product sigma-algebra

The product sigma-algebra is generated by measurable rectangles $A \times B$.

5.2 Product measures

On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles.

5.3 Sections of sets

For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory.

5.4 Sections of functions

Fixing one variable in a measurable function produces measurable sections, allowing iterated integration.

5.5 Tonelli theorem

Nonnegative measurable functions may be integrated in either order, with the common value possibly infinite.

Example 5.1 (Tonelli for a nonnegative product). For $f(x, y) = e^{-x-y}$ on $[0, \infty)^2$, Tonelli applies immediately because $f \geq 0$. Either iterated integral equals

$$\left(\int_0^\infty e^{-x} dx \right) \left(\int_0^\infty e^{-y} dy \right) = 1.$$

Nonnegativity legitimizes the interchange even before finiteness is known.

5.6 Fubini theorem

Absolutely integrable functions have integrable sections almost everywhere and finite iterated integrals equal to the product integral.

Example 5.2 (Fubini after checking absolute integrability). On $[0, 1]^2$, $f(x, y) = x - y$ satisfies $|f| \leq 1$, so $f \in L^1$. Fubini therefore applies and either integration order gives zero. The absolute-integrability check is the structural step that distinguishes a legal change of order from a formal rearrangement.

5.7 Changing order

Absolute integrability prevents order-dependent cancellation.

Example 5.3 (Changing order needs a theorem). For a signed function with divergent absolute integral, positive and negative mass may cancel differently in different iterated orders. Thus the diagnostic is to inspect the integral of $|f|$. If it is infinite, Fubini cannot be invoked; each iterated integral must be analyzed independently.

5.8 Applications

Areas, convolution kernels, expectations, and integral operators all rely on product integration.

5.9 Core structural results

Theorem 5.4 (Tonelli theorem). *For nonnegative measurable f on a sigma-finite product space, the product integral equals both iterated integrals, possibly with value ∞ .*

Proof. Prove first for rectangle indicators, then nonnegative simple functions, and finally pass to arbitrary nonnegative measurable functions by monotone convergence. $\square$

Theorem 5.5 (Fubini theorem). *If $f \in L^1(\mu \times \nu)$, then almost all sections are integrable and the iterated integrals equal $\int f$.*

Proof. Apply Tonelli to $|f|$ to obtain finite section integrals almost everywhere, then decompose f into positive and negative parts. $\square$

Theorem 5.6 (Rectangle rule). *For measurable rectangles, $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$.*

Proof. This defines the canonical premeasure on rectangles; sigma-finiteness gives the standard extension and uniqueness on the generated product sigma-algebra. $\square$

5.10 Worked examples

Example 5.7 (Product sigma-algebra diagnostic). Give a rigorous worked analysis of Product sigma-algebra in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The product sigma-algebra is generated by measurable rectangles $A \times B$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 5.8 (Product measures diagnostic). Give a rigorous worked analysis of Product measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 5.9 (Sections of sets diagnostic). Give a rigorous worked analysis of Sections of sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 5.10 (Sections of functions diagnostic). Give a rigorous worked analysis of Sections of functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Fixing one variable in a measurable function produces measurable sections, allowing iterated integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

5.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 5.11 (Product sigma-algebra diagnostic). Give a rigorous worked analysis of Product sigma-algebra in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The product sigma-algebra is generated by measurable rectangles $A \times B$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.12 (Product measures diagnostic). Give a rigorous worked analysis of Product measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.13 (Sections of sets diagnostic). Give a rigorous worked analysis of Sections of sets in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.14 (Sections of functions diagnostic). Give a rigorous worked analysis of Sections of functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fixing one variable in a measurable function produces measurable sections, allowing iterated integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.15 (Tonelli theorem diagnostic). Give a rigorous worked analysis of Tonelli theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Nonnegative measurable functions may be integrated in either order, with the common value possibly infinite. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.16 (Fubini theorem diagnostic). Give a rigorous worked analysis of Fubini theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Absolutely integrable functions have integrable sections almost everywhere and finite iterated integrals equal to the product integral. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.17 (Changing order diagnostic). Give a rigorous worked analysis of Changing order in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Absolute integrability prevents order-dependent cancellation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.18 (Applications diagnostic). Give a rigorous worked analysis of Applications in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Areas, convolution kernels, expectations, and integral operators all rely on product integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 5.19 (Tonelli theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For nonnegative measurable f on a sigma-finite product space, the product integral equals both iterated integrals, possibly with value ∞ .

Solution. Prove first for rectangle indicators, then nonnegative simple functions, and finally pass to arbitrary nonnegative measurable functions by monotone convergence. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 5.20 (Fubini theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^1(\mu \times \nu)$, then almost all sections are integrable and the iterated integrals equal $\int f$.

Solution. Apply Tonelli to $|f|$ to obtain finite section integrals almost everywhere, then decompose f into positive and negative parts. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 5.21 (Rectangle rule proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For measurable rectangles, $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$.

Solution. This defines the canonical premeasure on rectangles; sigma-finiteness gives the standard extension and uniqueness on the generated product sigma-algebra. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 5.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Product measures combine measurable coordinate spaces, while Tonelli and Fubini convert multidimensional integration into iterated integration. Absolute integrability is the key condition that permits changing order safely. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

5.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 5.23. State and apply the central principle behind Product sigma-algebra. Give one immediate consequence in the setting of this chapter.

Hint. On rectangles, start from measurable rectangles and extend to the product sigma-algebra before using product measure. $\square$

Solution. The product sigma-algebra is generated by measurable rectangles $A \times B$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 5.24. State and apply the central principle behind Product measures. Give one immediate consequence in the setting of this chapter.

Hint. Tonelli applies to nonnegative functions even when the integral is infinite; use it before worrying about absolute integrability. $\square$

Solution. On sigma-finite spaces, the product measure is characterized by $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$ on rectangles. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 5.25. State and apply the central principle behind Sections of sets. Give one immediate consequence in the setting of this chapter.

Hint. Fubini requires integrability; check $\int |f| < \infty$ before changing the integration order. $\square$

Solution. For measurable $E \subset X \times Y$, the section $E_x = \{y : (x, y) \in E\}$ is measurable for almost every relevant parameter under the standard product theory. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 5.26. State and apply the central principle behind Sections of functions. Give one immediate consequence in the setting of this chapter.

Hint. Compute the inner integral as a function of the outer variable and verify its measurability. $\square$

Solution. Fixing one variable in a measurable function produces measurable sections, allowing iterated integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 5.27. State and apply the central principle behind Tonelli theorem. Give one immediate consequence in the setting of this chapter.

Hint. For indicator functions, reduce Fubini/Tonelli to measuring sections of the underlying measurable set. $\square$

Solution. Nonnegative measurable functions may be integrated in either order, with the common value possibly infinite. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 5.28. State and apply the central principle behind Fubini theorem. Give one immediate consequence in the setting of this chapter.

Hint. Sigma-finiteness is structural in standard product-measure construction; identify the finite-measure pieces explicitly. $\square$

Solution. Absolutely integrable functions have integrable sections almost everywhere and finite iterated integrals equal to the product integral. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 5.29. State and apply the central principle behind Changing order. Give one immediate consequence in the setting of this chapter.

Hint. To expose failure of Fubini, look for conditionally convergent positive and negative parts with infinite total variation. □

Solution. Absolute integrability prevents order-dependent cancellation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 5.30. State and apply the central principle behind Applications. Give one immediate consequence in the setting of this chapter.

Hint. After swapping integrals, compare the two iterated expressions to the original product-space integral rather than only to each other. □

Solution. Areas, convolution kernels, expectations, and integral operators all rely on product integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

5.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 5.31 (Rectangle). Find the product measure of a rectangle with side measures 2 and 3.

Hint. Multiply. □

Solution. The measure is 6. □

Exercise 5.32 (Separable integral). Integrate xy over the unit square.

Hint. Factor the iterated integral. □

Solution. The value is $1/4$. □

Exercise 5.33 (Section). For $0 < y < x < 1$, describe the y -section at fixed x .

Hint. Fix x . □

Solution. It is $(0, x)$. □

Exercise 5.34 (Tonelli infinity). If a nonnegative iterated integral is infinite, what does Tonelli say about the product integral?

Hint. Use equality in the extended sense. □

Solution. It is infinite as well. □

Exercise 5.35 (Fubini criterion). Name a standard sufficient condition for changing order for a signed function.

Hint. Check absolute integrability. □

Solution. Membership in L^1 of the product space. □

Proofs

Exercise 5.36 (Rectangle rule). Explain the product rectangle rule.

Hint. It defines the product premeasure. □

Solution. Extension preserves $\mu(A)\nu(B)$ on rectangles. □

Exercise 5.37 (Tonelli proof). Outline Tonelli's extension steps.

Hint. Indicators, simple functions, then monotone convergence. □

Solution. These steps cover all nonnegative measurable functions. □

Exercise 5.38 (Fubini from Tonelli). Derive Fubini from Tonelli.

Hint. Apply Tonelli to $|f|$, then split signs. □

Solution. Absolute integrability makes positive and negative parts finite enough for subtraction. □

Exercise 5.39 (Section measurability). Why are measurable sections needed?

Hint. Iterated integrals require measurable one-variable functions. □

Solution. Product measurability supplies the section structure. □

Counterexamples and hypothesis testing

Exercise 5.40 (Conditional order). Why can changing order fail without absolute integrability?

Hint. Cancellation can depend on rearrangement. □

Solution. Positive and negative infinite masses can regroup differently. □

Exercise 5.41 (Sigma-finite caveat). Why is sigma-finiteness prominent in product theory?

Hint. It supports extension uniqueness and standard Fubini/Tonelli formulations. □

Solution. Without it, product-measure behavior can be subtler. □

Exercise 5.42 (Iterated existence). Why does existence of two iterated integrals not automatically prove Fubini?

Hint. Absolute integrability may still fail. □

Solution. Fubini is a stronger product-integral statement with hypotheses. □

Applications and investigations

Exercise 5.43 (Probability). How does Fubini appear in expectations of two variables?

Hint. Use a product probability space. □

Solution. Integrable joint functions can be averaged conditionally in either order. □

Exercise 5.44 (Convolution theorem). Why is Fubini used in proving the convolution theorem?

Hint. The transform of a convolution is a double integral. □

Solution. Absolute integrability permits swapping and factoring the integrals. □

Challenge problems

Exercise 5.45 (Triangle domain). Reverse the order in the integral over $0 < y < x < 1$.

Hint. Describe the same triangle from y first. □

Solution. It becomes y from 0 to 1 and x from y to 1. □

Exercise 5.46 (A.e. section integrability). Why are sections of an L^1 product function integrable almost everywhere?

Hint. Apply Tonelli to $|f|$. □

Solution. The function of the section integral is integrable, hence finite almost everywhere. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 6

Lp Spaces

L^p spaces turn integral size into geometry. After identifying functions equal almost everywhere, the L^p quantities become genuine norms for $p \geq 1$, and the resulting spaces are complete.

Learning goals

After this chapter, the reader should be able to:

- compute L^p norms and determine membership in L^p for explicit singular or decaying functions;
- distinguish equality almost everywhere from equality as pointwise functions in L^p spaces;
- prove completeness of L^p in the chapter's setting using Cauchy-sequence control;
- compare L^p spaces on finite- and infinite-measure domains with explicit inclusions or counterexamples;
- use truncation or simple-function approximation to construct dense subclasses;
- analyze limiting behavior in L^p without confusing norm convergence with pointwise convergence;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

6.1 Almost-everywhere classes

Two measurable functions equal almost everywhere represent the same L^p element.

6.2 L^p norms

For $1 \leq p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum.

Example 6.1 (Critical exponent for a singularity). On $(0, 1)$, let $f(x) = x^{-\alpha}$. Then

$$\|f\|_p^p = \int_0^1 x^{-\alpha p} dx,$$

which is finite exactly when $\alpha p < 1$. The local singularity therefore sets a sharp membership threshold across the L^p scale.

6.3 Essential supremum

The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound.

6.4 Vector-space structure

Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space.

6.5 Completeness

Every L^p Cauchy sequence converges in L^p for $1 \leq p \leq \infty$.

6.6 Finite-measure inclusions

On finite-measure spaces, $L^q \subset L^p$ when $q > p$, with an explicit measure factor.

Example 6.2 (Finite measure gives L^q inside L^p). If $\mu(X) < \infty$ and $q > p$, Hölder applied to $|f|^p \cdot 1$ gives

$$\|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q.$$

On a probability space the measure factor is one. The finite-measure hypothesis is essential; this inclusion can fail on infinite-measure spaces.

6.7 Convergence in measure

For finite p , L^p convergence implies convergence in measure by Chebyshev's inequality.

6.8 Approximation

Truncation, finite-measure cutoffs, and simple functions provide dense approximations in L^p for $p < \infty$.

Example 6.3 (Truncation approximates in L^p). For $f \in L^p$, $p < \infty$, let $f_n = f 1_{\{|f| \leq n\}}$. Then

$$|f - f_n|^p = |f|^p 1_{\{|f| > n\}} \rightarrow 0$$

pointwise and is dominated by $|f|^p \in L^1$. DCT yields $\|f - f_n\|_p \rightarrow 0$. Thus unbounded L^p functions can first be reduced to bounded approximants.

6.9 Core structural results

Theorem 6.4 (Norm-zero criterion). For $1 \leq p \leq \infty$, $\|f\|_p = 0$ iff $f = 0$ almost everywhere.

Proof. For finite p , $\int |f|^p = 0$ forces $|f|^p = 0$ a.e.; for $p = \infty$, essential supremum zero means $|f| = 0$ a.e. $\square$

Theorem 6.5 (Finite-measure inclusion). If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then $\|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q$.

Proof. For finite q , apply Hölder to $|f|^p \cdot 1$ with exponent q/p and its conjugate; the $q = \infty$ case follows directly. $\square$

Theorem 6.6 (Completeness of L^p). *For $1 \leq p \leq \infty$, L^p is complete.*

Proof. Extract a subsequence with summable norm increments, use the resulting pointwise absolutely summable majorant to construct an a.e. limit, prove subsequence norm convergence, and recover the full Cauchy sequence. $\square$

6.10 Worked examples

Example 6.7 (Almost-everywhere classes diagnostic). Give a rigorous worked analysis of Almost-everywhere classes in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Two measurable functions equal almost everywhere represent the same L^p element. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 6.8 (L^p norms diagnostic). Give a rigorous worked analysis of L^p norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For $1 \leq p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 6.9 (Essential supremum diagnostic). Give a rigorous worked analysis of Essential supremum in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 6.10 (Vector-space structure diagnostic). Give a rigorous worked analysis of Vector-space structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

6.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 6.11 (Almost-everywhere classes diagnostic). Give a rigorous worked analysis of Almost-everywhere classes in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Two measurable functions equal almost everywhere represent the same L^p element. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.12 (L^p norms diagnostic). Give a rigorous worked analysis of L^p norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For $1 \leq p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.13 (Essential supremum diagnostic). Give a rigorous worked analysis of Essential supremum in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.14 (Vector-space structure diagnostic). Give a rigorous worked analysis of Vector-space structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.15 (Completeness diagnostic). Give a rigorous worked analysis of Completeness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Every L^p Cauchy sequence converges in L^p for $1 \leq p \leq \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.16 (Finite-measure inclusions diagnostic). Give a rigorous worked analysis of Finite-measure inclusions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On finite-measure spaces, $L^q \subset L^p$ when $q > p$, with an explicit measure factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.17 (Convergence in measure diagnostic). Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For finite p , L^p convergence implies convergence in measure by Chebyshev's inequality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.18 (Approximation diagnostic). Give a rigorous worked analysis of Approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Truncation, finite-measure cutoffs, and simple functions provide dense approximations in L^p for $p < \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 6.19 (Norm-zero criterion proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p \leq \infty$, $\|f\|_p = 0$ iff $f = 0$ almost everywhere.

Solution. For finite p , $\int |f|^p = 0$ forces $|f|^p = 0$ a.e.; for $p = \infty$, essential supremum zero means $|f| = 0$ a.e. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 6.20 (Finite-measure inclusion proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then $\|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q$.

Solution. For finite q , apply Hölder to $|f|^p \cdot 1$ with exponent q/p and its conjugate; the $q = \infty$ case follows directly. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 6.21 (Completeness of L^p proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p \leq \infty$, L^p is complete.

Solution. Extract a subsequence with summable norm increments, use the resulting pointwise absolutely summable majorant to construct an a.e. limit, prove subsequence norm convergence, and recover the full Cauchy sequence. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 6.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. L^p spaces turn integral size into geometry. After identifying functions equal almost everywhere, the L^p quantities become genuine norms for $p \geq 1$, and the resulting spaces are complete. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

6.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 6.23. State and apply the central principle behind Almost-everywhere classes. Give one immediate consequence in the setting of this chapter.

Hint. Compute $\int |f|^p$ and examine the singularity and tail separately; both regions must be integrable. $\square$

Solution. Two measurable functions equal almost everywhere represent the same L^p element. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.24. State and apply the central principle behind L^p norms. Give one immediate consequence in the setting of this chapter.

Hint. Remember that L^p elements are equivalence classes: changing a function on a null set does not change its class. $\square$

Solution. For $1 \leq p < \infty$, use $\|f\|_p = (\int |f|^p)^{1/p}$; for $p = \infty$, use essential supremum. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.25. State and apply the central principle behind Essential supremum. Give one immediate consequence in the setting of this chapter.

Hint. For a Cauchy sequence in L^p , extract a rapidly Cauchy subsequence and use summable norm differences to build the limit. $\square$

Solution. The L^∞ norm ignores exceptional values on null sets and records the least almost-everywhere uniform bound. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.26. State and apply the central principle behind Vector-space structure. Give one immediate consequence in the setting of this chapter.

Hint. On finite-measure spaces, apply Hölder to compare L^q and L^p norms when $q > p$. $\square$

Solution. Hölder and Minkowski control sums and scalar multiples, making L^p a normed vector space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.27. State and apply the central principle behind Completeness. Give one immediate consequence in the setting of this chapter.

Hint. On infinite-measure spaces, test inclusions with power-law examples instead of assuming the finite-measure relation survives. $\square$

Solution. Every L^p Cauchy sequence converges in L^p for $1 \leq p \leq \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.28. State and apply the central principle behind Finite-measure inclusions. Give one immediate consequence in the setting of this chapter.

Hint. Use truncation in size and support to approximate an L^p function by bounded finite-support functions. $\square$

Solution. On finite-measure spaces, $L^q \subset L^p$ when $q > p$, with an explicit measure factor. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.29. State and apply the central principle behind Convergence in measure. Give one immediate consequence in the setting of this chapter.

Hint. Norm convergence does not imply pointwise convergence of the whole sequence; look for an almost-everywhere convergent subsequence if needed. $\square$

Solution. For finite p , L^p convergence implies convergence in measure by Chebyshev's inequality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 6.30. State and apply the central principle behind Approximation. Give one immediate consequence in the setting of this chapter.

Hint. Check the endpoint $p = \infty$ separately: the essential supremum replaces an integral formula. $\square$

Solution. Truncation, finite-measure cutoffs, and simple functions provide dense approximations in L^p for $p < \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

6.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 6.31 (Power singularity). For which p is $x^{-1/3}$ in $L_p(0,1)$?

Hint. Require $p/3 < 1$. $\square$

Solution. Exactly $p < 3$. $\square$

Exercise 6.32 (Constant norm). What is the L_p norm of 1 on a set of measure M ?

Hint. Integrate 1. $\square$

Solution. $M^{1/p}$ for finite p . $\square$

Exercise 6.33 (Essential sup). A function is 7 at one point and 0 elsewhere on $[0,1]$. What is its Linfinity norm?

Hint. Ignore null exceptions. $\square$

Solution. It is 0. $\square$

Exercise 6.34 (Finite inclusion). On $[0,1]$, what follows from f in L_4 ?

Hint. Use finite-measure inclusion. $\square$

Solution. f is in L_2 and its L_2 norm is at most its L_4 norm. $\square$

Exercise 6.35 (Chebyshev). If the L_p norm of f is 2, bound the measure of $|f| > t$.

Hint. Use Chebyshev. $\square$

Solution. At most $2^p/t^p$. $\square$

Proofs

Exercise 6.36 (Norm zero). Prove zero L_p norm means zero almost everywhere.

Hint. Apply the zero-integral criterion to $|f|^p$. □

Solution. The p -th power is zero almost everywhere. □

Exercise 6.37 (Finite inclusion). Prove L_q is contained in L_p on finite measure when $q > p$.

Hint. Apply Holder to $|f|^p$ times 1. □

Solution. The exponent computation gives the stated measure factor. □

Exercise 6.38 (L_p to measure). Prove L_p convergence implies convergence in measure.

Hint. Use Chebyshev on the difference. □

Solution. Bad-set measure is bounded by norm error to the p divided by ϵ^p . □

Exercise 6.39 (Truncation). Prove bounded truncations converge in L_p .

Hint. Use DCT on the p -th power of the error. □

Solution. The tail indicator tends to zero under the integrable dominator $|f|^p$. □

Counterexamples and hypothesis testing

Exercise 6.40 (Infinite measure). Show finite-measure L_q -to- L_p inclusion can fail on an infinite-measure domain.

Hint. Use power decay on $[1, \infty)$. □

Solution. Choose an exponent integrable for one power but not the other. □

Exercise 6.41 (Pointwise not norm). Give pointwise convergence without L_1 convergence.

Hint. Use moving spikes. □

Solution. n on $(0, 1/n)$ tends to zero almost everywhere but has L_1 norm 1. □

Exercise 6.42 (Norm not pointwise). Can L_p convergence fail to imply pointwise convergence of the full sequence?

Hint. Use a typewriter sequence of moving supports. □

Solution. Yes; norm can tend to zero while points are revisited infinitely often. □

Applications and investigations

Exercise 6.43 (Energy). Why is L_2 natural for signal energy?

Hint. Squared magnitude and orthogonality behave well. □

Solution. L_2 is a Hilbert space and Plancherel preserves its norm. □

Exercise 6.44 (Worst case). Why does L_∞ encode worst-case magnitude only almost everywhere?

Hint. It uses essential supremum. □

Solution. Null spikes are ignored. □

Challenge problems

Exercise 6.45 (L1-Linfinity interpolation). If f is in L^1 and L^∞ , prove $\|f\|_p \leq \|f\|_1^{1/p} \|f\|_\infty^{(p-1)/p}$.

Hint. Factor $\|f\|_p^p$. □

Solution. Bound $p-1$ factors by the essential supremum and integrate. □

Exercise 6.46 (Completeness strategy). Outline the subsequence proof that L^p is complete.

Hint. Choose a subsequence with summable norm increments. □

Solution. The telescoping increments converge almost everywhere and in norm to a candidate limit. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 7

Hölder, Minkowski and Interpolation

Hölder and Minkowski are the fundamental inequalities of L^p geometry. Interpolation then explains how information at two exponents controls intermediate exponents.

Learning goals

After this chapter, the reader should be able to:

- apply Hölder's inequality with correctly chosen conjugate exponents;
- derive Minkowski's inequality from Hölder and use it to prove the triangle inequality in L^p ;
- compute interpolation exponents from endpoint data in elementary interpolation inequalities;
- identify equality or near-equality cases in Hölder-type estimates;
- use Hölder and Minkowski to control products, sums, and integral operators;
- construct examples showing why exponent restrictions such as $1/p + 1/q = 1$ are structural rather than cosmetic;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

7.1 Conjugate exponents

For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$.

7.2 Young inequality

The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder.

7.3 Hölder inequality

Products of L^p and L^q functions lie in L^1 with sharp norm control.

Example 7.1 (Holder with equality). For $p = q = 2$, Holder is Cauchy–Schwarz. If $g = cf$, then

$$\int |fg| = |c| \|f\|_2^2 = \|f\|_2 \|g\|_2,$$

so equality holds. This agrees with the general equality condition that $|f|^p$ and $|g|^q$ are proportional almost everywhere.

7.4 Minkowski inequality

The L^p norm satisfies the triangle inequality.

7.5 Equality cases

Sharpness is governed by proportionality of the powers $|f|^p$ and $|g|^q$.

Example 7.2 (Equality structure). After normalizing nonzero f and g , equality in Young’s inequality occurs precisely when the normalized powers agree. Hence Holder equality is governed by

$$\frac{|f|^p}{\|f\|_p^p} = \frac{|g|^q}{\|g\|_q^q} \quad \text{a.e.}$$

This identifies extremizers, not just the sharp constant.

7.6 Discrete versions

Counting measure recovers the familiar finite-sum Hölder and Minkowski inequalities.

7.7 Interpolation of norms

Intermediate L^r norms are logarithmically convex between endpoint norms.

Example 7.3 (Interpolating L^3 between L^2 and L^6). Solve

$$\frac{1}{3} = \frac{\theta}{2} + \frac{1-\theta}{6},$$

so $\theta = 1/2$. The log-convex interpolation inequality gives

$$\|f\|_3 \leq \|f\|_2^{1/2} \|f\|_6^{1/2}.$$

The weights act on reciprocal exponents, which is the key bookkeeping rule.

7.8 Operator interpolation preview

Linear operators bounded at endpoint exponents often inherit intermediate bounds under an interpolation theorem.

7.9 Core structural results

Theorem 7.4 (Hölder inequality). *If $1 < p, q < \infty$ are conjugate, then $\int |fg| \leq \|f\|_p \|g\|_q$.*

Proof. Normalize the two norms to one, apply Young's inequality pointwise to the normalized product, integrate, and rescale. $\square$

Theorem 7.5 (Minkowski inequality). *For $1 \leq p < \infty$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.*

Proof. For $p > 1$, estimate $\int |f + g|^p$ by pairing $|f + g|^{p-1}$ with $|f| + |g|$ and apply Hölder; the case $p = 1$ is immediate. $\square$

Theorem 7.6 (Log-convex interpolation). *If $1/r = \theta/p + (1 - \theta)/q$, then $\|f\|_r \leq \|f\|_p^\theta \|f\|_q^{1-\theta}$.*

Proof. Factor $|f|^r$ into suitable powers of $|f|^p$ and $|f|^q$ and apply Hölder with reciprocal weights. $\square$

7.10 Worked examples

Example 7.7 (Conjugate exponents diagnostic). Give a rigorous worked analysis of Conjugate exponents in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 7.8 (Young inequality diagnostic). Give a rigorous worked analysis of Young inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 7.9 (Hölder inequality diagnostic). Give a rigorous worked analysis of Hölder inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Products of L^p and L^q functions lie in L^1 with sharp norm control. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 7.10 (Minkowski inequality diagnostic). Give a rigorous worked analysis of Minkowski inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The L^p norm satisfies the triangle inequality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

7.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 7.11 (Conjugate exponents diagnostic). Give a rigorous worked analysis of Conjugate exponents in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.12 (Young inequality diagnostic). Give a rigorous worked analysis of Young inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.13 (Hölder inequality diagnostic). Give a rigorous worked analysis of Hölder inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Products of L^p and L^q functions lie in L^1 with sharp norm control. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.14 (Minkowski inequality diagnostic). Give a rigorous worked analysis of Minkowski inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The L^p norm satisfies the triangle inequality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.15 (Equality cases diagnostic). Give a rigorous worked analysis of Equality cases in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sharpness is governed by proportionality of the powers $|f|^p$ and $|g|^q$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.16 (Discrete versions diagnostic). Give a rigorous worked analysis of Discrete versions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Counting measure recovers the familiar finite-sum Hölder and Minkowski inequalities. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.17 (Interpolation of norms diagnostic). Give a rigorous worked analysis of Interpolation of norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Intermediate L^r norms are logarithmically convex between endpoint norms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.18 (Operator interpolation preview diagnostic). Give a rigorous worked analysis of Operator interpolation preview in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Linear operators bounded at endpoint exponents often inherit intermediate bounds under an interpolation theorem. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 7.19 (Hölder inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $1 < p, q < \infty$ are conjugate, then $\int |fg| \leq \|f\|_p \|g\|_q$.

Solution. Normalize the two norms to one, apply Young's inequality pointwise to the normalized product, integrate, and rescale. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 7.20 (Minkowski inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p < \infty$, $\|f+g\|_p \leq \|f\|_p + \|g\|_p$.

Solution. For $p > 1$, estimate $\int |f+g|^p$ by pairing $|f+g|^{p-1}$ with $|f| + |g|$ and apply Hölder; the case $p = 1$ is immediate. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 7.21 (Log-convex interpolation proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $1/r = \theta/p + (1-\theta)/q$, then $\|f\|_r \leq \|f\|_p^\theta \|f\|_q^{1-\theta}$.

Solution. Factor $|f|^r$ into suitable powers of $|f|^p$ and $|f|^q$ and apply Hölder with reciprocal weights. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 7.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Hölder and Minkowski are the fundamental inequalities of L^p geometry. Interpolation then explains how information at two exponents controls intermediate exponents. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

7.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 7.23. State and apply the central principle behind Conjugate exponents. Give one immediate consequence in the setting of this chapter.

Hint. Choose conjugate exponents with $1/p + 1/q = 1$ before applying Hölder; the exponent relation is part of the theorem. $\square$

Solution. For $1 < p < \infty$, the conjugate exponent q satisfies $1/p + 1/q = 1$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.24. State and apply the central principle behind Young inequality. Give one immediate consequence in the setting of this chapter.

Hint. Normalize the functions by their L^p and L^q norms if you want a clean proof of Hölder. $\square$

Solution. The scalar bound $ab \leq a^p/p + b^q/q$ encodes the convex duality behind Hölder. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.25. State and apply the central principle behind Hölder inequality. Give one immediate consequence in the setting of this chapter.

Hint. Apply Hölder to $|f + g|^{p-1}|f|$ and $|f + g|^{p-1}|g|$ to derive Minkowski. $\square$

Solution. Products of L^p and L^q functions lie in L^1 with sharp norm control. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.26. State and apply the central principle behind Minkowski inequality. Give one immediate consequence in the setting of this chapter.

Hint. Interpolation exponents are determined by solving a convex combination of reciprocal exponents. $\square$

Solution. The L^p norm satisfies the triangle inequality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.27. State and apply the central principle behind Equality cases. Give one immediate consequence in the setting of this chapter.

Hint. Equality in Hölder requires proportionality of the appropriate powers of $|f|$ and $|g|$. $\square$

Solution. Sharpness is governed by proportionality of the powers $|f|^p$ and $|g|^q$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.28. State and apply the central principle behind Discrete versions. Give one immediate consequence in the setting of this chapter.

Hint. Use Hölder to bound an integral operator by separating the kernel factor from the input function. $\square$

Solution. Counting measure recovers the familiar finite-sum Hölder and Minkowski inequalities. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.29. State and apply the central principle behind Interpolation of norms. Give one immediate consequence in the setting of this chapter.

Hint. At $p = 1$ or $p = \infty$, check the endpoint form rather than dividing by a vanishing conjugate exponent. $\square$

Solution. Intermediate L^r norms are logarithmically convex between endpoint norms. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 7.30. State and apply the central principle behind Operator interpolation preview. Give one immediate consequence in the setting of this chapter.

Hint. Track every exponent through the estimate; a correct inequality with the wrong target exponent is not an interpolation result. $\square$

Solution. Linear operators bounded at endpoint exponents often inherit intermediate bounds under an interpolation theorem. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

7.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 7.31 (Conjugate). Find the conjugate exponent of 3.

Hint. Solve $1/3 + 1/q = 1$. $\square$

Solution. $q = 3/2$. $\square$

Exercise 7.32 (Holder bound). If L^2 norms are 3 and 4, bound the integral of $|fg|$.

Hint. Use Holder. $\square$

Solution. At most 12. $\square$

Exercise 7.33 (Interpolation). Solve $1/4 = \theta/2 + (1-\theta)/6$.

Hint. Solve linearly. $\square$

Solution. $\theta = 1/4$. $\square$

Exercise 7.34 (Minkowski). If L^p norms are 2 and 5, bound the L^p norm of $f+g$.

Hint. Use triangle inequality. $\square$

Solution. At most 7. $\square$

Exercise 7.35 (Young $p=2$). State Young's inequality for $p=q=2$.

Hint. Use the quadratic case. $\square$

Solution. $ab \leq a^2/2 + b^2/2$. $\square$

Proofs

Exercise 7.36 (Holder proof). Prove Holder from Young.

Hint. Normalize both functions. $\square$

Solution. Apply Young pointwise to normalized magnitudes, integrate, then rescale. $\square$

Exercise 7.37 (Minkowski proof). Derive Minkowski for $p > 1$.

Hint. Pair $|f+g|^{p-1}$ with $|f|+|g|$. $\square$

Solution. Holder yields the triangle inequality after division. $\square$

Exercise 7.38 (Interpolation proof). Prove the log-convex norm inequality.

Hint. Factor $|f|^r$ into two powers. □

Solution. Apply Holder with reciprocal interpolation weights. □

Exercise 7.39 (Equality). Explain the equality condition in Holder.

Hint. Trace equality in Young. □

Solution. Normalized p -th and q -th powers must match almost everywhere. □

Counterexamples and hypothesis testing

Exercise 7.40 (Wrong exponents). Why is conjugacy structural in the L_1 product estimate?

Hint. The powers must combine to exponent one. □

Solution. Without $1/p + 1/q = 1$ the standard Holder factorization does not target L_1 . □

Exercise 7.41 (p below one). Why is the p -expression not a norm for $p < 1$?

Hint. Triangle inequality fails. □

Solution. The geometry is only a quasi-norm because t^p is concave. □

Exercise 7.42 (Equality rare). Why is Holder usually strict?

Hint. Equality requires proportional powered magnitudes. □

Solution. Generic function pairs do not satisfy that relation. □

Applications and investigations

Exercise 7.43 (Moments). How does Holder bound $E|XY|$?

Hint. Use probability measure and conjugate moments. □

Solution. It is bounded by the product of the p and q moment norms. □

Exercise 7.44 (Kernel operator). How does Holder bound an integral operator at fixed x ?

Hint. Pair the kernel section with f . □

Solution. The output is bounded by the L_q norm of the kernel section times the L_p norm of f . □

Challenge problems

Exercise 7.45 (Dual extremizer). What function g should nearly attain the dual L_p norm of f ?

Hint. Use $\text{sign}(f)|f|^{p-1}$ and normalize. □

Solution. That choice has L_q norm 1 and attains Holder equality. □

Exercise 7.46 (L_1 - L_∞ to L_2). If $\|f\|_1 = 1$ and $\|f\|_\infty = 9$, bound $\|f\|_2$.

Hint. Interpolate. □

Solution. At most 3. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 8

Egorov, Vitali and Weak- L^p Ideas

Egorov, Vitali, and weak- L^p theory compare convergence modes and exceptional-set control. They clarify how almost-everywhere convergence can become nearly uniform and how tail bounds encode borderline integrability.

Learning goals

After this chapter, the reader should be able to:

- apply Egorov's theorem on finite-measure sets and track the exceptional-set measure explicitly;
- use Vitali-type covering ideas to obtain quantitative measure estimates;
- compute weak- L^p distribution-function bounds for explicit functions;
- distinguish strong L^p membership from weak- L^p membership by examples;
- translate decay of level-set measure into weak-norm estimates and back again;
- construct examples showing why finite measure, covering hypotheses, or strong-integrability assumptions cannot be dropped silently;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

8.1 Convergence in measure

A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

8.2 Egorov theorem

On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure.

Example 8.1 (Egorov in a concrete sequence). On $[0, 1]$, $x^n \rightarrow 0$ almost everywhere. Remove $E = (1 - \eta, 1]$, whose measure is η . On the complement,

$$\sup x^n = (1 - \eta)^n \rightarrow 0.$$

Thus convergence is uniform outside a set of arbitrarily small measure, exactly as Egorov predicts.

8.3 Uniform integrability

A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure.

8.4 Vitali convergence

Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses.

8.5 Weak- L^p

Weak- L^p controls tails through $\sup_{t>0} t^p \mu(|f| > t)$ rather than directly integrating $|f|^p$.

Example 8.2 (A weak- L^p critical singularity). On $(0, 1)$, let $f(x) = x^{-1/p}$. For $t \geq 1$,

$$\mu(|f| > t) = t^{-p}.$$

Hence $t^p \mu(|f| > t) = 1$, so f is weak- L^p . But

$$\int_0^1 |f|^p dx = \int_0^1 x^{-1} dx = \infty,$$

so it is not strong L^p .

8.6 Chebyshev inequality

Strong L^p bounds imply weak- L^p tail estimates.

8.7 Subsequence extraction

Convergence in measure admits almost-everywhere convergent subsequences under standard assumptions.

8.8 Borderline behavior

Power singularities often lie in weak- L^p at the exact exponent where strong L^p fails.

Example 8.3 (Borderline power behavior). For $x^{-\alpha}$ on $(0, 1)$, strong L^p requires $\alpha p < 1$. At the critical relation $\alpha p = 1$, the strong integral diverges logarithmically while the distribution function decays like t^{-p} . Weak- L^p therefore records the exact critical endpoint of the power-law scale.

8.9 Core structural results

Theorem 8.4 (Egorov theorem). *If $\mu(X) < \infty$ and $f_n \rightarrow f$ a.e., then for every $\varepsilon > 0$ there is measurable E with $\mu(E) < \varepsilon$ such that convergence is uniform on $X \setminus E$.*

Proof. For each accuracy level, form decreasing tail-exception sets. Their intersection is null; continuity from above on the finite-measure space yields a late stage with small measure, and a diagonal choice handles all accuracy levels. $\square$

Theorem 8.5 (Chebyshev inequality). *If $f \in L^p$ and $t > 0$, then $\mu(|f| > t) \leq t^{-p} \|f\|_p^p$.*

Proof. On the superlevel set, $|f|^p \geq t^p$; integrate and rearrange. $\square$

Theorem 8.6 (Vitali convergence principle). *On a finite-measure space, convergence in measure plus uniform integrability implies L^1 convergence.*

Proof. Split the error integral into the region where the error is small and its exceptional set; convergence in measure makes the exceptional set small and uniform integrability controls the mass there. $\square$

8.10 Worked examples

Example 8.7 (Convergence in measure diagnostic). Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 8.8 (Egorov theorem diagnostic). Give a rigorous worked analysis of Egorov theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 8.9 (Uniform integrability diagnostic). Give a rigorous worked analysis of Uniform integrability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 8.10 (Vitali convergence diagnostic). Give a rigorous worked analysis of Vitali convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

8.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 8.11 (Convergence in measure diagnostic). Give a rigorous worked analysis of Convergence in measure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.12 (Egorov theorem diagnostic). Give a rigorous worked analysis of Egorov theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.13 (Uniform integrability diagnostic). Give a rigorous worked analysis of Uniform integrability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.14 (Vitali convergence diagnostic). Give a rigorous worked analysis of Vitali convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.15 (Weak- L^p diagnostic). Give a rigorous worked analysis of Weak- L^p in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Weak- L^p controls tails through $\sup_{t>0} t^p \mu(|f| > t)$ rather than directly integrating $|f|^p$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.16 (Chebyshev inequality diagnostic). Give a rigorous worked analysis of Chebyshev inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Strong L^p bounds imply weak- L^p tail estimates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.17 (Subsequence extraction diagnostic). Give a rigorous worked analysis of Subsequence extraction in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convergence in measure admits almost-everywhere convergent subsequences under standard assumptions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.18 (Borderline behavior diagnostic). Give a rigorous worked analysis of Borderline behavior in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Power singularities often lie in weak- L^p at the exact exponent where strong L^p fails. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 8.19 (Egorov theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $\mu(X) < \infty$ and $f_n \rightarrow f$ a.e., then for every $\varepsilon > 0$ there is measurable E with $\mu(E) < \varepsilon$ such that convergence is uniform on $X \setminus E$.

Solution. For each accuracy level, form decreasing tail-exception sets. Their intersection is null; continuity from above on the finite-measure space yields a late stage with small measure, and a diagonal choice handles all accuracy levels. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 8.20 (Chebyshev inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^p$ and $t > 0$, then $\mu(|f| > t) \leq t^{-p} \|f\|_p^p$.

Solution. On the superlevel set, $|f|^p \geq t^p$; integrate and rearrange. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 8.21 (Vitali convergence principle proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: On a finite-measure space, convergence in measure plus uniform integrability implies L^1 convergence.

Solution. Split the error integral into the region where the error is small and its exceptional set; convergence in measure makes the exceptional set small and uniform integrability controls the mass there. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 8.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Egorov, Vitali, and weak- L^p theory compare convergence modes and exceptional-set control. They clarify how almost-everywhere convergence can become nearly uniform and how tail bounds encode borderline integrability. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

8.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 8.23. State and apply the central principle behind Convergence in measure. Give one immediate consequence in the setting of this chapter.

Hint. Egorov needs finite measure; choose the exceptional set first and make uniform convergence hold on its complement. $\square$

Solution. A sequence converges in measure when $\mu(|f_n - f| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.24. State and apply the central principle behind Egorov theorem. Give one immediate consequence in the setting of this chapter.

Hint. Quantify the exceptional set by summing the measures of the bad sets in the proof. $\square$

Solution. On finite-measure spaces, almost-everywhere convergence becomes uniform after removing a set of arbitrarily small measure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.25. State and apply the central principle behind Uniform integrability. Give one immediate consequence in the setting of this chapter.

Hint. For a Vitali covering argument, extract a disjoint subfamily and compare the original set with an expanded union of those balls. $\square$

Solution. A family is uniformly integrable when integral mass cannot concentrate on sets of very small measure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.26. State and apply the central principle behind Vitali convergence. Give one immediate consequence in the setting of this chapter.

Hint. Compute the level set $\{|f| > \lambda\}$ explicitly before evaluating the weak- L^p quantity. $\square$

Solution. Convergence in measure plus uniform integrability upgrades to L^1 convergence under the standard finite-measure hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.27. State and apply the central principle behind Weak- L^p . Give one immediate consequence in the setting of this chapter.

Hint. Weak- L^p control is $\sup_{\lambda>0} \lambda^p \mu(|f| > \lambda)$; optimize in λ only after deriving the level-set measure. $\square$

Solution. Weak- L^p controls tails through $\sup_{t>0} t^p \mu(|f| > t)$ rather than directly integrating $|f|^p$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.28. State and apply the central principle behind Chebyshev inequality. Give one immediate consequence in the setting of this chapter.

Hint. To separate weak and strong L^p , use the borderline power $|x|^{-n/p}$ on a suitable domain. $\square$

Solution. Strong L^p bounds imply weak- L^p tail estimates. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.29. State and apply the central principle behind Subsequence extraction. Give one immediate consequence in the setting of this chapter.

Hint. Strong L^p implies weak L^p by Chebyshev's inequality; write that one-line estimate first. $\square$

Solution. Convergence in measure admits almost-everywhere convergent subsequences under standard assumptions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 8.30. State and apply the central principle behind Borderline behavior. Give one immediate consequence in the setting of this chapter.

Hint. Do not use Egorov on an infinite-measure space without first restricting to a finite-measure subset. $\square$

Solution. Power singularities often lie in weak- L^p at the exact exponent where strong L^p fails. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

8.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 8.31 (Chebyshev). If $\|f\|_p=3$, bound the measure of $|f|>t$.

Hint. Use Chebyshev. $\square$

Solution. At most $3^p/t^p$. $\square$

Exercise 8.32 (Weak exponent). If a distribution tail is bounded by $4/t^2$, which weak space is indicated?

Hint. Match the exponent. $\square$

Solution. Weak L^2 . $\square$

Exercise 8.33 (Critical example). Is $x^{-1/2}$ in $L^2(0,1)$?

Hint. Integrate x^{-1} . $\square$

Solution. No, but it is weak L^2 . $\square$

Exercise 8.34 (Egorov hypothesis). What measure hypothesis does Egorov require?

Hint. Check the theorem. $\square$

Solution. Finite total measure. $\square$

Exercise 8.35 (Measure convergence). State convergence in measure.

Hint. Use bad sets. $\square$

Solution. For every ϵ , the measure of $|f_n - f| > \epsilon$ tends to zero. $\square$

Proofs

Exercise 8.36 (Chebyshev proof). Prove Chebyshev.

Hint. On the superlevel set, $|f|_p \geq t$. □

Solution. Integrate over that set and rearrange. □

Exercise 8.37 (Egorov mechanism). Explain the exceptional-set construction.

Hint. Track points where the tail is not yet uniformly accurate. □

Solution. A.e. convergence makes the ultimate bad set null; finite measure plus continuity from above gives small late bad sets. □

Exercise 8.38 (Strong to weak). Prove L_p is contained in weak L_p .

Hint. Use Chebyshev. □

Solution. The weak tail quantity is bounded by the strong L_p norm to the p . □

Exercise 8.39 (Vitali split). Explain the Vitali error split.

Hint. Separate small pointwise error from a small exceptional set. □

Solution. Finite measure controls the small-error region; uniform integrability controls the exceptional region. □

Counterexamples and hypothesis testing

Exercise 8.40 (Infinite measure Egorov). Why can Egorov fail on infinite measure?

Hint. Slow convergence can drift outward. □

Solution. No one small-measure exceptional set need contain all bad tails. □

Exercise 8.41 (Weak not strong). Give a weak- L_p function not in strong L_p .

Hint. Use $x^{-1/p}$ on $(0,1)$. □

Solution. Its tail has critical t^{-p} decay while its p -th power integral diverges. □

Exercise 8.42 (Measure not L_1). Give convergence in measure without L_1 convergence.

Hint. Use moving spikes. □

Solution. n on $(0,1/n)$ converges in measure to zero but has L_1 norm 1. □

Applications and investigations

Exercise 8.43 (Critical PDE tails). Why do weak spaces appear at critical PDE exponents?

Hint. They preserve scale-invariant tail control when strong integrability fails. □

Solution. They capture borderline kernels and singular solutions. □

Exercise 8.44 (Uniform integrability). Why is uniform integrability a concentration condition?

Hint. Small sets cannot hold large integral mass uniformly. □

Solution. It rules out moving spike behavior. □

Challenge problems

Exercise 8.45 (Weak quasi-norm). Compute the weak-L_p tail quantity for $x^{(-1/p)}$ on $(0,1)$.

Hint. Handle $t < 1$ and $t \geq 1$. □

Solution. The supremum of t^p times the superlevel measure is 1. □

Exercise 8.46 (A.e. subsequence). Why can convergence in measure yield an a.e.-convergent subsequence?

Hint. Choose a subsequence with summable bad-set measures. □

Solution. A Borel-Cantelli/diagonal argument gives eventual accuracy almost everywhere. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Part II

Fourier Analysis

Chapter 9

Fourier Series

Fourier series decompose periodic functions into orthogonal trigonometric modes. Coefficients are projections, partial sums are finite-dimensional approximations, and convergence depends on both function regularity and the geometry of the trigonometric system.

Learning goals

After this chapter, the reader should be able to:

- compute Fourier coefficients of elementary periodic functions using symmetry before integration;
- derive partial sums and identify their relation to convolution with Dirichlet-type kernels;
- apply convergence criteria at continuity points and jump discontinuities with the correct limiting value;
- use orthogonality to prove coefficient identities and energy estimates;
- distinguish pointwise, uniform, and L^2 convergence of Fourier series in representative examples;
- construct examples illustrating slow convergence or Gibbs-type behavior near jumps;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

9.1 Periodic functions

A 2π -periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates.

9.2 Orthogonality

Distinct sine and cosine modes are orthogonal on a full period.

9.3 Fourier coefficients

Coefficients are obtained by integrating against the corresponding orthogonal modes.

Example 9.1 (Sawtooth Fourier coefficients). For the periodic extension of $f(x) = x$ on $(-\pi, \pi)$, odd symmetry kills the constant and cosine coefficients. Integration by parts gives

$$b_n = \frac{2(-1)^{n+1}}{n},$$

hence

$$x \sim 2 \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \sin(nx).$$

Symmetry supplies an immediate check on the coefficient calculation.

9.4 Partial sums

The N th Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N .

9.5 Bessel inequality

Finite coefficient energy cannot exceed the L^2 energy of the function.

9.6 Dirichlet kernel

Partial sums can be written as convolution with the Dirichlet kernel, making convergence an approximate-identity problem with oscillatory kernels.

Example 9.2 (Dirichlet kernel as convolution). The partial sum is

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t) dt, \quad D_N(t) = \frac{\sin((N + \frac{1}{2})t)}{\sin(t/2)}.$$

Thus Fourier convergence is an oscillatory convolution problem, not merely a formal coefficient expansion.

9.7 Pointwise convergence

Piecewise smooth functions converge to their value at continuity points and to midpoint values at jumps under classical Dirichlet hypotheses.

9.8 Gibbs phenomenon

Near jumps, overshoot persists in relative height while its spatial width shrinks.

Example 9.3 (Gibbs overshoot). For a jump discontinuity, Fourier partial sums overshoot near the jump. As the number of modes grows, the oscillatory region narrows, but the relative overshoot height does not vanish. Consequently convergence cannot be uniform on any neighborhood containing the jump even though pointwise convergence holds away from it.

9.9 Core structural results

Theorem 9.4 (Orthogonality relations). *Distinct trigonometric modes are orthogonal in $L^2[-\pi, \pi]$.*

Proof. Use product-to-sum identities; every nonconstant integer-frequency sine or cosine integrates to zero over a full period. $\square$

Theorem 9.5 (Bessel inequality). *The square-sum of Fourier coefficients over any finite set of orthonormal modes is bounded by $\|f\|_2^2$.*

Proof. Subtract the finite orthogonal projection from f and expand the nonnegative squared norm using orthogonality. $\square$

Theorem 9.6 (Dirichlet convergence principle). *Under standard piecewise smooth hypotheses, Fourier partial sums converge at x to $(f(x^-) + f(x^+))/2$.*

Proof. Express the partial sum by the Dirichlet kernel, split the integral into a small neighborhood and its complement, use local one-sided regularity near zero and oscillatory cancellation away from zero. $\square$

9.10 Worked examples

Example 9.7 (Periodic functions diagnostic). Give a rigorous worked analysis of Periodic functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A 2π -periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 9.8 (Orthogonality diagnostic). Give a rigorous worked analysis of Orthogonality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Distinct sine and cosine modes are orthogonal on a full period. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 9.9 (Fourier coefficients diagnostic). Give a rigorous worked analysis of Fourier coefficients in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Coefficients are obtained by integrating against the corresponding orthogonal modes. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 9.10 (Partial sums diagnostic). Give a rigorous worked analysis of Partial sums in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The N th Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

9.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 9.11 (Periodic functions diagnostic). Give a rigorous worked analysis of Periodic functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A 2π -periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.12 (Orthogonality diagnostic). Give a rigorous worked analysis of Orthogonality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distinct sine and cosine modes are orthogonal on a full period. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.13 (Fourier coefficients diagnostic). Give a rigorous worked analysis of Fourier coefficients in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Coefficients are obtained by integrating against the corresponding orthogonal modes. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.14 (Partial sums diagnostic). Give a rigorous worked analysis of Partial sums in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The N th Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.15 (Bessel inequality diagnostic). Give a rigorous worked analysis of Bessel inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Finite coefficient energy cannot exceed the L^2 energy of the function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.16 (Dirichlet kernel diagnostic). Give a rigorous worked analysis of Dirichlet kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Partial sums can be written as convolution with the Dirichlet kernel, making convergence an approximate-identity problem with oscillatory kernels. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.17 (Pointwise convergence diagnostic). Give a rigorous worked analysis of Pointwise convergence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Piecewise smooth functions converge to their value at continuity points and to midpoint values at jumps under classical Dirichlet hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.18 (Gibbs phenomenon diagnostic). Give a rigorous worked analysis of Gibbs phenomenon in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Near jumps, overshoot persists in relative height while its spatial width shrinks. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 9.19 (Orthogonality relations proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Distinct trigonometric modes are orthogonal in $L^2[-\pi, \pi]$.

Solution. Use product-to-sum identities; every nonconstant integer-frequency sine or cosine integrates to zero over a full period. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 9.20 (Bessel inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: The square-sum of Fourier coefficients over any finite set of orthonormal modes is bounded by $\|f\|_2^2$.

Solution. Subtract the finite orthogonal projection from f and expand the nonnegative squared norm using orthogonality. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 9.21 (Dirichlet convergence principle proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Under standard piecewise smooth hypotheses, Fourier partial sums converge at x to $(f(x^-) + f(x^+))/2$.

Solution. Express the partial sum by the Dirichlet kernel, split the integral into a small neighborhood and its complement, use local one-sided regularity near zero and oscillatory cancellation away from zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 9.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Fourier series decompose periodic functions into orthogonal trigonometric modes. Coefficients are projections, partial sums are finite-dimensional approximations, and convergence depends on both function regularity and the geometry of the trigonometric system. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

9.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 9.23. State and apply the central principle behind Periodic functions. Give one immediate consequence in the setting of this chapter.

Hint. Use parity and periodicity before computing any Fourier coefficient; many integrals vanish immediately. $\square$

Solution. A 2π -periodic function can be viewed as a function on the circle, making trigonometric modes natural coordinates. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.24. State and apply the central principle behind Orthogonality. Give one immediate consequence in the setting of this chapter.

Hint. Write the partial sum as convolution with the Dirichlet kernel to see where localization and oscillation enter. $\square$

Solution. Distinct sine and cosine modes are orthogonal on a full period. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.25. State and apply the central principle behind Fourier coefficients. Give one immediate consequence in the setting of this chapter.

Hint. At a jump discontinuity, the limit is the average of the two one-sided limits under the standard convergence hypotheses. $\square$

Solution. Coefficients are obtained by integrating against the corresponding orthogonal modes. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.26. State and apply the central principle behind Partial sums. Give one immediate consequence in the setting of this chapter.

Hint. Orthogonality of exponentials or sines/cosines turns coefficient computations into inner-product calculations. $\square$

Solution. The N th Fourier partial sum is the orthogonal projection onto trigonometric polynomials of degree at most N . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.27. State and apply the central principle behind Bessel inequality. Give one immediate consequence in the setting of this chapter.

Hint. Parseval-type identities require the correct normalization of the basis and the period; check both constants. $\square$

Solution. Finite coefficient energy cannot exceed the L^2 energy of the function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.28. State and apply the central principle behind Dirichlet kernel. Give one immediate consequence in the setting of this chapter.

Hint. To compare pointwise and L^2 convergence, separate what is controlled by norm from what is controlled at each point. $\square$

Solution. Partial sums can be written as convolution with the Dirichlet kernel, making convergence an approximate-identity problem with oscillatory kernels. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.29. State and apply the central principle behind Pointwise convergence. Give one immediate consequence in the setting of this chapter.

Hint. Gibbs behavior is localized near jumps; inspect overshoot in scaled coordinates rather than expecting uniform convergence. $\square$

Solution. Piecewise smooth functions converge to their value at continuity points and to midpoint values at jumps under classical Dirichlet hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 9.30. State and apply the central principle behind Gibbs phenomenon. Give one immediate consequence in the setting of this chapter.

Hint. For coefficient decay, integrate by parts only when the function has the needed regularity and periodic boundary compatibility. $\square$

Solution. Near jumps, overshoot persists in relative height while its spatial width shrinks. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

9.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 9.31 (Odd parity). Which coefficients vanish for an odd periodic function?

Hint. Use parity. $\square$

Solution. The constant and cosine coefficients. $\square$

Exercise 9.32 (Single mode). What is the Fourier series of $\cos(4x)$?

Hint. It is already a basis mode. $\square$

Solution. Only the fourth cosine coefficient is nonzero. $\square$

Exercise 9.33 (Jump value). If one-sided limits are 1 and 5, what is the Dirichlet limit?

Hint. Average. $\square$

Solution. 3. $\square$

Exercise 9.34 (Orthogonality). What is the integral over a full symmetric period of $\cos(nx)\sin(mx)$?

Hint. Use parity or product identities. $\square$

Solution. 0. $\square$

Exercise 9.35 (Bessel). What does Bessel inequality compare?

Hint. Coefficient energy to function energy. ☐

Solution. Squared Fourier coefficients are bounded by the L2 energy. ☐

Proofs

Exercise 9.36 (Parity proof). Prove cosine coefficients vanish for odd f .

Hint. Odd times even is odd. ☐

Solution. The symmetric integral is zero. ☐

Exercise 9.37 (Projection). Why is $S_N f$ the best L2 degree- N trigonometric approximation?

Hint. It is an orthogonal projection. ☐

Solution. Pythagoras minimizes the residual. ☐

Exercise 9.38 (Bessel proof). Prove finite Bessel inequality.

Hint. Expand squared projection residual. ☐

Solution. Nonnegativity yields the coefficient bound. ☐

Exercise 9.39 (Midpoint principle). Explain the jump midpoint value.

Hint. Use symmetric kernel concentration. ☐

Solution. The local contribution averages the two one-sided limits. ☐

Counterexamples and hypothesis testing

Exercise 9.40 (Uniform discontinuity). Why cannot continuous partial sums converge uniformly to a discontinuous target?

Hint. Uniform limits of continuous functions are continuous. ☐

Solution. Contradiction. ☐

Exercise 9.41 (Bessel not Parseval). Why does Bessel not imply equality automatically?

Hint. Completeness is needed. ☐

Solution. Energy may remain outside the chosen orthogonal span. ☐

Exercise 9.42 (Gibbs persistence). Why does more resolution not remove relative overshoot?

Hint. The kernel has a universal rescaled jump profile. ☐

Solution. Width shrinks but normalized peak height persists. ☐

Applications and investigations

Exercise 9.43 (Spectral compression). Why do smooth periodic functions often need few Fourier modes?

Hint. Smoothness gives coefficient decay. ☐

Solution. High-frequency energy becomes small. ☐

Exercise 9.44 (Edge ringing). Why does Gibbs matter numerically?

Hint. Jumps cause persistent ringing. ☐

Solution. Reconstructed edges can overshoot even at high resolution. ☐

Challenge problems

Exercise 9.45 (Parseval identity). Use the sawtooth coefficients to obtain the classic sum of reciprocal squares.

Hint. Apply Parseval to $f(x)=x$. ☐

Solution. The resulting identity is $\sum 1/n^2 = \pi^2/6$. ☐

Exercise 9.46 (Fejer improvement). Why do Fejer means converge more robustly?

Hint. They average Dirichlet kernels into positive kernels. ☐

Solution. The Fejer kernel behaves as a genuine approximate identity. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 10

Convolution and Approximate Identities

Convolution averages one function against translates of another. It interacts cleanly with translation, integration, differentiation, and Fourier transformation, and approximate identities use it to smooth and recover functions.

Learning goals

After this chapter, the reader should be able to:

- compute convolutions of elementary functions and determine their support;
- prove basic algebraic properties of convolution under the required integrability hypotheses;
- use Young-type estimates to control convolution norms;
- show how approximate identities recover functions in the appropriate topology;
- distinguish smoothing by convolution from pointwise multiplication;
- construct mollifier-based approximations while tracking support, normalization, and convergence mode;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

10.1 Definition of convolution

For suitable functions, $(f * g)(x) = \int f(x - y)g(y)dy$.

Example 10.1 (Two interval indicators). For $f = g = 1_{[0,1]}$, the convolution equals the length of $[0, 1] \cap [x - 1, x]$. Hence it is x on $[0, 1]$, $2 - x$ on $[1, 2]$, and zero elsewhere. The triangular graph also verifies the support rule: the new support is contained in $[0, 1] + [0, 1] = [0, 2]$.

10.2 Commutativity and associativity

Under absolute-integrability hypotheses, changes of variables and Fubini prove $f * g = g * f$ and associativity.

10.3 Young inequalities

Convolution maps certain L^p pairs into controlled L^r spaces.

10.4 Translation viewpoint

Convolution is an average of translates of one factor weighted by the other.

10.5 Approximate identities

Kernels concentrating mass near zero recover functions in norm or pointwise under appropriate hypotheses.

Example 10.2 (Approximate identity recovery). For a normalized concentrating kernel ϕ_ε ,

$$f * \phi_\varepsilon(x) - f(x) = \int [f(x-y) - f(x)] \phi_\varepsilon(y) dy.$$

Near $y = 0$, continuity or translation-continuity makes the bracket small; away from zero, the kernel mass becomes small. This near/far split is the standard recovery proof.

10.6 Smoothing

Convolution with a smooth compactly supported kernel produces smooth functions.

10.7 Differentiation

Derivatives pass to the smooth kernel: $D^\alpha(f * \phi) = f * D^\alpha\phi$ under standard assumptions.

10.8 Support

For compact supports, $\text{supp}(f * g)$ lies in the Minkowski sum of the two supports.

Example 10.3 (Support under convolution). If the supports of f and g lie in A and B , a nonzero integrand $f(x-y)g(y)$ requires $x-y \in A$ and $y \in B$, hence $x \in A+B$. Therefore the support of $f * g$ is contained in the Minkowski sum of the original supports.

10.9 Core structural results

Theorem 10.4 (Commutativity). *If the convolution integral is absolutely meaningful, then $f * g = g * f$.*

Proof. Substitute $z = x - y$ in the defining integral. □

Theorem 10.5 (Young L^1 inequality). *If $f, g \in L^1(\mathbb{R}^n)$, then $f * g \in L^1$ and $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$.*

Proof. Use Tonelli on $|f(x-y)g(y)|$ and translation invariance of Lebesgue measure. □

Theorem 10.6 (Approximate-identity recovery). *If ϕ_ε is an approximate identity and $f \in L^p$, $1 \leq p < \infty$, then $f * \phi_\varepsilon \rightarrow f$ in L^p under standard normalization.*

Proof. Write the difference as an average of translations $\tau_y f - f$, use continuity of translations in L^p , concentrate kernel mass near zero, and control the far tail by kernel mass. □

10.10 Worked examples

Example 10.7 (Definition of convolution diagnostic). Give a rigorous worked analysis of Definition of convolution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For suitable functions, $(f * g)(x) = \int f(x - y)g(y)dy$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 10.8 (Commutativity and associativity diagnostic). Give a rigorous worked analysis of Commutativity and associativity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Under absolute-integrability hypotheses, changes of variables and Fubini prove $f * g = g * f$ and associativity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 10.9 (Young inequalities diagnostic). Give a rigorous worked analysis of Young inequalities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convolution maps certain L^p pairs into controlled L^r spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 10.10 (Translation viewpoint diagnostic). Give a rigorous worked analysis of Translation viewpoint in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convolution is an average of translates of one factor weighted by the other. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

10.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 10.11 (Definition of convolution diagnostic). Give a rigorous worked analysis of Definition of convolution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For suitable functions, $(f * g)(x) = \int f(x - y)g(y)dy$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.12 (Commutativity and associativity diagnostic). Give a rigorous worked analysis of Commutativity and associativity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Under absolute-integrability hypotheses, changes of variables and Fubini prove $f * g = g * f$ and associativity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.13 (Young inequalities diagnostic). Give a rigorous worked analysis of Young inequalities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution maps certain L^p pairs into controlled L^r spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.14 (Translation viewpoint diagnostic). Give a rigorous worked analysis of Translation viewpoint in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution is an average of translates of one factor weighted by the other. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.15 (Approximate identities diagnostic). Give a rigorous worked analysis of Approximate identities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Kernels concentrating mass near zero recover functions in norm or pointwise under appropriate hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.16 (Smoothing diagnostic). Give a rigorous worked analysis of Smoothing in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution with a smooth compactly supported kernel produces smooth functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.17 (Differentiation diagnostic). Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Derivatives pass to the smooth kernel: $D^\alpha(f * \phi) = f * D^\alpha\phi$ under standard assumptions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.18 (Support diagnostic). Give a rigorous worked analysis of Support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For compact supports, $\text{supp}(f * g)$ lies in the Minkowski sum of the two supports. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 10.19 (Commutativity proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If the convolution integral is absolutely meaningful, then $f * g = g * f$.

Solution. Substitute $z = x - y$ in the defining integral. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 10.20 (Young L^1 inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f, g \in L^1(\mathbb{R}^n)$, then $f * g \in L^1$ and $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$.

Solution. Use Tonelli on $|f(x-y)g(y)|$ and translation invariance of Lebesgue measure. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 10.21 (Approximate-identity recovery proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If ϕ_ε is an approximate identity and $f \in L^p$, $1 \leq p < \infty$, then $f * \phi_\varepsilon \rightarrow f$ in L^p under standard normalization.

Solution. Write the difference as an average of translations $\tau_y f - f$, use continuity of translations in L^p , concentrate kernel mass near zero, and control the far tail by kernel mass. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 10.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Convolution averages one function against translates of another. It interacts cleanly with translation, integration, differentiation, and Fourier transformation, and approximate identities use it to smooth and recover functions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

10.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 10.23. State and apply the central principle behind Definition of convolution. Give one immediate consequence in the setting of this chapter.

Hint. Start from $(f * g)(x) = \int f(x-y)g(y)dy$ and check integrability before changing variables. $\square$

Solution. For suitable functions, $(f * g)(x) = \int f(x-y)g(y)dy$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.24. State and apply the central principle behind Commutativity and associativity. Give one immediate consequence in the setting of this chapter.

Hint. The support of a convolution lies in the Minkowski sum of supports; prove inclusion before looking for equality. $\square$

Solution. Under absolute-integrability hypotheses, changes of variables and Fubini prove $f * g = g * f$ and associativity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.25. State and apply the central principle behind Young inequalities. Give one immediate consequence in the setting of this chapter.

Hint. Use Fubini/Tonelli to justify associativity or commutativity rather than rearranging integrals formally. $\square$

Solution. Convolution maps certain L^p pairs into controlled L^r spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.26. State and apply the central principle behind Translation viewpoint. Give one immediate consequence in the setting of this chapter.

Hint. Young's inequality controls $\|f * g\|_r$; choose exponents satisfying the correct reciprocal relation. $\square$

Solution. Convolution is an average of translates of one factor weighted by the other. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.27. State and apply the central principle behind Approximate identities. Give one immediate consequence in the setting of this chapter.

Hint. For an approximate identity, verify unit mass, concentration near zero, and uniform L^1 control. $\square$

Solution. Kernels concentrating mass near zero recover functions in norm or pointwise under appropriate hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.28. State and apply the central principle behind Smoothing. Give one immediate consequence in the setting of this chapter.

Hint. Split the approximation error into near and far regions so continuity handles one part and concentration handles the other. $\square$

Solution. Convolution with a smooth compactly supported kernel produces smooth functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.29. State and apply the central principle behind Differentiation. Give one immediate consequence in the setting of this chapter.

Hint. A mollifier smooths because derivatives fall on the smooth kernel; differentiate under the integral after checking justification. $\square$

Solution. Derivatives pass to the smooth kernel: $D^\alpha(f * \phi) = f * D^\alpha\phi$ under standard assumptions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 10.30. State and apply the central principle behind Support. Give one immediate consequence in the setting of this chapter.

Hint. Keep support control explicit when using mollifiers on a domain; convolution near the boundary may leave the domain. $\square$

Solution. For compact supports, $\text{supp}(f * g)$ lies in the Minkowski sum of the two supports. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

10.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 10.31 (Support sum). What support bound holds for indicators of $[0,2]$ and $[0,3]$?

Hint. Add supports. ☐

Solution. The convolution is supported in $[0,5]$. ☐

Exercise 10.32 (Commutativity). What substitution proves convolution commutativity?

Hint. Use $z=x-y$. ☐

Solution. It interchanges the factors. ☐

Exercise 10.33 (Young L1). If L1 norms are 2 and 5, bound the convolution L1 norm.

Hint. Use Young. ☐

Solution. At most 10. ☐

Exercise 10.34 (Mollifier mass). What is the integral of a normalized scaled mollifier?

Hint. Use change of variables. ☐

Solution. 1. ☐

Exercise 10.35 (Derivative). Where does a derivative go in f convolved with a smooth ϕ ?

Hint. Move it to the smooth factor. ☐

Solution. $D(f*\phi)=f*(D\phi)$, under standard hypotheses. ☐

Proofs

Exercise 10.36 (Commutativity proof). Prove $f*g=g*f$.

Hint. Change variables. ☐

Solution. Substitute $z=x-y$. ☐

Exercise 10.37 (Young proof). Prove the L1 convolution inequality.

Hint. Use Tonelli and translation invariance. ☐

Solution. The double absolute integral factors into the product of L1 norms. ☐

Exercise 10.38 (Support proof). Prove support inclusion.

Hint. A nonzero integrand imposes two support constraints. ☐

Solution. Those constraints imply x is in $A+B$. ☐

Exercise 10.39 (Recovery proof). Outline approximate-identity convergence in L_p .

Hint. Average translation differences. ☐

Solution. Use L_p continuity of translations near zero and small far kernel mass. ☐

Counterexamples and hypothesis testing

Exercise 10.40 (Product versus convolution). Why is convolution not pointwise multiplication?

Hint. Convolution averages translates. ☐

Solution. It mixes nearby locations and broadens support. ☐

Exercise 10.41 (Associativity hypotheses). Why are absolute-integrability hypotheses important for associativity?

Hint. A triple integral is rearranged. ☐

Solution. Fubini must be justified. ☐

Exercise 10.42 (Support equality). Why may support inclusion be strict?

Hint. Cancellation can occur. ☐

Solution. Contributions may cancel on part of the Minkowski sum. ☐

Applications and investigations

Exercise 10.43 (Smoothing). Why does a mollifier smooth rough data?

Hint. It averages against a smooth kernel. ☐

Solution. Derivatives fall on the kernel, creating smooth outputs. ☐

Exercise 10.44 (Weak solutions). Why use mollification in PDE?

Hint. It produces smooth approximants. ☐

Solution. One proves estimates in the smooth setting and passes to a limit. ☐

Challenge problems

Exercise 10.45 (Uniform error). For uniformly continuous f and compactly supported mollifiers, how can sup error be bounded?

Hint. Use modulus of continuity on the small kernel support. ☐

Solution. The error is at most $\omega_f(C\epsilon)$, which tends to zero. ☐

Exercise 10.46 (Associativity). Prove convolution associativity for L^1 functions.

Hint. Write the double integral and use Fubini. ☐

Solution. Absolute integrability allows regrouping and variable changes. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 11

The Fourier Transform

The Fourier transform converts translation and differentiation into multiplication, convolution into products, and spatial localization into frequency information. It is the central transform chapter of Volume III and is developed with extra structural depth.

Learning goals

After this chapter, the reader should be able to:

- compute Fourier transforms of elementary functions under the chapter's normalization convention;
- derive translation, modulation, scaling, and differentiation identities from the transform definition;
- identify when the transform is continuous, bounded, or decaying from integrability and regularity assumptions;
- compute inverse transforms in examples where an inversion theorem applies;
- use convolution-multiplication duality to simplify transform calculations;
- check transform formulas by dimensional, symmetry, or inverse-transform consistency tests rather than memorization alone;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

11.1 Definition on L^1

Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx$ for integrable functions.

11.2 Boundedness and continuity

The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \leq \|f\|_1$.

11.3 Riemann–Lebesgue lemma

Fourier transforms of L^1 functions vanish at infinity.

11.4 Translation and modulation

Translation in physical space creates a phase factor in frequency; modulation shifts the transform.

Example 11.1 (Translation versus modulation). If $g(x) = f(x - a)$, then

$$\widehat{g}(\xi) = e^{-2\pi i a \xi} \widehat{f}(\xi).$$

Translation changes phase but not transform magnitude. By contrast multiplying f by $e^{2\pi i b x}$ shifts the transform to $\widehat{f}(\xi - b)$. These two dual identities are best remembered together.

11.5 Scaling

Dilation in space inversely dilates frequency and contributes the Jacobian factor.

Example 11.2 (Scaling rule). For $g(x) = f(ax)$,

$$\widehat{g}(\xi) = |a|^{-1} \widehat{f}(\xi/a).$$

The Jacobian factor and reciprocal frequency scaling are both essential. Checking the formula at $\xi = 0$ immediately verifies the factor: $\widehat{g}(0) = |a|^{-1} \int f$.

11.6 Differentiation and moments

Differentiation becomes multiplication by $2\pi i \xi$, while multiplication by x becomes differentiation in frequency.

Example 11.3 (Differentiation becomes multiplication). Under sufficient decay,

$$\widehat{f'}(\xi) = 2\pi i \xi \widehat{f}(\xi).$$

Integration by parts proves the identity. Thus constant-coefficient differential operators become algebraic multipliers in frequency, which is the basic reason Fourier transform is powerful for PDE.

11.7 Convolution theorem

Convolution transforms into pointwise multiplication, and products transform into convolution when hypotheses permit.

11.8 Inversion

Under suitable integrability or Schwartz hypotheses, applying the inverse transform reconstructs the original function.

11.9 Symmetry and reality

Even, odd, real, and radial structures impose corresponding symmetry on the transform.

11.10 Uncertainty viewpoint

Strong concentration in space conflicts with strong concentration in frequency, foreshadowing quantitative uncertainty principles.

11.11 Core structural results

Theorem 11.4 (L^1 bound). *For $f \in L^1$, $|\widehat{f}(\xi)| \leq \|f\|_1$ for every ξ .*

Proof. Take absolute values inside the integral; the complex exponential has modulus one. $\square$

Theorem 11.5 (Riemann–Lebesgue lemma). *If $f \in L^1(\mathbb{R}^n)$, then $\widehat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.*

Proof. First prove it for smooth compactly supported functions by integration by parts or oscillatory cancellation, then approximate an arbitrary L^1 function in L^1 and use the uniform transform bound. $\square$

Theorem 11.6 (Convolution theorem). *If $f, g \in L^1$, then $\widehat{f * g} = \widehat{f} \widehat{g}$.*

Proof. Insert the convolution integral, use Fubini justified by absolute integrability, change variables $u = x - y$, and factor the two resulting integrals. $\square$

Theorem 11.7 (Fourier inversion on Schwartz space). *For $f \in \mathcal{S}(\mathbb{R}^n)$, inverse transformation recovers f .*

Proof. Regularize with a Gaussian multiplier, use Fubini to identify the regularized inverse as convolution with an approximate identity, and pass to the limit in the Schwartz or uniform topology. $\square$

11.12 Worked examples

Example 11.8 (Definition on L^1 diagnostic). Give a rigorous worked analysis of Definition on L^1 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx$ for integrable functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 11.9 (Boundedness and continuity diagnostic). Give a rigorous worked analysis of Boundedness and continuity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \leq \|f\|_1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 11.10 (Riemann–Lebesgue lemma diagnostic). Give a rigorous worked analysis of Riemann–Lebesgue lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Fourier transforms of L^1 functions vanish at infinity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 11.11 (Translation and modulation diagnostic). Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Translation in physical space creates a phase

factor in frequency; modulation shifts the transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

11.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 11.12 (Definition on L^1 diagnostic). Give a rigorous worked analysis of Definition on L^1 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fix the convention $\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x \cdot \xi} dx$ for integrable functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.13 (Boundedness and continuity diagnostic). Give a rigorous worked analysis of Boundedness and continuity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The transform of an L^1 function is bounded and uniformly continuous, with $\|\hat{f}\|_\infty \leq \|f\|_1$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.14 (Riemann–Lebesgue lemma diagnostic). Give a rigorous worked analysis of Riemann–Lebesgue lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fourier transforms of L^1 functions vanish at infinity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.15 (Translation and modulation diagnostic). Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Translation in physical space creates a phase factor in frequency; modulation shifts the transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.16 (Scaling diagnostic). Give a rigorous worked analysis of Scaling in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Dilation in space inversely dilates frequency and contributes the Jacobian factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.17 (Differentiation and moments diagnostic). Give a rigorous worked analysis of Differentiation and moments in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Differentiation becomes multiplication by $2\pi i\xi$, while multiplication by x becomes differentiation in frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.18 (Convolution theorem diagnostic). Give a rigorous worked analysis of Convolution theorem in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution transforms into pointwise multiplication, and products transform into convolution when hypotheses permit. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 11.19 (L^1 bound proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $f \in L^1$, $|\widehat{f}(\xi)| \leq \|f\|_1$ for every ξ .

Solution. Take absolute values inside the integral; the complex exponential has modulus one. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 11.20 (Riemann–Lebesgue lemma proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^1(\mathbb{R}^n)$, then $\widehat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

Solution. First prove it for smooth compactly supported functions by integration by parts or oscillatory cancellation, then approximate an arbitrary L^1 function in L^1 and use the uniform transform bound. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 11.21 (Convolution theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f, g \in L^1$, then $\widehat{f * g} = \widehat{f} \widehat{g}$.

Solution. Insert the convolution integral, use Fubini justified by absolute integrability, change variables $u = x - y$, and factor the two resulting integrals. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 11.22 (Fourier inversion on Schwartz space proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $f \in \mathcal{S}(\mathbb{R}^n)$, inverse transformation recovers f .

Solution. Regularize with a Gaussian multiplier, use Fubini to identify the regularized inverse as convolution with an approximate identity, and pass to the limit in the Schwartz or uniform topology. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 11.23 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Fourier transform converts translation and differentiation into multiplication, convolution into products, and spatial localization into frequency information. It is the central transform chapter of Volume III and is developed with extra structural depth. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

11.14 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 11.24. State and apply the central principle behind Definition on L^1 . Give one immediate consequence in the setting of this chapter.

Hint. Fix the transform convention first; every scaling and inversion constant depends on it. $\square$

Solution. Fix the convention $\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x \cdot \xi} dx$ for integrable functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.25. State and apply the central principle behind Boundedness and continuity. Give one immediate consequence in the setting of this chapter.

Hint. For translation, substitute $y = x - a$ and keep the resulting exponential phase factor. $\square$

Solution. The transform of an L^1 function is bounded and uniformly continuous, with $\|\widehat{f}\|_\infty \leq \|f\|_1$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.26. State and apply the central principle behind Riemann–Lebesgue lemma. Give one immediate consequence in the setting of this chapter.

Hint. For modulation, multiply by the exponential first and read the frequency shift directly from the kernel. $\square$

Solution. Fourier transforms of L^1 functions vanish at infinity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.27. State and apply the central principle behind Translation and modulation. Give one immediate consequence in the setting of this chapter.

Hint. For scaling, change variables carefully and track the absolute determinant or scalar factor. $\square$

Solution. Translation in physical space creates a phase factor in frequency; modulation shifts the transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.28. State and apply the central principle behind Scaling. Give one immediate consequence in the setting of this chapter.

Hint. Integration by parts moves differentiation in space to multiplication by frequency, with boundary terms killed by the hypotheses. $\square$

Solution. Dilation in space inversely dilates frequency and contributes the Jacobian factor. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.29. State and apply the central principle behind Differentiation and moments. Give one immediate consequence in the setting of this chapter.

Hint. Convolution becomes multiplication after transforming; use this before attempting a direct convolution integral. $\square$

Solution. Differentiation becomes multiplication by $2\pi i\xi$, while multiplication by x becomes differentiation in frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.30. State and apply the central principle behind Convolution theorem. Give one immediate consequence in the setting of this chapter.

Hint. Check inversion only under the theorem's hypotheses; a formal inverse integral need not converge pointwise. $\square$

Solution. Convolution transforms into pointwise multiplication, and products transform into convolution when hypotheses permit. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 11.31. State and apply the central principle behind Inversion. Give one immediate consequence in the setting of this chapter.

Hint. Use symmetry and reality conditions to predict whether the transform is even, odd, real, or conjugate-symmetric. $\square$

Solution. Under suitable integrability or Schwartz hypotheses, applying the inverse transform reconstructs the original function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

11.15 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 11.32 (L1 bound). If the L1 norm is 7, bound the transform sup norm.

Hint. Use the L1 transform bound. $\square$

Solution. At most 7. $\square$

Exercise 11.33 (Translation). Transform $f(x-a)$.

Hint. Use translation. $\square$

Solution. Multiply the transform by the phase $\exp(-2\pi i a \xi)$. $\square$

Exercise 11.34 (Modulation). Transform $\exp(2\pi i b x) f(x)$.

Hint. Use modulation. $\square$

Solution. Shift the transform to $\hat{f}(\xi-b)$. $\square$

Exercise 11.35 (Scaling). Transform $f(2x)$.

Hint. Use scaling. ☐

Solution. One half times $\hat{f}(\xi/2)$. ☐

Exercise 11.36 (Derivative). Transform f' .

Hint. Integrate by parts. ☐

Solution. $2\pi i \xi$ times $\hat{f}$. ☐

Proofs

Exercise 11.37 (L1 bound proof). Prove the transform L1 bound.

Hint. Take absolute values inside the integral. ☐

Solution. The complex exponential has modulus one. ☐

Exercise 11.38 (Translation proof). Derive the translation rule.

Hint. Change variables. ☐

Solution. A constant phase factor remains. ☐

Exercise 11.39 (Convolution theorem). Prove transform of convolution equals product.

Hint. Use Fubini and a change of variables. ☐

Solution. The double integral factors into the two transforms. ☐

Exercise 11.40 (Derivative proof). Derive the derivative identity.

Hint. Integrate by parts. ☐

Solution. Boundary terms vanish under the decay hypotheses. ☐

Counterexamples and hypothesis testing

Exercise 11.41 (Riemann-Lebesgue converse). Is every continuous function vanishing at infinity the transform of an L1 function?

Hint. The lemma is one-way. ☐

Solution. No; the Fourier image of L1 has additional constraints. ☐

Exercise 11.42 (Scaling error). Why is omitting the Jacobian in scaling impossible?

Hint. Check the transform at zero. ☐

Solution. The integral scales by $|a|^{-1}$. ☐

Exercise 11.43 (Inversion hypothesis). Why is pointwise inversion not automatic for every L1 function?

Hint. Inversion requires extra hypotheses. ☐

Solution. L1 alone gives a bounded continuous transform, not unconditional pointwise recovery. ☐

Applications and investigations

Exercise 11.44 (Filtering). Why does convolution filtering become multiplication in frequency?

Hint. Use the convolution theorem. ☐

Solution. The filter kernel becomes a frequency multiplier. ☐

Exercise 11.45 (Constant-coefficient PDE). Why does Fourier transform simplify such PDE?

Hint. Derivatives become polynomial multipliers. ☐

Solution. The differential equation becomes algebraic in frequency. ☐

Challenge problems

Exercise 11.46 (Moment identity). Derive the frequency derivative identity involving $x f(x)$.

Hint. Differentiate under the integral. ☐

Solution. Differentiating the exponential produces $-2\pi i x$. ☐

Exercise 11.47 (Uncertainty scaling). Explain the narrow-space/broad-frequency heuristic.

Hint. Use reciprocal scaling. ☐

Solution. Compressing f in x evaluates its transform at ξ/a and spreads the frequency profile. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 12

The Gaussian and Transform Calculus

The Gaussian is the model function of transform calculus: it is smooth, rapidly decaying, and mapped to another Gaussian. Differentiation, scaling, and tensor-product arguments then generate a large transform table.

Learning goals

After this chapter, the reader should be able to:

- derive the Fourier transform of a Gaussian and track normalization constants correctly;
- use differentiation and multiplication identities to compute transforms of Gaussian-weighted polynomials;
- apply completion-of-the-square arguments in transform integrals;
- derive heat-kernel and semigroup formulas from Gaussian transform calculus;
- compare spatial and frequency widths quantitatively for Gaussian families;
- construct transform identities by differentiating with respect to parameters when justified;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

12.1 Gaussian integrals

The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization.

12.2 Self-reciprocity

With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform.

Example 12.1 (Gaussian self-transform). Let $G(x) = e^{-\pi x^2}$ and $F = \widehat{G}$. Differentiating F and integrating by parts gives

$$F'(\xi) = -2\pi\xi F(\xi).$$

Hence $F(\xi) = Ce^{-\pi\xi^2}$. The Gaussian integral gives $F(0) = 1$, so $C = 1$. Therefore the standard Gaussian is its own Fourier transform.

12.3 Scaling

Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor.

Example 12.2 (Scaled Gaussian). For $a > 0$,

$$e^{-\pi ax^2} = G(\sqrt{a}x),$$

so Fourier scaling yields

$$\widehat{e^{-\pi ax^2}}(\xi) = a^{-1/2}e^{-\pi\xi^2/a}.$$

Spatial narrowing and frequency broadening are reciprocal.

12.4 Differentiated Gaussians

Polynomial times Gaussian functions arise by differentiating in space or frequency.

12.5 Heat kernels

Gaussian kernels are the fundamental averaging kernels of the heat equation.

Example 12.3 (Heat semigroup). The heat kernel has Fourier multiplier

$$e^{-4\pi^2 t |\xi|^2}.$$

Multipliers at times s and t multiply to the one at time $s + t$, so Fourier inversion gives

$$G_t * G_s = G_{t+s}.$$

This is the semigroup property of heat flow.

12.6 Tensor products

The n -dimensional transform of a product of one-dimensional Gaussians factors coordinate-wise.

12.7 Hermite structure

Repeated differentiation of Gaussians produces Hermite-polynomial factors and a transform-invariant basis structure.

12.8 Transform tables

Translation, modulation, scaling, differentiation, and convolution generate many explicit transform pairs from a few seeds.

12.9 Core structural results

Theorem 12.4 (Gaussian transform). *Under the convention $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \xi} dx$, $e^{-\pi x^2}$ transforms to $e^{-\pi \xi^2}$.*

Proof. Differentiate the transform with respect to ξ , integrate by parts to obtain an ODE, solve the ODE, and determine the constant from the Gaussian integral at zero. $\square$

Theorem 12.5 (Scaling rule). *If $f_a(x) = f(ax)$ with $a \neq 0$, then $\widehat{f}_a(\xi) = |a|^{-1} \widehat{f}(\xi/a)$ in one dimension.*

Proof. Substitute $u = ax$ in the transform integral and account for the Jacobian. $\square$

Theorem 12.6 (Heat-kernel semigroup). *Gaussian heat kernels satisfy $G_t * G_s = G_{t+s}$.*

Proof. Fourier transformation turns convolution into multiplication, while $\widehat{G}_t$ is an exponential multiplier whose product at times t, s equals the multiplier at $t + s$; invert the transform. $\square$

12.10 Worked examples

Example 12.7 (Gaussian integrals diagnostic). Give a rigorous worked analysis of Gaussian integrals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 12.8 (Self-reciprocity diagnostic). Give a rigorous worked analysis of Self-reciprocity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 12.9 (Scaling diagnostic). Give a rigorous worked analysis of Scaling in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 12.10 (Differentiated Gaussians diagnostic). Give a rigorous worked analysis of Differentiated Gaussians in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Polynomial times Gaussian functions arise by differentiating in space or frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

12.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 12.11 (Gaussian integrals diagnostic). Give a rigorous worked analysis of Gaussian integrals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.12 (Self-reciprocity diagnostic). Give a rigorous worked analysis of Self-reciprocity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.13 (Scaling diagnostic). Give a rigorous worked analysis of Scaling in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.14 (Differentiated Gaussians diagnostic). Give a rigorous worked analysis of Differentiated Gaussians in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Polynomial times Gaussian functions arise by differentiating in space or frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.15 (Heat kernels diagnostic). Give a rigorous worked analysis of Heat kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Gaussian kernels are the fundamental averaging kernels of the heat equation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.16 (Tensor products diagnostic). Give a rigorous worked analysis of Tensor products in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The n -dimensional transform of a product of one-dimensional Gaussians factors coordinatewise. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.17 (Hermite structure diagnostic). Give a rigorous worked analysis of Hermite structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Repeated differentiation of Gaussians produces Hermite-polynomial factors and a transform-invariant basis structure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.18 (Transform tables diagnostic). Give a rigorous worked analysis of Transform tables in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Translation, modulation, scaling, differentiation, and convolution generate many explicit transform pairs from a few seeds. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 12.19 (Gaussian transform proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Under the convention $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \xi} dx$, $e^{-\pi x^2}$ transforms to $e^{-\pi \xi^2}$.

Solution. Differentiate the transform with respect to ξ , integrate by parts to obtain an ODE, solve the ODE, and determine the constant from the Gaussian integral at zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 12.20 (Scaling rule proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f_a(x) = f(ax)$ with $a \neq 0$, then $\widehat{f_a}(\xi) = |a|^{-1} \widehat{f}(\xi/a)$ in one dimension.

Solution. Substitute $u = ax$ in the transform integral and account for the Jacobian. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 12.21 (Heat-kernel semigroup proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Gaussian heat kernels satisfy $G_t * G_s = G_{t+s}$.

Solution. Fourier transformation turns convolution into multiplication, while $\widehat{G}_t$ is an exponential multiplier whose product at times t, s equals the multiplier at $t + s$; invert the transform. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 12.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Gaussian is the model function of transform calculus: it is smooth, rapidly decaying, and mapped to another Gaussian. Differentiation, scaling, and tensor-product arguments then generate a large transform table. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

12.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 12.23. State and apply the central principle behind Gaussian integrals. Give one immediate consequence in the setting of this chapter.

Hint. Complete the square in the exponent and compare the transformed Gaussian with the original Gaussian family. $\square$

Solution. The integral of $e^{-\pi x^2}$ over the line is one under the standard normalization. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.24. State and apply the central principle behind Self-reciprocity. Give one immediate consequence in the setting of this chapter.

Hint. Differentiate the Gaussian transform identity with respect to frequency to generate transforms of $x^k e^{-ax^2}$. $\square$

Solution. With the 2π Fourier convention, $e^{-\pi|x|^2}$ is its own Fourier transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.25. State and apply the central principle behind Scaling. Give one immediate consequence in the setting of this chapter.

Hint. Multiplication by x in physical space becomes differentiation in frequency; apply the rule instead of reintegrating from scratch. $\square$

Solution. Scaled Gaussians transform to reciprocal-width Gaussians with the appropriate determinant factor. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.26. State and apply the central principle behind Differentiated Gaussians. Give one immediate consequence in the setting of this chapter.

Hint. Differentiate in the Gaussian parameter when justified to produce moments or heat-kernel identities. $\square$

Solution. Polynomial times Gaussian functions arise by differentiating in space or frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.27. State and apply the central principle behind Heat kernels. Give one immediate consequence in the setting of this chapter.

Hint. For the heat kernel, transform the PDE so the time evolution becomes a scalar ODE in frequency. $\square$

Solution. Gaussian kernels are the fundamental averaging kernels of the heat equation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.28. State and apply the central principle behind Tensor products. Give one immediate consequence in the setting of this chapter.

Hint. Check mass conservation by integrating the Gaussian kernel or evaluating its transform at zero frequency. $\square$

Solution. The n -dimensional transform of a product of one-dimensional Gaussians factors coordinatewise. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.29. State and apply the central principle behind Hermite structure. Give one immediate consequence in the setting of this chapter.

Hint. Spatial narrowing corresponds to frequency widening; derive the scaling law quantitatively. $\square$

Solution. Repeated differentiation of Gaussians produces Hermite-polynomial factors and a transform-invariant basis structure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 12.30. State and apply the central principle behind Transform tables. Give one immediate consequence in the setting of this chapter.

Hint. Normalize constants by testing the formula at a parameter value whose integral is already known. $\square$

Solution. Translation, modulation, scaling, differentiation, and convolution generate many explicit transform pairs from a few seeds. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

12.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 12.31 (Gaussian integral). What is the normalized Gaussian integral?

Hint. Use the chapter normalization. $\square$

Solution. Integral of $\exp(-\pi x^2)$ is 1. $\square$

Exercise 12.32 (Self transform). What is the transform of $\exp(-\pi x^2)$?

Hint. Use self-reciprocity. $\square$

Solution. It is $\exp(-\pi \xi^2)$. $\square$

Exercise 12.33 (Scaled Gaussian). Transform $\exp(-4 \pi x^2)$.

Hint. Use $a=4$. $\square$

Solution. One half times $\exp(-\pi \xi^2/4)$. $\square$

Exercise 12.34 (Derivative). Differentiate $\exp(-\pi x^2)$.

Hint. Use chain rule. $\square$

Solution. $-2 \pi x \exp(-\pi x^2)$. $\square$

Exercise 12.35 (Heat multiplier). State the heat multiplier.

Hint. Recall transform solution. $\square$

Solution. $\exp(-4 \pi^2 t |\xi|^2)$. $\square$

Proofs

Exercise 12.36 (Gaussian ODE). Derive the ODE for the Gaussian transform.

Hint. Differentiate in frequency and integrate by parts. □

Solution. $F' = -2\pi i \xi F$. □

Exercise 12.37 (Scaled formula). Prove the scaled transform formula.

Hint. Apply Fourier scaling to $G(\sqrt{a}x)$. □

Solution. It gives $a^{-1/2} G(\xi/\sqrt{a})$. □

Exercise 12.38 (Semigroup proof). Prove the heat-kernel semigroup.

Hint. Transform convolution. □

Solution. The exponential multipliers multiply and times add. □

Exercise 12.39 (Tensor product). Why does the multidimensional Gaussian transform factor?

Hint. Both Gaussian and kernel split by coordinates. □

Solution. Fubini turns the integral into a product of one-dimensional transforms. □

Counterexamples and hypothesis testing

Exercise 12.40 (Normalization). Why do Gaussian constants depend on Fourier convention?

Hint. The placement of 2π changes scaling. □

Solution. Self-reciprocity in this exact form is convention-dependent. □

Exercise 12.41 (Differentiate under integral). Why must the step be justified?

Hint. Need integrable control on moments. □

Solution. Without domination the formal derivative may be invalid. □

Exercise 12.42 (Width). Why is every scaled Gaussian not literally self-identical under transform?

Hint. Scaling is reciprocal. □

Solution. Only the normalized unit-width case has the same width. □

Applications and investigations

Exercise 12.43 (Heat smoothing). Why does heat flow suppress high frequencies?

Hint. Inspect the multiplier. □

Solution. Large frequencies are exponentially damped. □

Exercise 12.44 (Gaussian filtering). Why is repeated Gaussian smoothing still Gaussian?

Hint. Use the semigroup law. □

Solution. The time/variance parameters add. □

Challenge problems

Exercise 12.45 (Hermite structure). Why do repeated Gaussian derivatives remain polynomial times Gaussian?

Hint. Induct with product rule. □

Solution. Each derivative changes the polynomial and adds a linear Gaussian factor. □

Exercise 12.46 (Parameter derivative). How can differentiating in a generate transforms involving x^2 times a Gaussian?

Hint. Differentiate the Gaussian family before and after transform. □

Solution. The a derivative inserts a factor proportional to $-x^2$, generating the transform identity. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 13

Plancherel and L^2 Fourier Theory

Plancherel theory extends the Fourier transform from integrable functions to a unitary operator on L^2 . Orthogonality and energy conservation become the main language, allowing transform methods beyond pointwise integral formulas.

Learning goals

After this chapter, the reader should be able to:

- apply Plancherel's theorem to convert L^2 norm or inner-product problems into frequency-space calculations;
- extend the Fourier transform from an integrable dense subclass to L^2 ;
- use Parseval identities to compute energies and orthogonality relations;
- distinguish L^1 -based pointwise transform statements from L^2 -equivalence-class statements;
- prove uniqueness results from vanishing Fourier transform in L^2 ;
- solve elementary convolution or multiplier problems by exploiting the unitary L^2 transform structure;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

13.1 Parseval identity

For suitable functions, the inner product is preserved by the Fourier transform.

13.2 Plancherel extension

Density of Schwartz functions extends the transform uniquely to all of L^2 .

Example 13.1 (Extension by completion). Choose Schwartz $f_n \rightarrow f$ in L^2 . On Schwartz space,

$$\|\widehat{f_n} - \widehat{f_m}\|_2 = \|f_n - f_m\|_2,$$

so the transformed sequence is Cauchy. Define $\widehat{f}$ as its L^2 limit. The isometry also proves independence of the approximating sequence, giving a unique unitary extension.

13.3 Unitarity

The L^2 transform preserves norm and inner product.

13.4 Inverse transform in L^2

Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable.

13.5 Convolution in L^2 settings

Combining L^1 and L^2 hypotheses yields useful convolution bounds and multiplier formulas.

13.6 Multipliers

Bounded frequency multipliers define bounded operators on L^2 .

Example 13.2 (Bounded multiplier). For

$$Tf = \mathcal{F}^{-1}(m\hat{f}), \quad m \in L^\infty,$$

Plancherel gives

$$\|Tf\|_2 = \|m\hat{f}\|_2 \leq \|m\|_\infty \|f\|_2.$$

Thus a frequency-domain sup bound is exactly the operator control needed in L^2 .

13.7 Energy methods

Differentiation identities translate Sobolev-type derivative energy into weighted frequency energy.

13.8 Approximation

Frequency cutoffs approximate L^2 functions and provide smooth or band-limited regularizations.

Example 13.3 (Frequency cutoffs approximate energy). Let

$$\hat{f}_R = 1_{\{|\xi| \leq R\}} \hat{f}.$$

Then

$$\|f - f_R\|_2^2 = \int_{|\xi| > R} |\hat{f}(\xi)|^2 d\xi \rightarrow 0.$$

Band-limited functions therefore approximate arbitrary L^2 functions in energy, even when pointwise transform formulas are unavailable.

13.9 Core structural results

Theorem 13.4 (Plancherel theorem). *The Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$.*

Proof. Prove norm preservation on Schwartz functions, use density of Schwartz space in L^2 to define the extension by completion, and preserve the norm in the limit. $\square$

Theorem 13.5 (Parseval identity). *For $f, g \in L^2$, $\langle \widehat{f}, \widehat{g} \rangle = \langle f, g \rangle$.*

Proof. Use polarization of the norm identity obtained from Plancherel. $\square$

Theorem 13.6 (L^2 multiplier bound). *If $m \in L^\infty$ and $Tf = \mathcal{F}^{-1}(m\widehat{f})$, then $\|Tf\|_2 \leq \|m\|_\infty \|f\|_2$.*

Proof. Apply Plancherel and the pointwise bound $|m\widehat{f}| \leq \|m\|_\infty |\widehat{f}|$. $\square$

13.10 Worked examples

Example 13.7 (Parseval identity diagnostic). Give a rigorous worked analysis of Parseval identity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For suitable functions, the inner product is preserved by the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 13.8 (Plancherel extension diagnostic). Give a rigorous worked analysis of Plancherel extension in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Density of Schwartz functions extends the transform uniquely to all of L^2 . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 13.9 (Unitarity diagnostic). Give a rigorous worked analysis of Unitarity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The L^2 transform preserves norm and inner product. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 13.10 (Inverse transform in L^2 diagnostic). Give a rigorous worked analysis of Inverse transform in L^2 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

13.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 13.11 (Parseval identity diagnostic). Give a rigorous worked analysis of Parseval identity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For suitable functions, the inner product is preserved by the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.12 (Plancherel extension diagnostic). Give a rigorous worked analysis of Plancherel extension in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Density of Schwartz functions extends the transform uniquely to all of L^2 . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.13 (Unitarity diagnostic). Give a rigorous worked analysis of Unitarity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The L^2 transform preserves norm and inner product. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.14 (Inverse transform in L^2 diagnostic). Give a rigorous worked analysis of Inverse transform in L^2 in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.15 (Convolution in L^2 settings diagnostic). Give a rigorous worked analysis of Convolution in L^2 settings in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Combining L^1 and L^2 hypotheses yields useful convolution bounds and multiplier formulas. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.16 (Multipliers diagnostic). Give a rigorous worked analysis of Multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Bounded frequency multipliers define bounded operators on L^2 . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.17 (Energy methods diagnostic). Give a rigorous worked analysis of Energy methods in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Differentiation identities translate Sobolev-type derivative energy into weighted frequency energy. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.18 (Approximation diagnostic). Give a rigorous worked analysis of Approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Frequency cutoffs approximate L^2 functions and provide smooth or band-limited regularizations. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 13.19 (Plancherel theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: The Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$.

Solution. Prove norm preservation on Schwartz functions, use density of Schwartz space in L^2 to define the extension by completion, and preserve the norm in the limit. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 13.20 (Parseval identity proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $f, g \in L^2$, $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$.

Solution. Use polarization of the norm identity obtained from Plancherel. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 13.21 (L^2 multiplier bound proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $m \in L^\infty$ and $Tf = \mathcal{F}^{-1}(m\hat{f})$, then $\|Tf\|_2 \leq \|m\|_\infty \|f\|_2$.

Solution. Apply Plancherel and the pointwise bound $|m\hat{f}| \leq \|m\|_\infty |\hat{f}|$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 13.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Plancherel theory extends the Fourier transform from integrable functions to a unitary operator on L^2 . Orthogonality and energy conservation become the main language, allowing transform methods beyond pointwise integral formulas. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

13.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 13.23. State and apply the central principle behind Parseval identity. Give one immediate consequence in the setting of this chapter.

Hint. Use Plancherel to move the inner product to frequency space before estimating the transformed expression. $\square$

Solution. For suitable functions, the inner product is preserved by the Fourier transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.24. State and apply the central principle behind Plancherel extension. Give one immediate consequence in the setting of this chapter.

Hint. Define the L^2 transform by density from Schwartz or $L^1 \cap L^2$ functions, not by assuming the integral exists pointwise. $\square$

Solution. Density of Schwartz functions extends the transform uniquely to all of L^2 . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.25. State and apply the central principle behind Unitarity. Give one immediate consequence in the setting of this chapter.

Hint. Parseval is the inner-product form of Plancherel; polarize the norm identity if necessary. $\square$

Solution. The L^2 transform preserves norm and inner product. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.26. State and apply the central principle behind Inverse transform in L^2 . Give one immediate consequence in the setting of this chapter.

Hint. Two L^2 functions that differ on a null set represent the same transform class, so avoid pointwise claims unless extra regularity is present. $\square$

Solution. Inversion holds in the Hilbert-space sense even when neither function nor transform is integrable. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.27. State and apply the central principle behind Convolution in L^2 settings. Give one immediate consequence in the setting of this chapter.

Hint. If the Fourier transform vanishes in L^2 , Plancherel forces the original function to have zero L^2 norm. $\square$

Solution. Combining L^1 and L^2 hypotheses yields useful convolution bounds and multiplier formulas. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.28. State and apply the central principle behind Multipliers. Give one immediate consequence in the setting of this chapter.

Hint. Multipliers bounded in frequency define bounded L^2 operators; estimate by the essential supremum of the multiplier. $\square$

Solution. Bounded frequency multipliers define bounded operators on L^2 . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.29. State and apply the central principle behind Energy methods. Give one immediate consequence in the setting of this chapter.

Hint. For convolution in L^2 , verify the other factor provides enough integrability to make the product formula legitimate. $\square$

Solution. Differentiation identities translate Sobolev-type derivative energy into weighted frequency energy. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 13.30. State and apply the central principle behind Approximation. Give one immediate consequence in the setting of this chapter.

Hint. Check normalization constants by applying Plancherel to a Gaussian, where both sides can be computed independently. $\square$

Solution. Frequency cutoffs approximate L^2 functions and provide smooth or band-limited regularizations. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

13.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 13.31 (Energy). If $\|f\|_2=5$, what is $\|\hat{f}\|_2$?

Hint. Use Plancherel. $\square$

Solution. 5. $\square$

Exercise 13.32 (Multiplier). If $\|m\|_\infty=3$, bound $\|Tf\|_2$.

Hint. Use multiplier bound. $\square$

Solution. At most $3\|f\|_2$. $\square$

Exercise 13.33 (Parseval). What happens to inner products under Fourier transform in L^2 ?

Hint. Use Parseval. $\square$

Solution. They are preserved. $\square$

Exercise 13.34 (Cutoff error). Express the L^2 error of a sharp frequency cutoff.

Hint. Use Plancherel. $\square$

Solution. It is the L^2 mass of $\hat{f}$ outside the cutoff. $\square$

Exercise 13.35 (Zero transform). If $\hat{f}=0$ in L^2 , what is f ?

Hint. Use unitarity. $\square$

Solution. $f=0$ almost everywhere. $\square$

Proofs

Exercise 13.36 (Extension uniqueness). Why is the L^2 extension unique?

Hint. Schwartz space is dense and transform is continuous. $\square$

Solution. Two continuous extensions agreeing on a dense set agree everywhere. $\square$

Exercise 13.37 (Parseval from norm). Derive inner-product preservation from norm preservation.

Hint. Use polarization. ☐

Solution. Polarization reconstructs the inner product from norms. ☐

Exercise 13.38 (Multiplier proof). Prove the multiplier bound.

Hint. Use Plancherel and pointwise multiplier bound. ☐

Solution. Norm Tf equals norm $m \hat{f}$, bounded by $\|m\|_{\infty} \|\hat{f}\|_2$. ☐

Exercise 13.39 (Cutoff convergence). Prove sharp cutoffs converge in L^2 .

Hint. Use integrable tail of $|\hat{f}|^2$. ☐

Solution. The tail integral tends to zero. ☐

Counterexamples and hypothesis testing

Exercise 13.40 (L^2 not L^1). Why may an L^2 transform lack a pointwise integral formula?

Hint. L^2 does not imply L^1 on the whole real line. ☐

Solution. The transform is then defined as an equivalence class by completion. ☐

Exercise 13.41 (Unbounded multiplier). Why can an unbounded multiplier fail to be a bounded L^2 operator?

Hint. Concentrate spectral mass where m is large. ☐

Solution. Operator norms would become arbitrarily large. ☐

Exercise 13.42 (Pointwise identity). Why are L^2 transform identities generally almost-everywhere statements?

Hint. L^2 elements are equivalence classes. ☐

Solution. Point values are not canonical. ☐

Applications and investigations

Exercise 13.43 (Energy filter). How does a spectral mask change energy?

Hint. Use Plancherel. ☐

Solution. Output energy is the L^2 energy of the masked spectrum. ☐

Exercise 13.44 (Derivative energy). Why does gradient energy become weighted spectral energy?

Hint. Fourier derivative multiplies by frequency. ☐

Solution. The weight is proportional to $|\xi|^2$. ☐

Challenge problems

Exercise 13.45 (Projection). Show a sharp frequency cutoff is an orthogonal projection.

Hint. Its multiplier is a real indicator with $m^2=m$. □

Solution. The operator is self-adjoint and idempotent. □

Exercise 13.46 (Sobolev preview). Express L^2 plus gradient energy in frequency terms.

Hint. Use Plancherel and derivative rule. □

Solution. It is an integral of $(1+4\pi^2|\xi|^2)|\hat{f}|^2$. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 14

The Schwartz Space

The Schwartz space packages smoothness and rapid decay into a Fréchet-type test-function space stable under Fourier transformation. It is the natural common domain for classical transform calculus and tempered distributions.

Learning goals

After this chapter, the reader should be able to:

- verify that a function belongs to the Schwartz space by bounding all polynomially weighted derivatives;
- show that differentiation, multiplication by polynomials, and Fourier transformation preserve the Schwartz space;
- construct seminorm estimates controlling standard Schwartz-space operations;
- distinguish rapid decay from compact support and from mere integrability;
- use density of Schwartz functions in relevant L^p or L^2 settings where established;
- produce examples that lie in one standard function space but not another by comparing decay and smoothness requirements;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

14.1 Rapid decay

A Schwartz function decays faster than every polynomial together with all derivatives.

14.2 Schwartz seminorms

Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology.

Example 14.1 (Gaussian is Schwartz). Every derivative of e^{-x^2} is a polynomial times e^{-x^2} . Therefore for every pair of nonnegative integers m, k ,

$$\sup_x |x^m f^{(k)}(x)| < \infty,$$

because exponential decay dominates polynomial growth. This verifies all Schwartz seminorms directly.

14.3 Algebraic stability

Multiplication by polynomials and differentiation preserve Schwartz space.

14.4 Translation and modulation

Translations and frequency modulations preserve the class.

14.5 Fourier invariance

The Fourier transform maps Schwartz space bijectively to itself.

Example 14.2 (Fourier invariance through identities). Frequency differentiation corresponds to multiplication by powers of x , while frequency multiplication corresponds to spatial differentiation. Thus every weighted derivative of $\hat{f}$ is the transform of an integrable Schwartz expression built from f . The L^1 transform bound then controls all Schwartz seminorms of $\hat{f}$.

14.6 Convolution

Schwartz functions convolve to Schwartz functions; convolution with appropriate integrable functions smooths while retaining decay.

14.7 Density

Schwartz space is dense in L^p for finite p and in many Sobolev spaces.

14.8 Role in distribution theory

Its rapid decay makes pairing with polynomial-growth objects well-defined, leading to tempered distributions.

Example 14.3 (Why tempered distributions use Schwartz tests). A polynomial-growth function can be paired with a Schwartz test function because rapid decay beats every polynomial. For example

$$T(\phi) = \int (1 + x^2)^N \phi(x) dx$$

is finite and controlled by Schwartz seminorms. This lets distribution theory include slowly growing objects while preserving Fourier transform operations.

14.9 Core structural results

Theorem 14.4 (Fourier invariance). *If $f \in \mathcal{S}$, then $\hat{f} \in \mathcal{S}$.*

Proof. Differentiation in frequency corresponds to multiplication by powers of x , while multiplication by powers of ξ corresponds to derivatives of f ; all such weighted derivatives are bounded and integrable for Schwartz functions. $\square$

Theorem 14.5 (Algebraic stability). *If $f \in \mathcal{S}$, then $x^\alpha D^\beta f \in \mathcal{S}$.*

Proof. Every Schwartz seminorm of this expression is a finite linear combination of higher Schwartz seminorms of f . $\square$

Theorem 14.6 (Schwartz convolution closure). *If $f, g \in \mathcal{S}$, then $f * g \in \mathcal{S}$.*

Proof. Differentiate under the integral and distribute polynomial weights between $x - y$ and y ; rapid decay of both factors controls every resulting seminorm. $\square$

14.10 Worked examples

Example 14.7 (Rapid decay diagnostic). Give a rigorous worked analysis of Rapid decay in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A Schwartz function decays faster than every polynomial together with all derivatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 14.8 (Schwartz seminorms diagnostic). Give a rigorous worked analysis of Schwartz seminorms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 14.9 (Algebraic stability diagnostic). Give a rigorous worked analysis of Algebraic stability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Multiplication by polynomials and differentiation preserve Schwartz space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 14.10 (Translation and modulation diagnostic). Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Translations and frequency modulations preserve the class. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

14.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 14.11 (Rapid decay diagnostic). Give a rigorous worked analysis of Rapid decay in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A Schwartz function decays faster than every polynomial together with all derivatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.12 (Schwartz seminorms diagnostic). Give a rigorous worked analysis of Schwartz seminorms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.13 (Algebraic stability diagnostic). Give a rigorous worked analysis of Algebraic stability in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplication by polynomials and differentiation preserve Schwartz space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.14 (Translation and modulation diagnostic). Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Translations and frequency modulations preserve the class. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.15 (Fourier invariance diagnostic). Give a rigorous worked analysis of Fourier invariance in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Fourier transform maps Schwartz space bijectively to itself. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.16 (Convolution diagnostic). Give a rigorous worked analysis of Convolution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Schwartz functions convolve to Schwartz functions; convolution with appropriate integrable functions smooths while retaining decay. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.17 (Density diagnostic). Give a rigorous worked analysis of Density in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Schwartz space is dense in L^p for finite p and in many Sobolev spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.18 (Role in distribution theory diagnostic). Give a rigorous worked analysis of Role in distribution theory in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Its rapid decay makes pairing with polynomial-growth objects well-defined, leading to tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 14.19 (Fourier invariance proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in \mathcal{S}$, then $\widehat{f} \in \mathcal{S}$.

Solution. Differentiation in frequency corresponds to multiplication by powers of x , while multiplication by powers of ξ corresponds to derivatives of f ; all such weighted derivatives are bounded and integrable for Schwartz functions. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 14.20 (Algebraic stability proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in \mathcal{S}$, then $x^\alpha D^\beta f \in \mathcal{S}$.

Solution. Every Schwartz seminorm of this expression is a finite linear combination of higher Schwartz seminorms of f . This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 14.21 (Schwartz convolution closure proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f, g \in \mathcal{S}$, then $f * g \in \mathcal{S}$.

Solution. Differentiate under the integral and distribute polynomial weights between $x - y$ and y ; rapid decay of both factors controls every resulting seminorm. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 14.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Schwartz space packages smoothness and rapid decay into a Fréchet-type test-function space stable under Fourier transformation. It is the natural common domain for classical transform calculus and tempered distributions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

14.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 14.23. State and apply the central principle behind Rapid decay. Give one immediate consequence in the setting of this chapter.

Hint. To prove Schwartz decay, bound $x^\alpha \partial^\beta f(x)$ uniformly for every pair of multi-indices. $\square$

Solution. A Schwartz function decays faster than every polynomial together with all derivatives. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.24. State and apply the central principle behind Schwartz seminorms. Give one immediate consequence in the setting of this chapter.

Hint. Differentiation preserves the Schwartz class because it only changes the derivative index in the seminorms. $\square$

Solution. Quantities $\sup_x |x^\alpha D^\beta f(x)|$ encode the topology. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.25. State and apply the central principle behind Algebraic stability. Give one immediate consequence in the setting of this chapter.

Hint. Polynomial multiplication preserves Schwartz decay by shifting the polynomial weight index. $\square$

Solution. Multiplication by polynomials and differentiation preserve Schwartz space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.26. State and apply the central principle behind Translation and modulation. Give one immediate consequence in the setting of this chapter.

Hint. For the Fourier transform, integrate by parts to trade powers of frequency for derivatives of the original function. $\square$

Solution. Translations and frequency modulations preserve the class. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.27. State and apply the central principle behind Fourier invariance. Give one immediate consequence in the setting of this chapter.

Hint. Compact support implies rapid decay only after Fourier transformation; it is not the same property as being Schwartz in physical space. $\square$

Solution. The Fourier transform maps Schwartz space bijectively to itself. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.28. State and apply the central principle behind Convolution. Give one immediate consequence in the setting of this chapter.

Hint. Use the Schwartz seminorms to quantify convergence instead of relying on pointwise or uniform convergence alone. $\square$

Solution. Schwartz functions convolve to Schwartz functions; convolution with appropriate integrable functions smooths while retaining decay. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.29. State and apply the central principle behind Density. Give one immediate consequence in the setting of this chapter.

Hint. A function can be Schwartz without compact support; the Gaussian is the standard test case. $\square$

Solution. Schwartz space is dense in L^p for finite p and in many Sobolev spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 14.30. State and apply the central principle behind Role in distribution theory. Give one immediate consequence in the setting of this chapter.

Hint. To show a map is continuous on Schwartz space, estimate each output seminorm by finitely many input seminorms. $\square$

Solution. Its rapid decay makes pairing with polynomial-growth objects well-defined, leading to tempered distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

14.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 14.31 (Gaussian). Is $\exp(-x^2)$ a Schwartz function?

Hint. Polynomially weighted derivatives are bounded. ☐

Solution. Yes. ☐

Exercise 14.32 (Compact support). Is every smooth compactly supported function Schwartz?

Hint. Outside support it vanishes. ☐

Solution. Yes. ☐

Exercise 14.33 (Polynomial). Is x^2 Schwartz?

Hint. Check rapid decay. ☐

Solution. No. ☐

Exercise 14.34 (Derivative closure). If f is Schwartz, what about f' ?

Hint. Use seminorms. ☐

Solution. It is Schwartz. ☐

Exercise 14.35 (Polynomial multiplication). If f is Schwartz, what about $x^3 f$?

Hint. Shift polynomial weights. ☐

Solution. It is Schwartz. ☐

Proofs

Exercise 14.36 (Derivative stability). Prove differentiation preserves Schwartz space.

Hint. Each seminorm becomes another higher derivative seminorm. ☐

Solution. All remain finite. ☐

Exercise 14.37 (Polynomial stability). Prove multiplication by a polynomial preserves Schwartz space.

Hint. Combine polynomial weights. ☐

Solution. Every new seminorm is controlled by finitely many original seminorms. ☐

Exercise 14.38 (Fourier invariance). Outline the seminorm proof.

Hint. Convert frequency moments and derivatives to spatial derivatives and moments. ☐

Solution. All resulting expressions are integrable and have bounded transforms. ☐

Exercise 14.39 (Convolution closure). Explain why convolution of two Schwartz functions is Schwartz.

Hint. Differentiate under the integral and distribute polynomial weights. ☐

Solution. Rapid decay of both factors controls every seminorm. ☐

Counterexamples and hypothesis testing

Exercise 14.40 (Rapid not compact). Give a Schwartz function without compact support.

Hint. Use a Gaussian. ☐

Solution. The Gaussian is nonzero everywhere but rapidly decaying. ☐

Exercise 14.41 (L1 not Schwartz). Give an integrable function that is not Schwartz.

Hint. Use an interval indicator. ☐

Solution. It is not smooth. ☐

Exercise 14.42 (Smooth not Schwartz). Give a smooth function that is not Schwartz.

Hint. Use the constant function 1. ☐

Solution. It has no decay. ☐

Applications and investigations

Exercise 14.43 (Test space). Why is Schwartz space natural for Fourier distributions?

Hint. It is smooth, rapidly decaying, and Fourier-invariant. ☐

Solution. All key operations remain in one stable test space. ☐

Exercise 14.44 (Regularization). Why approximate rough data by Schwartz functions?

Hint. They are dense in many Lp/Sobolev settings and easy to manipulate. ☐

Solution. Prove estimates for smooth approximants and pass to limits. ☐

Challenge problems

Exercise 14.45 (Translation). Why does translation preserve Schwartz space?

Hint. Expand powers of x using $x-a$ plus a . ☐

Solution. Translated seminorms are controlled by finite sums of original seminorms. ☐

Exercise 14.46 (Modulation). Why does multiplying by $\exp(2\pi i b x)$ preserve Schwartz space?

Hint. Differentiate with Leibniz; the modulation has unit magnitude. ☐

Solution. Each derivative is a finite sum of constants times derivatives of f , so weighted bounds remain finite. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Part III

Distribution Theory

Chapter 15

Test-Function Spaces

Test-function spaces provide compactly supported smooth probes. Their topology is stronger and more local than the Schwartz topology, and distributions are defined as continuous linear functionals on these probes.

Learning goals

After this chapter, the reader should be able to:

- verify that a smooth function has compact support inside the prescribed open set;
- construct test functions by rescaling, translating, multiplying, and patching standard bumps;
- track support under multiplication, differentiation, and coordinate changes;
- distinguish convergence in test-function space from pointwise or uniform convergence alone;
- build cutoff functions equal to one on a compact set and supported in a larger open set;
- explain why compact support eliminates boundary terms in distributional integration by parts;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

15.1 Smooth compact support

$C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω .

15.2 Bump functions

Smooth cutoffs allow localization without sharp discontinuities.

Example 15.1 (A normalized one-dimensional bump). Define

$$\eta(x) = \begin{cases} \exp\left(-\frac{1}{1-x^2}\right), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Every derivative tends to zero as $x \rightarrow \pm 1$, so the zero extension is C^∞ and has support $[-1, 1]$. Translating and scaling,

$$\eta_{a,r}(x) = \eta\left(\frac{x-a}{r}\right),$$

produces a smooth probe supported in $[a-r, a+r]$. This is the basic localization building block used throughout distribution theory.

15.3 Support control

A sequence converging in the test-function topology eventually has support inside one fixed compact set.

Example 15.2 (Why fixed support is part of test-function convergence). Let $\phi_j(x) = \eta(x-j)$, where η is a fixed nonzero bump. Every derivative has the same sup norm for all j , but the supports drift to infinity. Thus ϕ_j does not converge to zero in $C_c^\infty(\mathbb{R})$, even though on each fixed compact set it is eventually zero. Test-function convergence requires eventual containment of all supports in one compact set together with uniform convergence of every derivative there.

15.4 Derivative control

Within that compact set, every derivative converges uniformly.

15.5 Partitions of unity

Locally finite smooth partitions subordinate to open covers allow local constructions to be patched globally.

Example 15.3 (Patching local data with a partition of unity). Suppose U_1, U_2 cover a compact set and smooth cutoffs $\psi_1, \psi_2 \geq 0$ satisfy $\psi_1 + \psi_2 > 0$ on that set. Define

$$\chi_j = \frac{\psi_j}{\psi_1 + \psi_2}.$$

Then $\chi_1 + \chi_2 = 1$, each χ_j is smooth, and its support stays inside the corresponding localization region after the cutoffs are chosen subordinate to the cover. A global object can therefore be decomposed into local pieces and reconstructed by summation.

15.6 Localization

Multiplying by a test function isolates distributional behavior in a chosen region.

15.7 Comparison with Schwartz space

Compact support implies rapid decay after extension by zero, but the topologies and intended duals differ.

15.8 Dual space preview

The continuous dual of $C_c^\infty(\Omega)$ is the space of distributions on Ω .

15.9 Core structural results

Theorem 15.4 (Existence of bump functions). *Given compact $K \subset U$ with U open in $\mathbb{R}^n$, there is $\phi \in C_c^\infty(U)$ with $0 \leq \phi \leq 1$ and $\phi = 1$ near K .*

Proof. Convolve the indicator of a slightly enlarged neighborhood of K with a sufficiently small smooth mollifier, then truncate the geometric construction so the support stays inside U . $\square$

Theorem 15.5 (Partition of unity). *Every open cover of a paracompact smooth Euclidean domain admits a smooth locally finite partition of unity subordinate to it.*

Proof. Refine the cover by locally finite relatively compact sets, choose bump functions on the refinement, and normalize their locally finite positive sum. $\square$

Theorem 15.6 (Test-function convergence). *If $\phi_j \rightarrow 0$ in $C_c^\infty(\Omega)$, then all supports eventually lie in one compact $K \Subset \Omega$ and every derivative converges uniformly on K .*

Proof. This is the defining sequential behavior of the standard LF topology and is precisely the control needed for distributional continuity. $\square$

15.10 Worked examples

Example 15.7 (Smooth compact support diagnostic). Give a rigorous worked analysis of Smooth compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. $C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 15.8 (Bump functions diagnostic). Give a rigorous worked analysis of Bump functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Smooth cutoffs allow localization without sharp discontinuities. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 15.9 (Support control diagnostic). Give a rigorous worked analysis of Support control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A sequence converging in the test-function topology eventually has support inside one fixed compact set. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 15.10 (Derivative control diagnostic). Give a rigorous worked analysis of Derivative control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Within that compact set, every derivative converges uniformly. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

15.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 15.11 (Smooth compact support diagnostic). Give a rigorous worked analysis of Smooth compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.12 (Bump functions diagnostic). Give a rigorous worked analysis of Bump functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Smooth cutoffs allow localization without sharp discontinuities. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.13 (Support control diagnostic). Give a rigorous worked analysis of Support control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sequence converging in the test-function topology eventually has support inside one fixed compact set. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.14 (Derivative control diagnostic). Give a rigorous worked analysis of Derivative control in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Within that compact set, every derivative converges uniformly. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.15 (Partitions of unity diagnostic). Give a rigorous worked analysis of Partitions of unity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Locally finite smooth partitions subordinate to open covers allow local constructions to be patched globally. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.16 (Localization diagnostic). Give a rigorous worked analysis of Localization in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplying by a test function isolates distributional behavior in a chosen region. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.17 (Comparison with Schwartz space diagnostic). Give a rigorous worked analysis of Comparison with Schwartz space in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Compact support implies rapid decay after extension by zero, but the topologies and intended duals differ. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.18 (Dual space preview diagnostic). Give a rigorous worked analysis of Dual space preview in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The continuous dual of $C_c^\infty(\Omega)$ is the space of distributions on Ω . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 15.19 (Existence of bump functions proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Given compact $K \subset U$ with U open in $\mathbb{R}^n$, there is $\phi \in C_c^\infty(U)$ with $0 \leq \phi \leq 1$ and $\phi = 1$ near K .

Solution. Convolve the indicator of a slightly enlarged neighborhood of K with a sufficiently small smooth mollifier, then truncate the geometric construction so the support stays inside U . This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 15.20 (Partition of unity proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Every open cover of a paracompact smooth Euclidean domain admits a smooth locally finite partition of unity subordinate to it.

Solution. Refine the cover by locally finite relatively compact sets, choose bump functions on the refinement, and normalize their locally finite positive sum. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 15.21 (Test-function convergence proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $\phi_j \rightarrow 0$ in $C_c^\infty(\Omega)$, then all supports eventually lie in one compact $K \Subset \Omega$ and every derivative converges uniformly on K .

Solution. This is the defining sequential behavior of the standard LF topology and is precisely the control needed for distributional continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 15.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Test-function spaces provide compactly supported smooth probes. Their topology is stronger and more local than the Schwartz topology, and distributions are defined as continuous linear functionals on these probes. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

15.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 15.23. State and apply the central principle behind Smooth compact support. Give one immediate consequence in the setting of this chapter.

Hint. To construct a bump, start from the standard smooth function $e^{-1/(1-|x|^2)}$ inside the unit ball and zero outside. $\square$

Solution. $C_c^\infty(\Omega)$ consists of infinitely differentiable functions whose supports are compact subsets of the open set Ω . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.24. State and apply the central principle behind Bump functions. Give one immediate consequence in the setting of this chapter.

Hint. Rescaling and translating a bump moves its support; compute the transformed support explicitly. $\square$

Solution. Smooth cutoffs allow localization without sharp discontinuities. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.25. State and apply the central principle behind Support control. Give one immediate consequence in the setting of this chapter.

Hint. Products of test functions remain test functions, with support contained in the intersection of the supports. $\square$

Solution. A sequence converging in the test-function topology eventually has support inside one fixed compact set. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.26. State and apply the central principle behind Derivative control. Give one immediate consequence in the setting of this chapter.

Hint. Differentiation does not enlarge support; use this when preparing integration-by-parts arguments. $\square$

Solution. Within that compact set, every derivative converges uniformly. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.27. State and apply the central principle behind Partitions of unity. Give one immediate consequence in the setting of this chapter.

Hint. Convergence in C_c^∞ requires eventual support in one fixed compact set plus uniform convergence of every derivative. $\square$

Solution. Locally finite smooth partitions subordinate to open covers allow local constructions to be patched globally. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.28. State and apply the central principle behind Localization. Give one immediate consequence in the setting of this chapter.

Hint. To build a cutoff equal to one on $K \subset U$, cover K by finitely many smaller neighborhoods and patch bumps. $\square$

Solution. Multiplying by a test function isolates distributional behavior in a chosen region. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.29. State and apply the central principle behind Comparison with Schwartz space. Give one immediate consequence in the setting of this chapter.

Hint. Compact support kills boundary terms because the test function and all needed derivatives vanish outside a bounded region. $\square$

Solution. Compact support implies rapid decay after extension by zero, but the topologies and intended duals differ. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 15.30. State and apply the central principle behind Dual space preview. Give one immediate consequence in the setting of this chapter.

Hint. Do not confuse a sequence of supports shrinking or escaping to infinity with convergence in the test-function topology. $\square$

Solution. The continuous dual of $C_c^\infty(\Omega)$ is the space of distributions on Ω . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

15.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 15.31 (Scaled support). A bump η is supported in $[-1,1]$. What is the support of $\eta((x-a)/r)$ for $r>0$?

Hint. Solve $|(x-a)/r| \leq 1$. $\square$

Solution. It is contained in $[a-r, a+r]$. $\square$

Exercise 15.32 (Derivative support). How does differentiation affect support of a smooth compactly supported function?

Hint. A derivative vanishes wherever the function is identically zero on a neighborhood. $\square$

Solution. The derivative support is contained in the original support. $\square$

Exercise 15.33 (Product support). Where is the support of $\phi \psi$?

Hint. A nonzero product requires both factors nonzero. $\square$

Solution. It is contained in $\text{supp}(\phi) \cap \text{supp}(\psi)$. $\square$

Exercise 15.34 (Translation). Does translating a test function preserve test-function status?

Hint. Smoothness and compactness are invariant under translation. $\square$

Solution. Yes; support is translated by the same vector. $\square$

Exercise 15.35 (Compact cutoff). What should a cutoff for K compactly contained in U satisfy?

Hint. Recall the bump theorem. □

Solution. It should be smooth, compactly supported in U , and equal to one near K . □

Proofs

Exercise 15.36 (Boundary terms). Explain why integration by parts against a test function has no boundary term at infinity.

Hint. The test function and all derivatives vanish outside a compact set. □

Solution. The ordinary boundary contribution is zero outside that compact support. □

Exercise 15.37 (Product closure). Prove the product of two test functions is a test function.

Hint. Use smoothness and compact support. □

Solution. The product is smooth and its support lies in the compact intersection of the supports. □

Exercise 15.38 (Fixed-support criterion). Explain why a sequence with supports escaping to infinity cannot converge in test-function space.

Hint. Test convergence requires one common eventual compact support. □

Solution. No such compact set can contain all escaping supports. □

Exercise 15.39 (Partition identity). Prove a locally finite partition of unity can be multiplied into a test function and summed back to it.

Hint. Only finitely many terms meet the compact support. □

Solution. The sum is locally finite and ϕ times the partition sums to ϕ . □

Counterexamples and hypothesis testing

Exercise 15.40 (Pointwise not test convergence). Give a sequence tending pointwise to zero that does not converge in test-function topology.

Hint. Translate one fixed bump to infinity. □

Solution. The supports fail the fixed-compact requirement. □

Exercise 15.41 (Uniform alone insufficient). Why does uniform convergence of ϕ_j to zero not imply test-function convergence?

Hint. Derivatives also need uniform control and supports must remain fixed. □

Solution. A rapidly oscillating sequence can have small amplitude but large derivatives. □

Exercise 15.42 (Compact support not preserved by arbitrary composition). Why can composing a compactly supported function with a nonproper map lose compact support?

Hint. The inverse image of a compact support may be noncompact. □

Solution. Support control under coordinate changes needs a proper/local diffeomorphism type hypothesis. □

Applications and investigations

Exercise 15.43 (Localization). Why multiply a distribution by a cutoff equal to one near a point?

Hint. The cutoff removes distant behavior without changing local action. $\square$

Solution. It isolates the distribution near the point or compact region. $\square$

Exercise 15.44 (Numerical patching). Why are partitions of unity useful in local finite-element or manifold constructions?

Hint. Local basis functions can be assembled into a global object. $\square$

Solution. The partition ensures compatibility and exact reconstruction. $\square$

Challenge problems

Exercise 15.45 (Nested cutoffs). Given K compactly contained in U , explain how to choose cutoffs χ and ψ with $\chi=1$ near K and $\psi=1$ near $\text{supp}(\chi)$, both supported in U .

Hint. Use two nested compact neighborhoods between K and the boundary of U . $\square$

Solution. Apply the bump theorem twice to the nested neighborhoods. $\square$

Exercise 15.46 (Test topology versus Schwartz). Why is every compactly supported smooth function Schwartz after zero extension, yet the test-function topology is not the Schwartz topology?

Hint. Schwartz convergence allows supports to spread if weighted seminorms shrink, while test convergence fixes eventual compact support. $\square$

Solution. The underlying sets overlap, but the topologies encode different localization behavior and have different duals. $\square$

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 16

Distributions and Distributional Derivatives

Distributions extend the derivative and transform calculus to nonsmooth objects by duality. A distribution acts continuously and linearly on test functions, and derivatives are transferred onto the test function by integration by parts.

Learning goals

After this chapter, the reader should be able to:

- verify that a linear functional defines a distribution by checking continuity on test functions;
- embed locally integrable functions into distributions and compute their action on test functions;
- compute distributional derivatives by transferring derivatives to test functions with the correct sign;
- derive the derivative of the Heaviside function and related jump distributions;
- prove convergence of distributions by testing against an arbitrary fixed test function;
- apply smooth multiplication and differential operators to distributions without reverting to pointwise intuition;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

16.1 Distributions

A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$.

16.2 Regular distributions

Locally integrable functions define distributions by integration against test functions.

16.3 Dirac mass

Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function.

Example 16.1 (Dirac is continuous on test functions). For a fixed x_0 , define

$$\delta_{x_0}(\phi) = \phi(x_0).$$

If test functions $\phi_j \rightarrow 0$ with supports in one compact set, then

$$|\delta_{x_0}(\phi_j)| \leq \|\phi_j\|_\infty \rightarrow 0.$$

Thus point evaluation is a continuous linear functional and hence a distribution. It is singular because no locally integrable function can represent exact point evaluation against every test function.

16.4 Distributional derivatives

Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$.

Example 16.2 (The jump of Heaviside becomes a Dirac mass). Let $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x > 0$. For a test function ϕ ,

$$H'(\phi) = - \int_0^\infty \phi'(x) dx = \phi(0).$$

Therefore

$$H' = \delta_0$$

in the distributional sense. The derivative records the jump that the classical derivative misses away from the single discontinuity.

16.5 Integration by parts

For smooth functions, distributional differentiation agrees with classical differentiation because test functions have compact support.

16.6 Convergence of distributions

A sequence converges distributionally when it converges on every test function.

Example 16.3 (Mollifiers converge to Dirac distributionally). Let $\rho_\varepsilon(x) = \varepsilon^{-n} \rho(x/\varepsilon)$ with $\int \rho = 1$. For every test function ϕ ,

$$\langle \rho_\varepsilon, \phi \rangle = \int \rho(y) \phi(\varepsilon y) dy \rightarrow \phi(0)$$

by dominated convergence. Hence

$$\rho_\varepsilon \rightarrow \delta_0$$

as distributions. A singular source can therefore arise as the limit of ordinary smooth densities.

16.7 Multiplication by smooth functions

If $a \in C^\infty$, define $(aT)(\phi) = T(a\phi)$.

16.8 Linear differential operators

Smooth-coefficient differential operators act naturally on distributions by combining multiplication and distributional differentiation.

16.9 Core structural results

Theorem 16.4 (Agreement with classical derivatives). *If $f \in C^1$ defines a regular distribution, then its distributional derivative is the regular distribution defined by f' .*

Proof. For every test function, $D'_f(\phi) = -\int f\phi' = \int f'\phi$ by integration by parts and vanishing boundary terms. $\square$

Theorem 16.5 (Derivative of Heaviside). *The distributional derivative of the Heaviside step is δ_0 .*

Proof. For a test function ϕ , $H'(\phi) = -\int_0^\infty \phi'(x)dx = \phi(0)$. $\square$

Theorem 16.6 (Continuity of differentiation). *If $T_n \rightarrow T$ in distributions, then $\partial^\alpha T_n \rightarrow \partial^\alpha T$.*

Proof. Evaluate on a test function and move the derivative to that fixed test function; convergence of T_n then applies directly. $\square$

16.10 Worked examples

Example 16.7 (Distributions diagnostic). Give a rigorous worked analysis of Distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 16.8 (Regular distributions diagnostic). Give a rigorous worked analysis of Regular distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Locally integrable functions define distributions by integration against test functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 16.9 (Dirac mass diagnostic). Give a rigorous worked analysis of Dirac mass in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 16.10 (Distributional derivatives diagnostic). Give a rigorous worked analysis of Distributional derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|}T(\partial^\alpha \phi)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

16.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 16.11 (Distributions diagnostic). Give a rigorous worked analysis of Distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.12 (Regular distributions diagnostic). Give a rigorous worked analysis of Regular distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Locally integrable functions define distributions by integration against test functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.13 (Dirac mass diagnostic). Give a rigorous worked analysis of Dirac mass in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.14 (Distributional derivatives diagnostic). Give a rigorous worked analysis of Distributional derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.15 (Integration by parts diagnostic). Give a rigorous worked analysis of Integration by parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For smooth functions, distributional differentiation agrees with classical differentiation because test functions have compact support. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.16 (Convergence of distributions diagnostic). Give a rigorous worked analysis of Convergence of distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A sequence converges distributionally when it converges on every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.17 (Multiplication by smooth functions diagnostic). Give a rigorous worked analysis of Multiplication by smooth functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $a \in C^\infty$, define $(aT)(\phi) = T(a\phi)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.18 (Linear differential operators diagnostic). Give a rigorous worked analysis of Linear differential operators in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Smooth-coefficient differential operators act naturally on distributions by combining multiplication and distributional differentiation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 16.19 (Agreement with classical derivatives proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in C^1$ defines a regular distribution, then its distributional derivative is the regular distribution defined by f' .

Solution. For every test function, $D'_f(\phi) = -\int f\phi' = \int f'\phi$ by integration by parts and vanishing boundary terms. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 16.20 (Derivative of Heaviside proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: The distributional derivative of the Heaviside step is δ_0 .

Solution. For a test function ϕ , $H'(\phi) = -\int_0^\infty \phi'(x)dx = \phi(0)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 16.21 (Continuity of differentiation proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $T_n \rightarrow T$ in distributions, then $\partial^\alpha T_n \rightarrow \partial^\alpha T$.

Solution. Evaluate on a test function and move the derivative to that fixed test function; convergence of T_n then applies directly. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 16.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Distributions extend the derivative and transform calculus to nonsmooth objects by duality. A distribution acts continuously and linearly on test functions, and derivatives are transferred onto the test function by integration by parts. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

16.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 16.23. State and apply the central principle behind Distributions. Give one immediate consequence in the setting of this chapter.

Hint. To verify a distribution, check linearity and estimate its action by finitely many test-function seminorms on each compact set. $\square$

Solution. A distribution on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.24. State and apply the central principle behind Regular distributions. Give one immediate consequence in the setting of this chapter.

Hint. A locally integrable function acts by $T_f(\phi) = \int f\phi$; use local integrability because ϕ has compact support. $\square$

Solution. Locally integrable functions define distributions by integration against test functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.25. State and apply the central principle behind Dirac mass. Give one immediate consequence in the setting of this chapter.

Hint. The distributional derivative is defined by $T'(\phi) = -T(\phi')$; the minus sign comes from integration by parts. $\square$

Solution. Point evaluation $\delta_{x_0}(\phi) = \phi(x_0)$ is a distribution not represented by an ordinary locally integrable function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.26. State and apply the central principle behind Distributional derivatives. Give one immediate consequence in the setting of this chapter.

Hint. For Heaviside, integrate $-\int_0^\infty \phi'(x) dx$ and use compact support to evaluate the endpoint. $\square$

Solution. Define $\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.27. State and apply the central principle behind Integration by parts. Give one immediate consequence in the setting of this chapter.

Hint. To prove convergence of distributions, fix an arbitrary test function first and pass to the scalar limit of the pairings. $\square$

Solution. For smooth functions, distributional differentiation agrees with classical differentiation because test functions have compact support. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.28. State and apply the central principle behind Convergence of distributions. Give one immediate consequence in the setting of this chapter.

Hint. Multiplication by a smooth function is defined by moving the multiplier onto the test function: $(aT)(\phi) = T(a\phi)$. $\square$

Solution. A sequence converges distributionally when it converges on every test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.29. State and apply the central principle behind Multiplication by smooth functions. Give one immediate consequence in the setting of this chapter.

Hint. For a differential operator, move every derivative onto the test function with the appropriate sign before simplifying. $\square$

Solution. If $a \in C^\infty$, define $(aT)(\phi) = T(a\phi)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 16.30. State and apply the central principle behind Linear differential operators. Give one immediate consequence in the setting of this chapter.

Hint. Do not evaluate a general distribution at a point; only its action on test functions is defined without extra structure. $\square$

Solution. Smooth-coefficient differential operators act naturally on distributions by combining multiplication and distributional differentiation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

16.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 16.31 (Dirac action). What is $\delta_a(\phi)$?

Hint. Use point evaluation. $\square$

Solution. It is $\phi(a)$. $\square$

Exercise 16.32 (Dirac derivative). What is $\delta'_0(\phi)$?

Hint. Use the distributional derivative definition. $\square$

Solution. It is $-\phi'(0)$. $\square$

Exercise 16.33 (Heaviside derivative). What is the distributional derivative of the Heaviside step?

Hint. Integrate by parts on the positive half-line. $\square$

Solution. δ_0 . $\square$

Exercise 16.34 (Smooth multiplier). How is aT defined for smooth a ?

Hint. Move multiplication to the test function. □

Solution. $(aT)(\phi) = T(a\phi)$. □

Exercise 16.35 (Distributional convergence). What must be checked to prove T_n tends to T distributionally?

Hint. Fix an arbitrary test function. □

Solution. Show $T_n(\phi)$ tends to $T(\phi)$ for every test ϕ . □

Proofs

Exercise 16.36 (Regular derivative). Prove a C^1 function has the same classical and distributional derivative.

Hint. Integrate by parts. □

Solution. Compact support kills the boundary term. □

Exercise 16.37 (Derivative continuity). Prove differentiation is continuous on distributions.

Hint. Move the derivative to the fixed test function. □

Solution. Convergence of T_n on that derivative gives convergence of derivatives. □

Exercise 16.38 (Linearity). Prove the distributional derivative is linear.

Hint. Apply the definition to a linear combination. □

Solution. The sign and test derivative distribute linearly. □

Exercise 16.39 (Leibniz smooth factor). Derive $D(aT) = aDT + a'T$ in one dimension.

Hint. Evaluate on a test function and expand $(a\phi)'$. □

Solution. The two terms arise from the product rule. □

Counterexamples and hypothesis testing

Exercise 16.40 (Dirac not ordinary function). Why can no locally integrable f represent δ_0 ?

Hint. Test with bumps supported in shrinking neighborhoods of zero. □

Solution. Absolute continuity of the integral would force the actions to shrink, while δ_0 can stay one. □

Exercise 16.41 (Pointwise derivative failure). Why is $H' = 0$ away from zero not the full derivative in distributions?

Hint. Distributional differentiation tests across the jump. □

Solution. The missing jump contributes δ_0 . □

Exercise 16.42 (Weak convergence versus pointwise). Why does distributional convergence not imply pointwise convergence?

Hint. Distributions may be singular and have no point values. □

Solution. Mollifiers converge to δ_0 , which is not an ordinary pointwise function. □

Applications and investigations

Exercise 16.43 (Impulse model). Why is Dirac useful for point forces or instantaneous impulses?

Hint. It concentrates finite action at a lower-dimensional set. □

Solution. Differential equations can encode idealized sources while remaining linear in the distribution framework. □

Exercise 16.44 (Approximation of sources). Why replace delta by a narrow mollifier numerically?

Hint. It provides a smooth regularization with the same total mass. □

Solution. The regularized solutions can converge distributionally to the singular-source solution. □

Challenge problems

Exercise 16.45 (Jump formula). If a piecewise C^1 function has jumps J_k at points x_k , describe its distributional derivative.

Hint. Differentiate classically on smooth pieces and collect internal boundary terms. □

Solution. It is the classical piecewise derivative plus $\sum J_k \delta_{x_k}$. □

Exercise 16.46 (Second derivative of $|x|$). Compute the second distributional derivative of $|x|$.

Hint. First derivative is $\text{sign}(x)$; its jump at zero has size 2. □

Solution. The second derivative is $2 \delta_0$. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 17

Support and Singular Distributions

Support records where a distribution can actually detect test functions. Singular distributions concentrate action on lower-dimensional or discrete sets and provide canonical models of sources, jumps, and point interactions.

Learning goals

After this chapter, the reader should be able to:

- determine the support of regular and singular distributions from their action on test functions;
- compute supports of Dirac masses, derivatives of Dirac masses, and finite linear combinations;
- distinguish singular support from ordinary support in representative examples;
- use localization by test functions to prove that a distribution vanishes on an open set;
- analyze how multiplication by smooth functions changes support;
- construct distributions concentrated on lower-dimensional or discrete sets in elementary settings;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

17.1 Support of a distribution

The support is the complement of the largest open set on which the distribution vanishes on every test function.

17.2 Compact support

Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support.

17.3 Dirac support

The support of δ_{x_0} is the singleton $\{x_0\}$.

17.4 Derivatives of Dirac

$\delta^{(k)}$ acts by signed evaluation of the k th derivative of the test function.

Example 17.1 (Support survives Dirac differentiation). The derivative of Dirac acts by

$$\delta'_0(\phi) = -\phi'(0).$$

If ϕ is supported in an open set avoiding 0, then all derivatives of ϕ vanish near 0, so $\delta'_0(\phi) = 0$. Conversely a test function supported near 0 can have nonzero derivative at 0. Therefore

$$\text{supp } \delta'_0 = \{0\}.$$

Differentiation can change singular order without enlarging support.

17.5 Measures as distributions

Locally finite signed or complex measures define distributions by integration.

17.6 Jump singularities

Distributional derivatives of piecewise smooth functions include Dirac masses weighted by jump sizes.

Example 17.2 (A jump produces a singular source). Let

$$f(x) = \begin{cases} x, & x < 0, \\ x + 3, & x > 0. \end{cases}$$

The classical derivative is 1 away from zero, while the jump size is 3. The distributional derivative is therefore

$$Df = 1 + 3\delta_0.$$

Integrating by parts on the two half-lines verifies that the internal boundary terms combine to $3\phi(0)$.

17.7 Surface distributions

Integration over smooth hypersurfaces defines distributions supported on those hypersurfaces.

Example 17.3 (A hypersurface measure as a distribution). For a smooth compact hypersurface $\Sigma \subset \mathbb{R}^n$, define

$$T_\Sigma(\phi) = \int_\Sigma \phi dS.$$

This is linear and continuous on test functions, and it vanishes whenever ϕ is supported away from Σ . Hence

$$\text{supp } T_\Sigma \subseteq \Sigma,$$

with equality when every point of Σ is detected by a local nonnegative test function.

17.8 Localization

Multiplying by test functions or cutoffs isolates singular behavior near chosen points or sets.

17.9 Core structural results

Theorem 17.4 (Support of a derivative). $\text{supp}(\partial^\alpha T) \subseteq \text{supp } T$.

Proof. If T vanishes on an open set, then it vanishes on derivatives of all test functions supported there, so the derivative distribution also vanishes there. $\square$

Theorem 17.5 (Dirac derivatives). $\delta_0^{(k)}(\phi) = (-1)^k \phi^{(k)}(0)$.

Proof. Apply the definition of distributional derivative repeatedly to the point-evaluation functional. $\square$

Theorem 17.6 (Jump formula). *If a piecewise C^1 function has jump J at x_0 , its distributional derivative equals the classical derivative away from the jump plus $J\delta_{x_0}$.*

Proof. Integrate by parts separately on the two sides of the jump; the internal boundary terms combine to the jump size times $\phi(x_0)$. $\square$

17.10 Worked examples

Example 17.7 (Support of a distribution diagnostic). Give a rigorous worked analysis of Support of a distribution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The support is the complement of the largest open set on which the distribution vanishes on every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 17.8 (Compact support diagnostic). Give a rigorous worked analysis of Compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 17.9 (Dirac support diagnostic). Give a rigorous worked analysis of Dirac support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The support of δ_{x_0} is the singleton $\{x_0\}$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 17.10 (Derivatives of Dirac diagnostic). Give a rigorous worked analysis of Derivatives of Dirac in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. $\delta^{(k)}$ acts by signed evaluation of the k th derivative of the test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

17.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 17.11 (Support of a distribution diagnostic). Give a rigorous worked analysis of Support of a distribution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The support is the complement of the largest open set on which the distribution vanishes on every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.12 (Compact support diagnostic). Give a rigorous worked analysis of Compact support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.13 (Dirac support diagnostic). Give a rigorous worked analysis of Dirac support in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The support of δ_{x_0} is the singleton $\{x_0\}$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.14 (Derivatives of Dirac diagnostic). Give a rigorous worked analysis of Derivatives of Dirac in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $\delta^{(k)}$ acts by signed evaluation of the k th derivative of the test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.15 (Measures as distributions diagnostic). Give a rigorous worked analysis of Measures as distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Locally finite signed or complex measures define distributions by integration. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.16 (Jump singularities diagnostic). Give a rigorous worked analysis of Jump singularities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distributional derivatives of piecewise smooth functions include Dirac masses weighted by jump sizes. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.17 (Surface distributions diagnostic). Give a rigorous worked analysis of Surface distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Integration over smooth hypersurfaces defines distributions supported on those hypersurfaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.18 (Localization diagnostic). Give a rigorous worked analysis of Localization in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplying by test functions or cutoffs isolates singular behavior near chosen points or sets. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 17.19 (Support of a derivative proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: $\text{supp}(\partial^\alpha T) \subseteq \text{supp } T$.

Solution. If T vanishes on an open set, then it vanishes on derivatives of all test functions supported there, so the derivative distribution also vanishes there. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 17.20 (Dirac derivatives proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: $\delta_0^{(k)}(\phi) = (-1)^k \phi^{(k)}(0)$.

Solution. Apply the definition of distributional derivative repeatedly to the point-evaluation functional. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 17.21 (Jump formula proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If a piecewise C^1 function has jump J at x_0 , its distributional derivative equals the classical derivative away from the jump plus $J\delta_{x_0}$.

Solution. Integrate by parts separately on the two sides of the jump; the internal boundary terms combine to the jump size times $\phi(x_0)$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 17.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Support records where a distribution can actually detect test functions. Singular distributions concentrate action on lower-dimensional or discrete sets and provide canonical models of sources, jumps, and point interactions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

17.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 17.23. State and apply the central principle behind Support of a distribution. Give one immediate consequence in the setting of this chapter.

Hint. A point lies outside the support if there is a neighborhood on which the distribution annihilates every test function. $\square$

Solution. The support is the complement of the largest open set on which the distribution vanishes on every test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.24. State and apply the central principle behind Compact support. Give one immediate consequence in the setting of this chapter.

Hint. For δ_{x_0} , test functions supported away from x_0 vanish at the evaluation point. $\square$

Solution. Compactly supported distributions can act on broader classes of smooth functions by inserting a cutoff equal to one near the support. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.25. State and apply the central principle behind Dirac support. Give one immediate consequence in the setting of this chapter.

Hint. Derivatives of a Dirac mass have the same support as the Dirac mass even though their action involves derivatives of test functions. $\square$

Solution. The support of δ_{x_0} is the singleton $\{x_0\}$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.26. State and apply the central principle behind Derivatives of Dirac. Give one immediate consequence in the setting of this chapter.

Hint. To identify singular support, first remove the largest open set on which the distribution agrees with a smooth function. $\square$

Solution. $\delta^{(k)}$ acts by signed evaluation of the k th derivative of the test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.27. State and apply the central principle behind Measures as distributions. Give one immediate consequence in the setting of this chapter.

Hint. Multiplication by a smooth function can shrink support to where both the multiplier and distribution are active. $\square$

Solution. Locally finite signed or complex measures define distributions by integration. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.28. State and apply the central principle behind Jump singularities. Give one immediate consequence in the setting of this chapter.

Hint. Use a cutoff supported in a small neighborhood to localize a distributional identity. $\square$

Solution. Distributional derivatives of piecewise smooth functions include Dirac masses weighted by jump sizes. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.29. State and apply the central principle behind Surface distributions. Give one immediate consequence in the setting of this chapter.

Hint. Finite sums have support contained in the union of the individual supports; check whether cancellation can reduce it. $\square$

Solution. Integration over smooth hypersurfaces defines distributions supported on those hypersurfaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 17.30. State and apply the central principle behind Localization. Give one immediate consequence in the setting of this chapter.

Hint. For lower-dimensional concentrations, define the distribution by integration over the submanifold and inspect which test functions it can detect. $\square$

Solution. Multiplying by test functions or cutoffs isolates singular behavior near chosen points or sets. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

17.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 17.31 (Dirac support). What is the support of δ_a ?

Hint. It only evaluates at a . $\square$

Solution. The support is a . $\square$

Exercise 17.32 (Dirac derivative). What is $\delta_a^{(k)}(\phi)$?

Hint. Apply derivative definition repeatedly. $\square$

Solution. It is $(-1)^k \phi^{(k)}(a)$. $\square$

Exercise 17.33 (Jump weight). A piecewise function jumps upward by 5 at $x=2$. What singular term appears in its derivative?

Hint. Use the jump formula. $\square$

Solution. $5 \delta_2$. $\square$

Exercise 17.34 (Smooth multiplier support). How does $\text{supp}(aT)$ compare with $\text{supp}(T)$?

Hint. If a test lies where T vanishes, so does aT . $\square$

Solution. $\text{supp}(aT)$ is contained in $\text{supp}(T)$, and also in the closure of where a is nonzero. $\square$

Exercise 17.35 (Surface support). Where is the support of integration over a smooth curve?

Hint. The functional only sees test functions on the curve. $\square$

Solution. It is the curve, under the usual nondegeneracy. $\square$

Proofs

Exercise 17.36 (Derivative support). Prove $\text{supp}(DT)$ is contained in $\text{supp}(T)$.

Hint. Test on an open set where T vanishes. □

Solution. Derivatives of test functions remain supported there, so DT also vanishes. □

Exercise 17.37 (Dirac derivative formula). Prove the formula for delta derivatives.

Hint. Induct from the distributional derivative definition. □

Solution. Each derivative introduces one minus sign and one test derivative. □

Exercise 17.38 (Jump formula). Derive the jump contribution for a piecewise C^1 function.

Hint. Integrate by parts on each side of the jump. □

Solution. Internal boundary terms leave jump size times the test value. □

Exercise 17.39 (Localization). Prove a distribution vanishing on every test function supported in U is zero on U .

Hint. This is the definition of vanishing on an open set. □

Solution. The support is contained in the complement of U . □

Counterexamples and hypothesis testing

Exercise 17.40 (Support not pointwise support). Why is distribution support not defined by pointwise nonzero values?

Hint. A general distribution has no pointwise values. □

Solution. Support is defined through vanishing on open sets of test functions. □

Exercise 17.41 (Derivative support equality not guaranteed). Why can $\text{supp}(DT)$ be strictly smaller than $\text{supp}(T)$?

Hint. Differentiation may annihilate constants. □

Solution. For the constant distribution on a connected domain, derivative is zero while the original support is the whole domain. □

Exercise 17.42 (Smooth factor can shrink support). Give a way multiplication by a smooth function can reduce support.

Hint. Choose the multiplier to vanish near part of the support. □

Solution. The product distribution then cannot detect test functions in that vanished region. □

Applications and investigations

Exercise 17.43 (Point source). Why does support distinguish a point source from a distributed source?

Hint. It records the smallest closed set where action occurs. ☐

Solution. Dirac support is a point, while regular forcing may occupy a region. ☐

Exercise 17.44 (Interface PDE). Why do surface distributions model interface forces?

Hint. They act only on test values along a hypersurface. ☐

Solution. They encode lower-dimensional sources in ambient PDEs. ☐

Challenge problems

Exercise 17.45 (Singular support). Explain the difference between support and singular support for a piecewise smooth function with one jump.

Hint. Support records where the function/distribution is nonzero; singular support records where it is not smooth. ☐

Solution. The singular support may be just the jump point even if ordinary support is much larger. ☐

Exercise 17.46 (Derivative hierarchy). Show all derivatives of δ_0 have support 0.

Hint. Use derivative support inclusion and a test function with prescribed k th derivative at zero. ☐

Solution. The support cannot enlarge and is nonempty because suitable local tests give nonzero action. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 18

Tempered Distributions

Tempered distributions are continuous linear functionals on Schwartz space. They include polynomial-growth functions, measures, Dirac masses, and their derivatives while remaining stable under the Fourier transform.

Learning goals

After this chapter, the reader should be able to:

- verify that a distribution is tempered by establishing polynomial-growth bounds on Schwartz test functions;
- embed polynomially growing functions and finite measures into tempered distributions when justified;
- distinguish general distributions from tempered distributions by their admissible test-function spaces;
- apply differentiation and polynomial multiplication inside the tempered-distribution framework;
- construct examples of tempered singular distributions such as Dirac masses and their derivatives;
- identify growth behavior that prevents a distribution from being tempered;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

18.1 Tempered distributions

A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$.

18.2 Regular tempered distributions

Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions.

Example 18.1 (Polynomial growth is absorbed by Schwartz decay). Let $f(x) = (1 + |x|)^N$. For a Schwartz test function ϕ , choose $M > N + n + 1$. Then

$$|f(x)\phi(x)| \leq C(1 + |x|)^{N-M} p_M(\phi),$$

where $p_M(\phi) = \sup_x (1 + |x|)^M |\phi(x)|$. The remaining weight is integrable, so

$$\left| \int f\phi \right| \leq C_M p_M(\phi).$$

Thus polynomial growth defines a continuous tempered distribution.

18.3 Schwartz topology

Continuity is measured against finitely many weighted derivative seminorms.

18.4 Differentiation

Distributional derivatives preserve temperedness.

Example 18.2 (Differentiating a tempered distribution). For $T \in \mathcal{S}'$,

$$\partial^\alpha T(\phi) = (-1)^{|\alpha|} T(\partial^\alpha \phi).$$

Since differentiation is a continuous endomorphism of Schwartz space, the map $\phi \mapsto T(\partial^\alpha \phi)$ is continuous. Therefore every distributional derivative of a tempered distribution is again tempered.

18.5 Polynomial multiplication

Multiplication by coordinate polynomials preserves temperedness.

18.6 Dirac family

Dirac masses and all derivatives are tempered.

18.7 Measures

Many measures with polynomial growth define tempered distributions.

18.8 Fourier readiness

The Fourier invariance of Schwartz space lets the Fourier transform be transferred to its dual.

Example 18.3 (Why Fourier transform extends to the dual). The Fourier transform is a continuous bijection $\mathcal{S} \rightarrow \mathcal{S}$. Thus for $T \in \mathcal{S}'$, one may define the transformed functional by

$$\widehat{T}(\phi) = T(\widehat{\phi})$$

with the convention-compatible placement of inverse transforms. Continuity of the Schwartz transform immediately makes $\widehat{T}$ tempered. The test space was chosen precisely to make this dual transfer possible.

18.9 Core structural results

Theorem 18.4 (Polynomial-growth functions are tempered). *If $|f(x)| \leq C(1 + |x|)^N$ and f is locally integrable, then f defines a tempered distribution.*

Proof. Choose a Schwartz seminorm with decay power larger than $N + n$; it dominates the integral of $|f\phi|$ and yields continuity. $\square$

Theorem 18.5 (Derivatives preserve temperedness). *If $T \in \mathcal{S}'$, then $\partial^\alpha T \in \mathcal{S}'$.*

Proof. The map $\phi \mapsto \partial^\alpha \phi$ is continuous on Schwartz space, and composing it with the continuous functional T preserves continuity. $\square$

Theorem 18.6 (Polynomial multiplication preserves temperedness). *If $T \in \mathcal{S}'$ and p is a polynomial, then $pT \in \mathcal{S}'$.*

Proof. Multiplication by a polynomial is a continuous endomorphism of Schwartz space, so $pT(\phi) = T(p\phi)$ is continuous. $\square$

18.10 Worked examples

Example 18.7 (Tempered distributions diagnostic). Give a rigorous worked analysis of Tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 18.8 (Regular tempered distributions diagnostic). Give a rigorous worked analysis of Regular tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 18.9 (Schwartz topology diagnostic). Give a rigorous worked analysis of Schwartz topology in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Continuity is measured against finitely many weighted derivative seminorms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 18.10 (Differentiation diagnostic). Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Distributional derivatives preserve temperedness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

18.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 18.11 (Tempered distributions diagnostic). Give a rigorous worked analysis of Tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.12 (Regular tempered distributions diagnostic). Give a rigorous worked analysis of Regular tempered distributions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.13 (Schwartz topology diagnostic). Give a rigorous worked analysis of Schwartz topology in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Continuity is measured against finitely many weighted derivative seminorms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.14 (Differentiation diagnostic). Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distributional derivatives preserve temperedness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.15 (Polynomial multiplication diagnostic). Give a rigorous worked analysis of Polynomial multiplication in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplication by coordinate polynomials preserves temperedness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.16 (Dirac family diagnostic). Give a rigorous worked analysis of Dirac family in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Dirac masses and all derivatives are tempered. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.17 (Measures diagnostic). Give a rigorous worked analysis of Measures in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Many measures with polynomial growth define tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.18 (Fourier readiness diagnostic). Give a rigorous worked analysis of Fourier readiness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Fourier invariance of Schwartz space lets the Fourier transform be transferred to its dual. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 18.19 (Polynomial-growth functions are tempered proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $|f(x)| \leq C(1 + |x|)^N$ and f is locally integrable, then f defines a tempered distribution.

Solution. Choose a Schwartz seminorm with decay power larger than $N + n$; it dominates the integral of $|f\phi|$ and yields continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 18.20 (Derivatives preserve temperedness proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $T \in \mathcal{S}'$, then $\partial^\alpha T \in \mathcal{S}'$.

Solution. The map $\phi \mapsto \partial^\alpha \phi$ is continuous on Schwartz space, and composing it with the continuous functional T preserves continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 18.21 (Polynomial multiplication preserves temperedness proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $T \in \mathcal{S}'$ and p is a polynomial, then $pT \in \mathcal{S}'$.

Solution. Multiplication by a polynomial is a continuous endomorphism of Schwartz space, so $pT(\phi) = T(p\phi)$ is continuous. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 18.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Tempered distributions are continuous linear functionals on Schwartz space. They include polynomial-growth functions, measures, Dirac masses, and their derivatives while remaining stable under the Fourier transform. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

18.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 18.23. State and apply the central principle behind Tempered distributions. Give one immediate consequence in the setting of this chapter.

Hint. Temperedness means continuity on Schwartz space; estimate the pairing by a finite combination of Schwartz seminorms. $\square$

Solution. A tempered distribution is a continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.24. State and apply the central principle behind Regular tempered distributions. Give one immediate consequence in the setting of this chapter.

Hint. A polynomially growing function is tempered because rapid decay of test functions makes the pairing absolutely integrable. $\square$

Solution. Functions of at most polynomial growth define tempered distributions when paired with rapidly decaying Schwartz functions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.25. State and apply the central principle behind Schwartz topology. Give one immediate consequence in the setting of this chapter.

Hint. Finite measures are tempered when their growth is controlled sufficiently to pair with Schwartz functions. $\square$

Solution. Continuity is measured against finitely many weighted derivative seminorms. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.26. State and apply the central principle behind Differentiation. Give one immediate consequence in the setting of this chapter.

Hint. Dirac masses are tempered because point evaluation is bounded by a Schwartz seminorm. $\square$

Solution. Distributional derivatives preserve temperedness. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.27. State and apply the central principle behind Polynomial multiplication. Give one immediate consequence in the setting of this chapter.

Hint. Differentiation preserves temperedness by moving derivatives to the Schwartz test function. $\square$

Solution. Multiplication by coordinate polynomials preserves temperedness. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.28. State and apply the central principle behind Dirac family. Give one immediate consequence in the setting of this chapter.

Hint. Multiplication by polynomials preserves temperedness because polynomial weights are built into the Schwartz seminorms. $\square$

Solution. Dirac masses and all derivatives are tempered. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.29. State and apply the central principle behind Measures. Give one immediate consequence in the setting of this chapter.

Hint. To show a distribution is not tempered, build Schwartz functions whose pairing grows faster than any finite seminorm bound permits. $\square$

Solution. Many measures with polynomial growth define tempered distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 18.30. State and apply the central principle behind Fourier readiness. Give one immediate consequence in the setting of this chapter.

Hint. Keep the test-function space visible: a distribution may be valid on C_c^∞ yet fail to extend continuously to Schwartz space. $\square$

Solution. The Fourier invariance of Schwartz space lets the Fourier transform be transferred to its dual. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

18.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 18.31 (Polynomial). Does x^5 define a tempered distribution?

Hint. It has polynomial growth. $\square$

Solution. Yes. $\square$

Exercise 18.32 (Dirac). Is δ_0 tempered?

Hint. Point evaluation is controlled by a Schwartz seminorm. $\square$

Solution. Yes. $\square$

Exercise 18.33 (Derivative). If T is tempered, what about DT ?

Hint. Differentiation preserves temperedness. $\square$

Solution. DT is tempered. $\square$

Exercise 18.34 (Polynomial multiplier). If T is tempered, what about $x^3 T$?

Hint. Polynomial multiplication preserves the class. $\square$

Solution. It is tempered. $\square$

Exercise 18.35 (Finite measure). Does every finite Borel measure define a tempered distribution?

Hint. Pair with bounded Schwartz functions. $\square$

Solution. Yes; absolute action is bounded by total variation times sup norm. $\square$

Proofs

Exercise 18.36 (Polynomial-growth proof). Prove a locally integrable polynomial-growth function is tempered.

Hint. Use a Schwartz seminorm with decay power larger than growth plus dimension. □

Solution. The weighted integrand becomes integrable and gives a seminorm bound. □

Exercise 18.37 (Derivative stability). Prove derivatives preserve temperedness.

Hint. Compose T with continuous differentiation on Schwartz space. □

Solution. Continuity is preserved. □

Exercise 18.38 (Polynomial multiplication). Prove pT is tempered for polynomial p .

Hint. Multiplication by p is continuous on Schwartz space. □

Solution. Compose that map with T . □

Exercise 18.39 (Dirac continuity). Prove δ and its derivatives are tempered.

Hint. Point derivatives are controlled by Schwartz derivative seminorms. □

Solution. Each action is a continuous seminorm evaluation. □

Counterexamples and hypothesis testing

Exercise 18.40 (Exponential growth). Why does $e^{\hat{x}^2}$ not define a regular tempered distribution by integration over the real line?

Hint. Schwartz functions need not decay faster than every exponential. □

Solution. Polynomial-weight seminorms cannot control the pairing for all Schwartz tests. □

Exercise 18.41 (General distribution not tempered). Why is every distribution not automatically tempered?

Hint. Tempered distributions act on the larger Schwartz test space, not just compact supports. □

Solution. Growth at infinity can destroy continuity on Schwartz space. □

Exercise 18.42 (Locally integrable not enough). Why is local integrability alone insufficient for temperedness?

Hint. Tempered tests have noncompact support. □

Solution. Behavior at infinity must also be controlled. □

Applications and investigations

Exercise 18.43 (Frequency-ready sources). Why are tempered distributions suited to whole-space PDE?

Hint. They include singular/polynomial-growth sources and admit Fourier transform. □

Solution. Constant-coefficient PDE can be transformed even for nonsmooth data. □

Exercise 18.44 (Idealized signals). Why can impulses and polynomial trends coexist in S' ?

Hint. Dirac masses and polynomial-growth functions are both tempered. □

Solution. The common framework handles both under Fourier operations. □

Challenge problems

Exercise 18.45 (Order estimate). Show δ_0' is controlled by a first-derivative Schwartz seminorm.

Hint. Its action is $-\phi'(0)$. □

Solution. Absolute value is at most $\sup |\phi'|$. □

Exercise 18.46 (Growth threshold). If $|f(x)| \leq C(1+|x|)^N$, explain how large a decay seminorm is needed in dimension n .

Hint. Choose a weight exponent greater than $N+n$. □

Solution. Then $(1+|x|)^{-(N+M)}$ is integrable, giving continuity. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 19

Fourier Transform of Distributions

The Fourier transform extends to tempered distributions by duality. The same algebraic rules for differentiation, multiplication, translation, and convolution then survive far beyond integrable functions.

Learning goals

After this chapter, the reader should be able to:

- define and compute Fourier transforms of tempered distributions by duality;
- derive transform formulas for Dirac masses, derivatives, polynomials, and exponentials in distributional form;
- use differentiation and multiplication identities without assuming classical integrability;
- solve algebraic multiplier equations in frequency space at the distribution level;
- check constants and signs by testing transform identities against Schwartz functions;
- distinguish ordinary-function Fourier formulas from their distributional extensions;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

19.1 Dual definition

For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement.

19.2 Fourier automorphism

Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions.

19.3 Dirac transforms

Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention.

Example 19.1 (The transform of Dirac is the constant distribution). Under the chapter's 2π convention,

$$\widehat{\delta}_0 = 1.$$

Indeed for a Schwartz test function, the dual definition gives the action of δ_0 on the transformed test, which is the transform value at zero. Fourier inversion identifies this with integration of the original test function, exactly the regular distribution defined by the constant 1.

19.4 Differentiation

Distributional differentiation becomes multiplication by polynomial frequency factors.

Example 19.2 (Distributional differentiation keeps the classical multiplier rule). For $T \in \mathcal{S}'$,

$$\widehat{\partial^\alpha T} = (2\pi i\xi)^\alpha \widehat{T}.$$

The proof is dual: move the derivative to the test function, use the classical Schwartz-space transform identity, then move multiplication back to the transformed distribution. No pointwise differentiability or integrability of T is required.

19.5 Polynomial multiplication

Multiplication by x^α becomes differentiation in frequency.

19.6 Translation and modulation

The classical transform identities extend by duality.

19.7 Convolution with Schwartz functions

A tempered distribution can be convolved with a Schwartz function to produce a smooth function with controlled growth.

19.8 PDE multipliers

Constant-coefficient differential equations become algebraic multiplier equations in frequency space.

Example 19.3 (Solving a multiplier equation with a singular source). For the one-dimensional equation

$$-u'' = \delta_0,$$

Fourier transformation gives

$$4\pi^2 \xi^2 \widehat{u} = 1.$$

Formally,

$$\widehat{u} = (4\pi^2 \xi^2)^{-1},$$

which must be interpreted distributionally at $\xi = 0$. In physical space a fundamental solution is proportional to $|x|$. The example shows why distributional Fourier multipliers are needed precisely where the symbol vanishes.

19.9 Core structural results

Theorem 19.4 (Fourier transform of Dirac). *Under the 2π convention, $\widehat{\delta}_0 = 1$ as a tempered distribution.*

Proof. For a Schwartz test function, evaluate the dual definition and use Fourier inversion at zero to identify the resulting functional with integration against the constant one. $\square$

Theorem 19.5 (Derivative rule). $\widehat{\partial^\alpha T} = (2\pi i\xi)^\alpha \widehat{T}$.

Proof. Move the derivative onto the test function, use the classical Schwartz-space transform identity, and then return the multiplication to the transformed distribution. $\square$

Theorem 19.6 (Fourier automorphism). *The Fourier transform is a continuous bijection of $\mathcal{S}'$ with continuous inverse.*

Proof. Dualize the continuous bijection of Schwartz space and use the inverse Fourier transform for the inverse operator. $\square$

19.10 Worked examples

Example 19.7 (Dual definition diagnostic). Give a rigorous worked analysis of Dual definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 19.8 (Fourier automorphism diagnostic). Give a rigorous worked analysis of Fourier automorphism in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 19.9 (Dirac transforms diagnostic). Give a rigorous worked analysis of Dirac transforms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 19.10 (Differentiation diagnostic). Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Distributional differentiation becomes multiplication by polynomial frequency factors. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

19.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 19.11 (Dual definition diagnostic). Give a rigorous worked analysis of Dual definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.12 (Fourier automorphism diagnostic). Give a rigorous worked analysis of Fourier automorphism in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.13 (Dirac transforms diagnostic). Give a rigorous worked analysis of Dirac transforms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.14 (Differentiation diagnostic). Give a rigorous worked analysis of Differentiation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Distributional differentiation becomes multiplication by polynomial frequency factors. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.15 (Polynomial multiplication diagnostic). Give a rigorous worked analysis of Polynomial multiplication in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Multiplication by x^α becomes differentiation in frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.16 (Translation and modulation diagnostic). Give a rigorous worked analysis of Translation and modulation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The classical transform identities extend by duality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.17 (Convolution with Schwartz functions diagnostic). Give a rigorous worked analysis of Convolution with Schwartz functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A tempered distribution can be convolved with a Schwartz function to produce a smooth function with controlled growth. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.18 (PDE multipliers diagnostic). Give a rigorous worked analysis of PDE multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Constant-coefficient differential equations become algebraic multiplier equations in frequency space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 19.19 (Fourier transform of Dirac proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Under the 2π convention, $\widehat{\delta}_0 = 1$ as a tempered distribution.

Solution. For a Schwartz test function, evaluate the dual definition and use Fourier inversion at zero to identify the resulting functional with integration against the constant one. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 19.20 (Derivative rule proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: $\widehat{\partial^\alpha T} = (2\pi i\xi)^\alpha \widehat{T}$.

Solution. Move the derivative onto the test function, use the classical Schwartz-space transform identity, and then return the multiplication to the transformed distribution. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 19.21 (Fourier automorphism proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: The Fourier transform is a continuous bijection of $\mathcal{S}'$ with continuous inverse.

Solution. Dualize the continuous bijection of Schwartz space and use the inverse Fourier transform for the inverse operator. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 19.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. The Fourier transform extends to tempered distributions by duality. The same algebraic rules for differentiation, multiplication, translation, and convolution then survive far beyond integrable functions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

19.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 19.23. State and apply the central principle behind Dual definition. Give one immediate consequence in the setting of this chapter.

Hint. Define $\widehat{T}$ by duality: pair T with the Fourier transform of the Schwartz test function. $\square$

Solution. For $T \in \mathcal{S}'$, define $\widehat{T}(\phi) = T(\widehat{\phi})$ up to the convention-compatible inverse placement. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.24. State and apply the central principle behind Fourier automorphism. Give one immediate consequence in the setting of this chapter.

Hint. To transform a Dirac mass, evaluate the transformed test function at the support point and simplify the exponential. $\square$

Solution. Because the Fourier transform is a topological automorphism of Schwartz space, it induces an automorphism of tempered distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.25. State and apply the central principle behind Dirac transforms. Give one immediate consequence in the setting of this chapter.

Hint. Distributional differentiation becomes multiplication by the frequency variable, with constants determined by the transform convention. $\square$

Solution. Point masses transform into complex exponentials, while constants transform into Dirac masses under the standard convention. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.26. State and apply the central principle behind Differentiation. Give one immediate consequence in the setting of this chapter.

Hint. Multiplication by x becomes differentiation in frequency; derive it by duality rather than formal pointwise manipulation. $\square$

Solution. Distributional differentiation becomes multiplication by polynomial frequency factors. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.27. State and apply the central principle behind Polynomial multiplication. Give one immediate consequence in the setting of this chapter.

Hint. Polynomials transform into derivatives of Dirac masses, so expect singular frequency-space objects rather than ordinary functions. $\square$

Solution. Multiplication by x^α becomes differentiation in frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.28. State and apply the central principle behind Translation and modulation. Give one immediate consequence in the setting of this chapter.

Hint. To solve a constant-coefficient equation, transform it and divide by the symbol only where that operation is distributionally meaningful. $\square$

Solution. The classical transform identities extend by duality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.29. State and apply the central principle behind Convolution with Schwartz functions. Give one immediate consequence in the setting of this chapter.

Hint. Check signs and 2π factors by applying the identity to one Schwartz test function and tracing the convention. $\square$

Solution. A tempered distribution can be convolved with a Schwartz function to produce a smooth function with controlled growth. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 19.30. State and apply the central principle behind PDE multipliers. Give one immediate consequence in the setting of this chapter.

Hint. Do not require classical integrability of the original object; tempered-distribution duality is exactly what replaces it. $\square$

Solution. Constant-coefficient differential equations become algebraic multiplier equations in frequency space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

19.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 19.31 (Dirac transform). What is the Fourier transform of δ_0 under the chapter convention?

Hint. Use the dual definition. $\square$

Solution. The constant distribution 1. $\square$

Exercise 19.32 (Constant transform). What is the transform of the constant distribution 1?

Hint. Use Fourier duality/inversion. $\square$

Solution. δ_0 , with the chapter normalization. $\square$

Exercise 19.33 (Derivative). What is the transform of $\mathcal{D}T$?

Hint. Use the derivative rule. $\square$

Solution. $2\pi i \xi$ times $\hat{T}$. $\square$

Exercise 19.34 (Translated Dirac). What is the transform of δ_a ?

Hint. Use translation of δ . $\square$

Solution. The exponential $\exp(-2\pi i a \xi)$. $\square$

Exercise 19.35 (Modulated constant). What distribution transforms to δ at a nonzero frequency?

Hint. Use modulation. $\square$

Solution. A complex exponential in x . $\square$

Proofs

Exercise 19.36 (Dirac proof). Prove the transform of δ_0 is 1.

Hint. Evaluate the dual definition on a Schwartz test. □

Solution. The transformed test at zero equals the integral of the original test. □

Exercise 19.37 (Derivative rule). Prove the distributional derivative multiplier rule.

Hint. Move derivatives through duality. □

Solution. The Schwartz transform identity supplies the factor. □

Exercise 19.38 (Automorphism). Why is Fourier transform a bijection on tempered distributions?

Hint. Dualize the Schwartz automorphism. □

Solution. The inverse Schwartz transform induces the inverse on the dual. □

Exercise 19.39 (Translation rule). Derive the transform of δ_a .

Hint. Evaluate δ_a on the transformed test. □

Solution. It yields the phase exponential multiplying the integral functional. □

Counterexamples and hypothesis testing

Exercise 19.40 (Ordinary formula failure). Why can $\hat{\delta}$ not be computed by the ordinary L1 integral formula?

Hint. δ is not an L1 function. □

Solution. The dual distribution definition is essential. □

Exercise 19.41 (Division by zeros). Why is dividing by a PDE symbol delicate where the symbol vanishes?

Hint. The reciprocal may not be an ordinary function. □

Solution. One needs a distributional extension and may gain nonuniqueness from homogeneous solutions. □

Exercise 19.42 (Polynomial transform). Why do transforms of polynomials become derivatives of δ rather than ordinary decaying functions?

Hint. Multiplication by x becomes frequency differentiation. □

Solution. Starting from transform of $1=\delta$ generates δ derivatives. □

Applications and investigations

Exercise 19.43 (Impulse response). Why is Fourier transform of delta central to linear systems?

Hint. Delta is the identity for convolution. □

Solution. Transforming it to 1 makes convolution responses and transfer functions algebraic. □

Exercise 19.44 (PDE Green kernel). How does a distributional multiplier produce a fundamental solution?

Hint. Invert the operator symbol in the distribution sense and inverse transform. □

Solution. The resulting distribution E satisfies $P(D)E = \delta$. □

Challenge problems

Exercise 19.45 (Transform of delta prime). Compute the Fourier transform of δ'_0 .

Hint. Apply the derivative rule to delta. □

Solution. It is $2\pi i \xi$. □

Exercise 19.46 (Transform of x). Describe the Fourier transform of the regular tempered distribution x .

Hint. Use multiplication by x as frequency differentiation of transform of 1. □

Solution. It is a constant multiple of the derivative of δ_0 , with the sign and 2π factor fixed by the chapter convention. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Part IV

Sobolev and PDE Methods

Chapter 20

Weak Derivatives

Weak derivatives extend differentiation to L^1_{loc} functions by testing against compactly supported smooth functions. They are the bridge from distribution theory to Sobolev spaces and weak PDE formulations.

Learning goals

After this chapter, the reader should be able to:

- verify that a locally integrable function has a proposed weak derivative by testing the defining integration-by-parts identity;
- compute weak derivatives of piecewise smooth functions and identify jump contributions when they occur distributionally;
- distinguish weak differentiability from classical differentiability;
- prove uniqueness of weak derivatives up to almost-everywhere equality;
- use approximation by smooth functions to justify weak-derivative identities;
- construct functions with weak derivatives that fail to have classical derivatives at exceptional points;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

20.1 Weak derivative definition

A function g is the weak derivative $D_j f$ if $\int f D_j \phi = - \int g \phi$ for every test function.

Example 20.1 (Weak derivative of absolute value). Let $f(x) = |x|$. Its classical derivative fails at zero, but define

$$g(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

For every test function ϕ , integration by parts on the two half-lines gives

$$\int |x| \phi'(x) dx = - \int g(x) \phi(x) dx.$$

Hence g is the weak derivative. A single classical nondifferentiability does not prevent weak differentiability.

20.2 Agreement with classical derivatives

Classically differentiable functions have the same weak derivative.

20.3 Piecewise smooth functions

Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses.

Example 20.2 (A jump has no ordinary weak derivative). For Heaviside H , the distributional derivative is δ_0 . If an ordinary $g \in L^1_{\text{loc}}$ were a weak derivative, its associated regular distribution would have to equal δ_0 , impossible. Thus a jump can prevent membership in $W^{1,p}$ even though the function is piecewise smooth: the missing derivative is singular rather than an ordinary function.

20.4 Uniqueness

Weak derivatives are unique up to almost-everywhere equality.

20.5 Higher weak derivatives

Iterating the definition yields multi-index weak derivatives.

20.6 Locality

Weak derivatives depend only on local almost-everywhere behavior.

20.7 Integration by parts

The weak derivative definition is precisely the boundary-free integration-by-parts identity.

20.8 Approximation

Mollification converts weak derivatives into classical derivatives and commutes with differentiation on interior regions.

Example 20.3 (Mollification commutes with weak differentiation). If $D_j f = g$ weakly and ρ_ε is a mollifier, then in an interior region

$$D_j(f * \rho_\varepsilon) = g * \rho_\varepsilon.$$

The left side is now a classical derivative of a smooth function. This identity lets one prove estimates first for mollified approximants while retaining exact control of the weak derivative.

20.9 Core structural results

Theorem 20.4 (Uniqueness of weak derivative). *If $g, h \in L^1_{\text{loc}}$ both satisfy the weak derivative identity for f , then $g = h$ almost everywhere.*

Proof. Their difference integrates to zero against every test function; the fundamental lemma of distributions implies the associated regular distribution is zero, hence the functions agree a.e. $\square$

Theorem 20.5 (Classical implies weak). *If $f \in C^1$, then its weak derivative equals its classical derivative.*

Proof. Integrate by parts against compactly supported test functions; boundary terms vanish. $\square$

Theorem 20.6 (Mollifier commutation). *For locally integrable f with weak derivative $D_j f$, $D_j(f * \rho_\varepsilon) = (D_j f) * \rho_\varepsilon$ where defined.*

Proof. Differentiate the smooth convolution and transfer the derivative from the mollifier to f using the weak derivative identity. $\square$

20.10 Worked examples

Example 20.7 (Weak derivative definition diagnostic). Give a rigorous worked analysis of Weak derivative definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A function g is the weak derivative $D_j f$ if $\int f D_j \phi = - \int g \phi$ for every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 20.8 (Agreement with classical derivatives diagnostic). Give a rigorous worked analysis of Agreement with classical derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Classically differentiable functions have the same weak derivative. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 20.9 (Piecewise smooth functions diagnostic). Give a rigorous worked analysis of Piecewise smooth functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 20.10 (Uniqueness diagnostic). Give a rigorous worked analysis of Uniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Weak derivatives are unique up to almost-everywhere equality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

20.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 20.11 (Weak derivative definition diagnostic). Give a rigorous worked analysis of Weak derivative definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A function g is the weak derivative $D_j f$ if $\int f D_j \phi = - \int g \phi$ for every test function. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.12 (Agreement with classical derivatives diagnostic). Give a rigorous worked analysis of Agreement with classical derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Classically differentiable functions have the same weak derivative. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.13 (Piecewise smooth functions diagnostic). Give a rigorous worked analysis of Piecewise smooth functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.14 (Uniqueness diagnostic). Give a rigorous worked analysis of Uniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Weak derivatives are unique up to almost-everywhere equality. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.15 (Higher weak derivatives diagnostic). Give a rigorous worked analysis of Higher weak derivatives in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Iterating the definition yields multi-index weak derivatives. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.16 (Locality diagnostic). Give a rigorous worked analysis of Locality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Weak derivatives depend only on local almost-everywhere behavior. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.17 (Integration by parts diagnostic). Give a rigorous worked analysis of Integration by parts in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The weak derivative definition is precisely the boundary-free integration-by-parts identity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.18 (Approximation diagnostic). Give a rigorous worked analysis of Approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Mollification converts weak derivatives into classical derivatives and commutes with differentiation on interior regions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 20.19 (Uniqueness of weak derivative proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $g, h \in L^1_{\text{loc}}$ both satisfy the weak derivative identity for f , then $g = h$ almost everywhere.

Solution. Their difference integrates to zero against every test function; the fundamental lemma of distributions implies the associated regular distribution is zero, hence the functions agree a.e. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 20.20 (Classical implies weak proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in C^1$, then its weak derivative equals its classical derivative.

Solution. Integrate by parts against compactly supported test functions; boundary terms vanish. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 20.21 (Mollifier commutation proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For locally integrable f with weak derivative $D_j f$, $D_j(f * \rho_\varepsilon) = (D_j f) * \rho_\varepsilon$ where defined.

Solution. Differentiate the smooth convolution and transfer the derivative from the mollifier to f using the weak derivative identity. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 20.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Weak derivatives extend differentiation to L^1_{loc} functions by testing against compactly supported smooth functions. They are the bridge from distribution theory to Sobolev spaces and weak PDE formulations. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

20.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 20.23. State and apply the central principle behind Weak derivative definition. Give one immediate consequence in the setting of this chapter.

Hint. To verify a weak derivative g , test $\int f\phi' = -\int g\phi$ for every compactly supported smooth ϕ . $\square$

Solution. A function g is the weak derivative $D_j f$ if $\int f D_j \phi = -\int g \phi$ for every test function. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.24. State and apply the central principle behind Agreement with classical derivatives. Give one immediate consequence in the setting of this chapter.

Hint. For piecewise smooth functions, integrate by parts on each smooth piece and inspect the interface terms. $\square$

Solution. Classically differentiable functions have the same weak derivative. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.25. State and apply the central principle behind Piecewise smooth functions. Give one immediate consequence in the setting of this chapter.

Hint. A jump creates a Dirac contribution distributionally, so an L^p weak derivative may fail to exist even when a distributional derivative does. $\square$

Solution. Jumps usually prevent the weak derivative from being represented by an ordinary function because the distributional derivative contains Dirac masses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.26. State and apply the central principle behind Uniqueness. Give one immediate consequence in the setting of this chapter.

Hint. Weak derivatives are unique because a locally integrable function pairing to zero with every test function vanishes almost everywhere. $\square$

Solution. Weak derivatives are unique up to almost-everywhere equality. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.27. State and apply the central principle behind Higher weak derivatives. Give one immediate consequence in the setting of this chapter.

Hint. Approximate by smooth functions and pass to the limit in the integration-by-parts identity when norm control permits. $\square$

Solution. Iterating the definition yields multi-index weak derivatives. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.28. State and apply the central principle behind Locality. Give one immediate consequence in the setting of this chapter.

Hint. Classical differentiability implies weak differentiability under local integrability, but the converse only holds almost everywhere or in a weaker sense. $\square$

Solution. Weak derivatives depend only on local almost-everywhere behavior. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.29. State and apply the central principle behind Integration by parts. Give one immediate consequence in the setting of this chapter.

Hint. Use difference quotients when the chapter provides them as a criterion for Sobolev weak differentiability. $\square$

Solution. The weak derivative definition is precisely the boundary-free integration-by-parts identity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 20.30. State and apply the central principle behind Approximation. Give one immediate consequence in the setting of this chapter.

Hint. Keep boundary terms out of the definition by choosing compactly supported test functions inside the domain. $\square$

Solution. Mollification converts weak derivatives into classical derivatives and commutes with differentiation on interior regions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

20.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 20.31 (Absolute value). What is the weak derivative of $|x|$?

Hint. Differentiate away from zero and check no jump in the function itself. $\square$

Solution. $\text{sign}(x)$ almost everywhere. $\square$

Exercise 20.32 (Heaviside). Does Heaviside have an L^1_{loc} weak derivative?

Hint. Its distributional derivative is δ . $\square$

Solution. No ordinary weak derivative exists. $\square$

Exercise 20.33 (Constant). What is the weak derivative of a constant?

Hint. Use the defining identity. $\square$

Solution. Zero. $\square$

Exercise 20.34 (Classical function). If f is C^1 , what is its weak derivative?

Hint. Use integration by parts. $\square$

Solution. The classical derivative. □

Exercise 20.35 (Mollified derivative). If $Df=g$ weakly, what is $D(f*\rho_\epsilon)$?

Hint. Use commutation. □

Solution. $g*\rho_\epsilon$ by convolution. □

Proofs

Exercise 20.36 (Uniqueness). Prove weak derivatives are unique almost everywhere.

Hint. Subtract two candidates. □

Solution. Their difference defines the zero distribution, hence is zero almost everywhere. □

Exercise 20.37 (Classical implies weak). Prove a C^1 derivative is also the weak derivative.

Hint. Integrate by parts against compactly supported tests. □

Solution. Boundary terms vanish. □

Exercise 20.38 (Locality). Prove weak derivatives are local.

Hint. Use tests supported in a region where two functions agree almost everywhere. □

Solution. Their weak derivative identities agree there. □

Exercise 20.39 (Mollifier commutation). Derive weak derivative commutation with convolution.

Hint. Differentiate convolution and transfer derivative using the weak identity. □

Solution. The two expressions coincide on interior regions. □

Counterexamples and hypothesis testing

Exercise 20.40 (Jump obstruction). Why is a piecewise C^1 function with a nonzero jump not in $W^{1,p}$ locally across the jump?

Hint. Its distributional derivative contains a Dirac mass. □

Solution. A Dirac mass is not represented by an L^p function. □

Exercise 20.41 (Classical failure not weak failure). Give a function weakly differentiable but not classically differentiable at one point.

Hint. Use $|x|$. □

Solution. Its weak derivative is $\text{sign}(x)$. □

Exercise 20.42 (Point changes). Why can changing f at one point not change its weak derivative?

Hint. Weak identities use integrals and functions are a.e. classes. □

Solution. A single point is null. □

Applications and investigations

Exercise 20.43 (Finite elements). Why do weak derivatives fit piecewise polynomial approximations?

Hint. Classical derivatives exist on elements and integral identities tolerate lower regularity. □

Solution. Energy methods require only derivative integrability, not global classical smoothness. □

Exercise 20.44 (Image edges). Why do sharp jumps behave differently from corners in Sobolev regularity?

Hint. A corner such as $|x|$ has an L_p derivative, while a jump produces a Dirac. □

Solution. The singularity type determines whether the derivative remains an ordinary function. □

Challenge problems

Exercise 20.45 (Second weak derivative of $|x|$). What is the second distributional derivative of $|x|$, and why does this block $W^{2,p}$ membership near zero?

Hint. Differentiate $\text{sign}(x)$. □

Solution. It is $2\delta_0$, not an L_p function. □

Exercise 20.46 (Chain rule preview). For a Lipschitz scalar F and sufficiently regular Sobolev u , what form should a weak chain rule take?

Hint. Expect derivative of composition to multiply weak derivative. □

Solution. Under standard hypotheses, $D(F(u)) = F'(u)Du$ almost everywhere; the precise theorem requires the stated regularity assumptions. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 21

Sobolev Spaces

Sobolev spaces measure both a function and its weak derivatives in L^p . They provide the natural energy spaces for variational PDEs, compactness arguments, and weak boundary conditions.

Learning goals

After this chapter, the reader should be able to:

- compute Sobolev norms and determine membership in $W^{k,p}$ or H^k for explicit functions;
- use weak derivatives to formulate Sobolev-space regularity;
- apply basic embedding or inequality results only after checking dimension, exponent, and domain hypotheses;
- interpret traces or boundary values in settings where the chapter establishes them;
- use completeness and Hilbert-space structure of H^k in variational arguments;
- distinguish Sobolev regularity from pointwise smoothness by examples and counterexamples;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

21.1 Definition of $W^{k,p}$

A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p .

21.2 Sobolev norms

Sum the L^p norms of derivatives up to order k ; for $p = 2$ this gives a Hilbert-space norm.

Example 21.1 (Testing Sobolev membership of a cusp). On $(-1, 1)$, let $u(x) = |x|^\alpha$ with $\alpha > 0$. Away from zero,

$$u'(x) = \alpha \operatorname{sgn}(x)|x|^{\alpha-1}.$$

Thus $u \in W^{1,p}$ near zero exactly when both $|x|^{\alpha p}$ and $|x|^{(\alpha-1)p}$ are integrable there. The derivative condition is the restrictive one:

$$(\alpha - 1)p > -1.$$

This gives a sharp regularity threshold from a one-line power test.

21.3 H^k notation

$H^k = W^{k,2}$ is especially important because inner-product methods apply.

21.4 Completeness

Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces.

21.5 Zero-boundary spaces

$W_0^{1,p}$ is the closure of compactly supported smooth functions and encodes homogeneous Dirichlet boundary data.

21.6 Poincare inequality

On suitable bounded domains, the L^p norm of a zero-boundary function is controlled by its gradient.

Example 21.2 (Poincare turns gradient into a full norm). For $u \in H_0^1(\Omega)$ on a bounded Lipschitz domain,

$$\|u\|_2 \leq C \|\nabla u\|_2.$$

Therefore

$$\|u\|_{H^1}^2 = \|u\|_2^2 + \|\nabla u\|_2^2 \leq (C^2 + 1) \|\nabla u\|_2^2.$$

On the zero-boundary space, gradient energy alone controls the full Sobolev norm. This is the coercivity mechanism behind the Dirichlet weak problem.

21.7 Embeddings

Sufficient Sobolev regularity implies improved integrability or continuity.

Example 21.3 (Sobolev embedding requires the exponent-dimension check). In one dimension, $H^1(0,1)$ functions have continuous representatives and obey a bound obtained from the fundamental theorem plus Cauchy–Schwarz. In higher dimensions the same H^1 regularity need not imply continuity. Therefore an embedding statement must always be checked against dimension, derivative order, and integrability exponent rather than transferred by analogy.

21.8 Compactness

Rellich-type theorems turn bounded Sobolev sequences into strongly convergent subsequences in weaker norms.

21.9 Core structural results

Theorem 21.4 (Completeness of $W^{k,p}$). *For $1 \leq p \leq \infty$, $W^{k,p}$ is complete.*

Proof. A Cauchy sequence is Cauchy in L^p for every derivative component; pass to limits in each component and use distributional convergence to verify that the limiting components are the weak derivatives of the limiting function. $\square$

Theorem 21.5 (Poincaré inequality). *On a bounded Lipschitz domain, $\|u\|_{L^p} \leq C\|\nabla u\|_{L^p}$ for $u \in W_0^{1,p}$.*

Proof. Prove first for smooth compactly supported functions by integrating along suitable line segments or using contradiction and compactness, then extend by density. $\square$

Theorem 21.6 (Hilbert structure of H^k). *H^k is a Hilbert space with inner product obtained by summing L^2 inner products of weak derivatives through order k .*

Proof. Completeness follows from Sobolev completeness and the norm is induced by the stated positive-definite inner product. $\square$

21.10 Worked examples

Example 21.7 (Definition of $W^{k,p}$ diagnostic). Give a rigorous worked analysis of Definition of $W^{k,p}$ in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 21.8 (Sobolev norms diagnostic). Give a rigorous worked analysis of Sobolev norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sum the L^p norms of derivatives up to order k ; for $p = 2$ this gives a Hilbert-space norm. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 21.9 (H^k notation diagnostic). Give a rigorous worked analysis of H^k notation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. $H^k = W^{k,2}$ is especially important because inner-product methods apply. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 21.10 (Completeness diagnostic). Give a rigorous worked analysis of Completeness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

21.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 21.11 (Definition of $W^{k,p}$ diagnostic). Give a rigorous worked analysis of Definition of $W^{k,p}$ in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.12 (Sobolev norms diagnostic). Give a rigorous worked analysis of Sobolev norms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sum the L^p norms of derivatives up to order k ; for $p = 2$ this gives a Hilbert-space norm. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.13 (H^k notation diagnostic). Give a rigorous worked analysis of H^k notation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $H^k = W^{k,2}$ is especially important because inner-product methods apply. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.14 (Completeness diagnostic). Give a rigorous worked analysis of Completeness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.15 (Zero-boundary spaces diagnostic). Give a rigorous worked analysis of Zero-boundary spaces in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $W_0^{1,p}$ is the closure of compactly supported smooth functions and encodes homogeneous Dirichlet boundary data. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.16 (Poincare inequality diagnostic). Give a rigorous worked analysis of Poincare inequality in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On suitable bounded domains, the L^p norm of a zero-boundary function is controlled by its gradient. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.17 (Embeddings diagnostic). Give a rigorous worked analysis of Embeddings in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sufficient Sobolev regularity implies improved integrability or continuity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.18 (Compactness diagnostic). Give a rigorous worked analysis of Compactness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Rellich-type theorems turn bounded Sobolev sequences into strongly convergent subsequences in weaker norms. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 21.19 (Completeness of $W^{k,p}$ proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p \leq \infty$, $W^{k,p}$ is complete.

Solution. A Cauchy sequence is Cauchy in L^p for every derivative component; pass to limits in each component and use distributional convergence to verify that the limiting components are the weak derivatives of the limiting function. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 21.20 (Poincaré inequality proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: On a bounded Lipschitz domain, $\|u\|_{L^p} \leq C\|\nabla u\|_{L^p}$ for $u \in W_0^{1,p}$.

Solution. Prove first for smooth compactly supported functions by integrating along suitable line segments or using contradiction and compactness, then extend by density. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 21.21 (Hilbert structure of H^k proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: H^k is a Hilbert space with inner product obtained by summing L^2 inner products of weak derivatives through order k .

Solution. Completeness follows from Sobolev completeness and the norm is induced by the stated positive-definite inner product. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 21.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Sobolev spaces measure both a function and its weak derivatives in L^p . They provide the natural energy spaces for variational PDEs, compactness arguments, and weak boundary conditions. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

21.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 21.23. State and apply the central principle behind Definition of $W^{k,p}$. Give one immediate consequence in the setting of this chapter.

Hint. Compute every weak derivative required by the Sobolev norm before evaluating the norm itself. $\square$

Solution. A function lies in $W^{k,p}(\Omega)$ when all weak derivatives through order k belong to L^p . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.24. State and apply the central principle behind Sobolev norms. Give one immediate consequence in the setting of this chapter.

Hint. For H^k , use the Hilbert norm built from L^2 norms of derivatives up to order k . $\square$

Solution. Sum the L^p norms of derivatives up to order k ; for $p = 2$ this gives a Hilbert-space norm. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.25. State and apply the central principle behind H^k notation. Give one immediate consequence in the setting of this chapter.

Hint. Check dimension and exponent restrictions before invoking any Sobolev embedding; they determine the target space. $\square$

Solution. $H^k = W^{k,2}$ is especially important because inner-product methods apply. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.26. State and apply the central principle behind Completeness. Give one immediate consequence in the setting of this chapter.

Hint. Trace statements require boundary regularity and sufficient Sobolev exponent; verify both before assigning boundary values. $\square$

Solution. Sobolev spaces are Banach spaces, and H^k spaces are Hilbert spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.27. State and apply the central principle behind Zero-boundary spaces. Give one immediate consequence in the setting of this chapter.

Hint. Use completeness by showing a Cauchy sequence converges in each derivative component consistently. $\square$

Solution. $W_0^{1,p}$ is the closure of compactly supported smooth functions and encodes homogeneous Dirichlet boundary data. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.28. State and apply the central principle behind Poincare inequality. Give one immediate consequence in the setting of this chapter.

Hint. To prove an inequality, normalize by scaling first and see which exponent is forced by dimensional balance. $\square$

Solution. On suitable bounded domains, the L^p norm of a zero-boundary function is controlled by its gradient. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 21.29. State and apply the central principle behind Embeddings. Give one immediate consequence in the setting of this chapter.

Hint. Construct counterexamples near singular points to test whether a proposed function lies in $W^{1,p}$. □

Solution. Sufficient Sobolev regularity implies improved integrability or continuity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

Exercise 21.30. State and apply the central principle behind Compactness. Give one immediate consequence in the setting of this chapter.

Hint. Do not infer continuity from Sobolev membership unless the relevant embedding theorem places the space inside a continuous-function class. □

Solution. Rellich-type theorems turn bounded Sobolev sequences into strongly convergent subsequences in weaker norms. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. □

21.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 21.31 (H1 norm). What quantities enter the H1 norm?

Hint. Use the function and first weak derivatives. □

Solution. The L2 norm of u and the L2 norm of its gradient. □

Exercise 21.32 (Cusp threshold). When is $|x|^\alpha$ locally $W^{1,p}$ near zero?

Hint. Check integrability of $|x|^{(\alpha-1)p}$. □

Solution. When $(\alpha-1)p > -1$. □

Exercise 21.33 (Zero boundary). What does H_0^1 encode for Dirichlet problems?

Hint. It is the closure of compactly supported smooth functions. □

Solution. Homogeneous boundary data in the Sobolev trace sense. □

Exercise 21.34 (Poincare). What does Poincare control?

Hint. Use zero-boundary assumption. □

Solution. The function L_p norm by its gradient L_p norm. □

Exercise 21.35 (H^k). What is H^k ?

Hint. Recall notation. □

Solution. $W^{k,2}$. □

Proofs

Exercise 21.36 (Hilbert structure). Explain why H_k is a Hilbert space.

Hint. Sum L^2 inner products of all weak derivatives. □

Solution. The induced norm is complete by Sobolev completeness. □

Exercise 21.37 (Poincare norm equivalence). Show gradient norm is equivalent to H^1 norm on H_0^1 .

Hint. Use Poincare plus the trivial reverse bound. □

Solution. Each norm controls the other up to constants. □

Exercise 21.38 (Completeness strategy). Outline completeness of $W^{k,p}$.

Hint. Take L^p limits of every derivative component. □

Solution. Distributional convergence verifies the limiting components remain weak derivatives of the limit. □

Exercise 21.39 (Embedding hypothesis). Why must dimension and exponent be checked in Sobolev embedding?

Hint. The critical scaling depends on k, p versus n . □

Solution. Crossing the critical exponent changes whether one gets only integrability or actual continuity. □

Counterexamples and hypothesis testing

Exercise 21.40 (Smoothness not required). Give an H^1 function not C^1 .

Hint. Use $|x|$ on a bounded interval. □

Solution. It has square-integrable weak derivative but a corner. □

Exercise 21.41 (Boundary value not pointwise automatic). Why can a general Sobolev function's boundary value require a trace theorem?

Hint. Sobolev functions are a.e. equivalence classes. □

Solution. Pointwise boundary evaluation may not be defined without extra regularity. □

Exercise 21.42 (Unbounded-domain Poincare). Why can Poincare fail without domain/boundary hypotheses?

Hint. Translations/constants may escape gradient control. □

Solution. The L^p size need not be controlled by gradient alone on arbitrary domains. □

Applications and investigations

Exercise 21.43 (Energy method). Why is H_0^1 the natural energy space for Poisson Dirichlet problems?

Hint. Both u and gradient are square integrable and boundary data are built in. □

Solution. The bilinear form $\int \text{grad } u \cdot \text{grad } v$ is bounded and coercive there. □

Exercise 21.44 (Compactness). Why is Rellich compactness stronger than boundedness?

Hint. It provides strongly convergent subsequences in weaker norms. □

Solution. This is crucial for nonlinear limit passages. □

Challenge problems

Exercise 21.45 (Scaling criticality). Use dimensional scaling to explain why the relation $kp=n$ is critical for Sobolev embeddings.

Hint. Track how derivative L_p norms scale under $u(\lambda x)$. □

Solution. At $kp=n$ the natural norm becomes scale-invariant, marking the transition between integrability gain and continuity-type behavior. □

Exercise 21.46 (One-dimensional estimate). Derive a continuity estimate for u in $H^1(0,1)$.

Hint. Write $u(x)-u(y)$ as the integral of u' . □

Solution. Cauchy–Schwarz gives $|u(x)-u(y)| \leq \|u'\|_2 |x-y|^{1/2}$, for an absolutely continuous representative. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 22

Approximation and Density

Approximation and density justify replacing rough functions by smooth ones. Mollifiers, cutoffs, and extension arguments let proofs and numerical constructions move between Sobolev or L^p functions and smooth compactly supported models.

Learning goals

After this chapter, the reader should be able to:

- construct smooth approximations by mollification and quantify the mode of convergence;
- prove density of smooth or compactly supported smooth functions in the function spaces covered by the chapter;
- control boundary effects when approximating functions on domains rather than on all of $\mathbb{R}^n$;
- use cutoff functions and convolution in the correct order to preserve desired support properties;
- distinguish approximation in norm from pointwise approximation;
- identify spaces or boundary conditions for which naive mollification does not directly give the required dense subclass;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

22.1 Mollifiers

A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one.

Example 22.1 (Mollifying a discontinuous indicator). Let $f = 1_{[0,1]}$ on $\mathbb{R}$ and let ρ_ε be a standard mollifier. Then

$$f_\varepsilon = f * \rho_\varepsilon$$

is smooth, equals 1 away from an $O(\varepsilon)$ neighborhood of the endpoints, and transitions smoothly near 0 and 1. For every finite p ,

$$\|f_\varepsilon - f\|_p \rightarrow 0.$$

The discrepancy is confined to shrinking boundary layers.

22.2 Approximate identities

Scaled mollifiers concentrate at the origin and approximate functions in L^p .

22.3 Smooth approximation in L^p

Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \leq p < \infty$.

22.4 Sobolev approximation

Mollification approximates weak derivatives simultaneously on interior subdomains.

22.5 Boundary issues

Near the boundary, extension operators or geometry-aware cutoffs are needed to mollify without leaving the domain.

Example 22.2 (Why boundary-aware extension is needed). If f is defined only on a domain Ω , direct convolution samples values $f(x - y)$ that may lie outside Ω when x is near the boundary. One therefore extends f to a larger region or first localizes with cutoffs. This is not a technical nuisance: without an extension rule, the convolution is not even defined from the original data near the boundary.

22.6 Density of test functions

$C_c^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$ by definition and under standard characterizations.

22.7 Frequency truncation

Fourier cutoffs give an alternative smooth approximation method in whole-space Sobolev spaces.

Example 22.3 (Frequency cutoff smooths whole-space Sobolev data). For $u \in H^s(\mathbb{R}^n)$, choose a smooth frequency cutoff $\chi_R(\xi)$ supported in $|\xi| \lesssim R$ and equal to one on a smaller ball. Define

$$\hat{u}_R = \chi_R \hat{u}.$$

Then u_R is smooth and

$$\|u_R - u\|_{H^s} \rightarrow 0$$

by dominated convergence with the Sobolev weight. Frequency truncation is the spectral analogue of mollification.

22.8 Approximation consequences

Density transfers identities and inequalities proved for smooth functions to rough functions by norm limits.

22.9 Core structural results

Theorem 22.4 (L^p mollifier convergence). *If $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$, then $f * \rho_\varepsilon \rightarrow f$ in L^p .*

Proof. Write the convolution difference as an average of translations $\tau_y f - f$ and use continuity of translations in L^p together with concentration of the mollifier near zero. $\square$

Theorem 22.5 (Derivative commutation). *If $f \in W^{k,p}$, then $D^\alpha(f * \rho_\varepsilon) = (D^\alpha f) * \rho_\varepsilon$ for $|\alpha| \leq k$.*

Proof. Use the weak derivative identity to move derivatives between the function and mollifier, then apply convolution associativity on compactly supported kernels. $\square$

Theorem 22.6 (Density of C_c^∞ in L^p). *For $1 \leq p < \infty$, smooth compactly supported functions are dense in $L^p(\mathbb{R}^n)$.*

Proof. First truncate f spatially and in magnitude to obtain bounded finite-support approximants, then mollify them; each step has arbitrarily small L^p error. $\square$

22.10 Worked examples

Example 22.7 (Mollifiers diagnostic). Give a rigorous worked analysis of Mollifiers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 22.8 (Approximate identities diagnostic). Give a rigorous worked analysis of Approximate identities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Scaled mollifiers concentrate at the origin and approximate functions in L^p . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 22.9 (Smooth approximation in L^p diagnostic). Give a rigorous worked analysis of Smooth approximation in L^p in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \leq p < \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 22.10 (Sobolev approximation diagnostic). Give a rigorous worked analysis of Sobolev approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Mollification approximates weak derivatives simultaneously on interior subdomains. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

22.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 22.11 (Mollifiers diagnostic). Give a rigorous worked analysis of Mollifiers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.12 (Approximate identities diagnostic). Give a rigorous worked analysis of Approximate identities in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Scaled mollifiers concentrate at the origin and approximate functions in L^p . The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.13 (Smooth approximation in L^p diagnostic). Give a rigorous worked analysis of Smooth approximation in L^p in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \leq p < \infty$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.14 (Sobolev approximation diagnostic). Give a rigorous worked analysis of Sobolev approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Mollification approximates weak derivatives simultaneously on interior subdomains. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.15 (Boundary issues diagnostic). Give a rigorous worked analysis of Boundary issues in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Near the boundary, extension operators or geometry-aware cutoffs are needed to mollify without leaving the domain. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.16 (Density of test functions diagnostic). Give a rigorous worked analysis of Density of test functions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. $C_c^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$ by definition and under standard characterizations. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.17 (Frequency truncation diagnostic). Give a rigorous worked analysis of Frequency truncation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fourier cutoffs give an alternative smooth approximation method in whole-space Sobolev spaces. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.18 (Approximation consequences diagnostic). Give a rigorous worked analysis of Approximation consequences in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Density transfers identities and inequalities proved for smooth functions to rough functions by norm limits. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 22.19 (L^p mollifier convergence proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$, then $f * \rho_\varepsilon \rightarrow f$ in L^p .

Solution. Write the convolution difference as an average of translations $\tau_y f - f$ and use continuity of translations in L^p together with concentration of the mollifier near zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 22.20 (Derivative commutation proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $f \in W^{k,p}$, then $D^\alpha(f * \rho_\varepsilon) = (D^\alpha f) * \rho_\varepsilon$ for $|\alpha| \leq k$.

Solution. Use the weak derivative identity to move derivatives between the function and mollifier, then apply convolution associativity on compactly supported kernels. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 22.21 (Density of C_c^∞ in L^p proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $1 \leq p < \infty$, smooth compactly supported functions are dense in $L^p(\mathbb{R}^n)$.

Solution. First truncate f spatially and in magnitude to obtain bounded finite-support approximants, then mollify them; each step has arbitrarily small L^p error. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 22.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Approximation and density justify replacing rough functions by smooth ones. Mollifiers, cutoffs, and extension arguments let proofs and numerical constructions move between Sobolev or L^p functions and smooth compactly supported models. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

22.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 22.23. State and apply the central principle behind Mollifiers. Give one immediate consequence in the setting of this chapter.

Hint. Extend or localize the function first if naive convolution would sample outside the domain. $\square$

Solution. A standard mollifier is smooth, nonnegative, compactly supported, and normalized to integral one. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.24. State and apply the central principle behind Approximate identities. Give one immediate consequence in the setting of this chapter.

Hint. Choose a mollifier ρ_ε and use $f * \rho_\varepsilon$ as the smooth approximation. $\square$

Solution. Scaled mollifiers concentrate at the origin and approximate functions in L^p . As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.25. State and apply the central principle behind Smooth approximation in L^p . Give one immediate consequence in the setting of this chapter.

Hint. Norm convergence follows by translation continuity plus averaging; write the convolution error as an average of translates. $\square$

Solution. Convolution plus cutoff proves density of C_c^∞ in L^p for $1 \leq p < \infty$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.26. State and apply the central principle behind Sobolev approximation. Give one immediate consequence in the setting of this chapter.

Hint. Use a cutoff after mollification when compact support inside the domain is required. $\square$

Solution. Mollification approximates weak derivatives simultaneously on interior subdomains. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.27. State and apply the central principle behind Boundary issues. Give one immediate consequence in the setting of this chapter.

Hint. Track weak derivatives through convolution: $D^\alpha(f * \rho_\varepsilon) = (D^\alpha f) * \rho_\varepsilon$ in the weak sense. $\square$

Solution. Near the boundary, extension operators or geometry-aware cutoffs are needed to mollify without leaving the domain. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.28. State and apply the central principle behind Density of test functions. Give one immediate consequence in the setting of this chapter.

Hint. Boundary conditions may not survive ordinary mollification; modify the construction rather than assuming they do. $\square$

Solution. $C_c^\infty(\Omega)$ is dense in $W_0^{1,p}(\Omega)$ by definition and under standard characterizations. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.29. State and apply the central principle behind Frequency truncation. Give one immediate consequence in the setting of this chapter.

Hint. Density in norm is stronger than pointwise approximation; estimate the actual target-space norm. $\square$

Solution. Fourier cutoffs give an alternative smooth approximation method in whole-space Sobolev spaces. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 22.30. State and apply the central principle behind Approximation consequences. Give one immediate consequence in the setting of this chapter.

Hint. Choose the approximation scale after fixing the error tolerance so each localization and smoothing error is controlled separately. $\square$

Solution. Density transfers identities and inequalities proved for smooth functions to rough functions by norm limits. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

22.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 22.31 (Mollifier mass). What is the integral of ρ_ϵ ?

Hint. Scale variables. $\square$

Solution. 1. $\square$

Exercise 22.32 (Mollifier support). If ρ is supported in the unit ball, where is ρ_ϵ supported?

Hint. Scale by ϵ . $\square$

Solution. In the ball of radius ϵ . $\square$

Exercise 22.33 (Smoothness). Why is $f * \rho_\epsilon$ smooth for locally integrable f ?

Hint. Derivatives fall on ρ_ϵ . $\square$

Solution. All derivatives are given by convolution with smooth compactly supported kernel derivatives. $\square$

Exercise 22.34 (L_p convergence). For finite p , what happens to $f * \rho_\epsilon$ for f in L_p ?

Hint. Use approximate identity theorem. $\square$

Solution. It converges to f in L_p . $\square$

Exercise 22.35 (Weak derivative). If f is $W^{1,p}$, how does derivative of its mollification relate to Df ?

Hint. Use commutation. $\square$

Solution. $D(f * \rho_\epsilon) = (Df) * \rho_\epsilon$. $\square$

Proofs

Exercise 22.36 (Approximate identity proof). Outline L_p mollifier convergence.

Hint. Write the difference as an average of translations. □

Solution. Translation continuity and kernel concentration give the result. □

Exercise 22.37 (Derivative commutation). Prove weak derivatives commute with mollification.

Hint. Transfer derivatives using the weak identity. □

Solution. The smooth derivative equals convolution with the weak derivative. □

Exercise 22.38 (Density). Explain the cutoff-plus-mollifier proof of density of C_c -infinity in L_p .

Hint. First truncate spatially, then mollify. □

Solution. Each step contributes arbitrarily small norm error. □

Exercise 22.39 (Boundary need). Why does naive zero extension sometimes fail to preserve W_{1p} regularity?

Hint. A nonzero trace creates a jump across the boundary. □

Solution. That jump contributes a singular derivative; extension operators or zero-trace hypotheses are needed. □

Counterexamples and hypothesis testing

Exercise 22.40 (Linfinity density). Why is C_c -infinity not dense in L_∞ on the whole line?

Hint. Uniform norm cannot approximate a nonvanishing constant by compact support. □

Solution. Outside every compact support the error from constant 1 is 1. □

Exercise 22.41 (Pointwise not norm). Why does pointwise convergence of mollifications not alone prove L_p density?

Hint. Norm convergence requires integrated error control. □

Solution. Approximate-identity estimates supply that stronger control. □

Exercise 22.42 (Boundary leakage). Why can direct mollification violate domain support or boundary conditions?

Hint. Convolution spreads support by the kernel radius. □

Solution. Values can cross the boundary unless cutoffs/extensions are used. □

Applications and investigations

Exercise 22.43 (Finite elements). Why is density important in variational numerics?

Hint. Smooth/finite-dimensional test functions approximate energy-space functions. ☐

Solution. Identities proved on simple functions extend to weak solutions by norm limits. ☐

Exercise 22.44 (Spectral regularization). Why use frequency cutoff rather than physical mollification in whole-space PDE?

Hint. It diagonalizes Sobolev norms and operator multipliers. ☐

Solution. Regularity and convergence can be read directly from weighted spectral tails. ☐

Challenge problems

Exercise 22.45 (Two-scale approximation). Describe how to approximate an L_p function by a smooth compactly supported one within epsilon.

Hint. Choose a compact cutoff making the tail small, then choose mollifier scale making local smoothing error small. ☐

Solution. Triangle inequality combines the two errors below epsilon. ☐

Exercise 22.46 (Sobolev density transfer). Why does derivative commutation allow mollifier convergence in W_{kp} rather than just L_p ?

Hint. Apply the L_p approximate-identity theorem to every weak derivative. ☐

Solution. Each derivative component converges, so the summed Sobolev norm converges. ☐

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 23

Weak Boundary-Value Problems

Weak boundary-value problems replace pointwise differential equations by integral identities tested against admissible functions. This is the natural formulation for rough coefficients, Sobolev solutions, and variational existence theory.

Learning goals

After this chapter, the reader should be able to:

- derive the weak formulation of a boundary-value problem by multiplying by a test function and integrating by parts;
- identify the natural trial and test spaces for elementary weak PDE problems;
- verify boundedness and coercivity of bilinear forms in representative examples;
- apply an existence theorem such as Lax–Milgram only after checking its hypotheses;
- interpret weak boundary conditions and distinguish them from pointwise boundary equalities;
- recover additional regularity or classical equations from a weak solution when justified;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

23.1 Strong and weak formulations

A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts.

Example 23.1 (Deriving the weak Poisson problem). For

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega,$$

multiply by $v \in C_c^\infty(\Omega)$ and integrate by parts:

$$\int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f v.$$

The weak problem asks for $u \in H_0^1(\Omega)$ satisfying this identity for every $v \in H_0^1(\Omega)$. The boundary condition is encoded in the space, not imposed pointwise.

23.2 Energy spaces

Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions.

23.3 Bilinear forms

Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_{\Omega} A \nabla u \cdot \nabla v + cuv$.

23.4 Linear functionals

Forcing terms act as continuous linear functionals on the chosen energy space.

23.5 Coercivity

A coercive estimate $a(v, v) \geq c\|v\|_V^2$ supplies control of the solution norm.

Example 23.2 (Poincare turns Poisson energy into coercivity). For the Poisson bilinear form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v$$

on $H_0^1(\Omega)$,

$$a(v, v) = \|\nabla v\|_2^2.$$

Poincare gives $\|v\|_2 \leq C\|\nabla v\|_2$, hence the gradient norm is equivalent to the full H^1 norm. Therefore $a(v, v)$ controls $\|v\|_{H^1}^2$ up to a constant, providing Lax–Milgram coercivity.

23.6 Weak boundary conditions

Dirichlet conditions are built into H_0^1 ; Neumann data typically arise through boundary terms in integration by parts.

23.7 Galerkin approximation

Finite-dimensional subspaces convert the weak problem into linear algebra and support constructive approximation.

Example 23.3 (Galerkin turns the weak PDE into linear algebra). Choose a finite-dimensional space $V_h = \text{span}\{\phi_1, \dots, \phi_m\}$ inside the energy space and write

$$u_h = \sum_{j=1}^m c_j \phi_j.$$

Testing with each basis function gives

$$\sum_j a(\phi_j, \phi_i) c_j = F(\phi_i), \quad i = 1, \dots, m.$$

Thus the stiffness matrix $A_{ij} = a(\phi_j, \phi_i)$ is the finite-dimensional representation of the weak operator. Coercivity makes the discrete system stable and nonsingular.

23.8 Uniqueness by energy

Testing the difference of two solutions against itself often proves uniqueness from positivity or coercivity.

23.9 Core structural results

Theorem 23.4 (Lax–Milgram theorem). *If V is a Hilbert space, a is bounded and coercive, and $F \in V^*$, then there is a unique $u \in V$ with $a(u, v) = F(v)$ for all $v \in V$.*

Proof. Use Riesz representation to write $a(u, v) = \langle Au, v \rangle$ for a bounded operator A . Coercivity makes A bounded below, hence injective with closed range; orthogonality to the range would contradict coercivity for the adjoint, so the range is all of V . $\square$

Theorem 23.5 (Energy uniqueness). *If a is coercive and $a(u, v) = 0$ for all v , then $u = 0$.*

Proof. Choose $v = u$. Then $0 = a(u, u) \geq c\|u\|_V^2$, hence $u = 0$. $\square$

Theorem 23.6 (Galerkin stability). *For coercive a , the Galerkin solution u_h satisfies $\|u_h\|_V \leq c^{-1}\|F\|_{V^*}$.*

Proof. Test the discrete equation with u_h and combine coercivity with the dual norm bound. $\square$

23.10 Worked examples

Example 23.7 (Strong and weak formulations diagnostic). Give a rigorous worked analysis of Strong and weak formulations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 23.8 (Energy spaces diagnostic). Give a rigorous worked analysis of Energy spaces in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 23.9 (Bilinear forms diagnostic). Give a rigorous worked analysis of Bilinear forms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_{\Omega} A \nabla u \cdot \nabla v + cuv$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 23.10 (Linear functionals diagnostic). Give a rigorous worked analysis of Linear functionals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Forcing terms act as continuous linear functionals on the chosen energy space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

23.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 23.11 (Strong and weak formulations diagnostic). Give a rigorous worked analysis of Strong and weak formulations in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.12 (Energy spaces diagnostic). Give a rigorous worked analysis of Energy spaces in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.13 (Bilinear forms diagnostic). Give a rigorous worked analysis of Bilinear forms in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_{\Omega} A \nabla u \cdot \nabla v + cuv$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.14 (Linear functionals diagnostic). Give a rigorous worked analysis of Linear functionals in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Forcing terms act as continuous linear functionals on the chosen energy space. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.15 (Coercivity diagnostic). Give a rigorous worked analysis of Coercivity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A coercive estimate $a(v, v) \geq c\|v\|_V^2$ supplies control of the solution norm. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.16 (Weak boundary conditions diagnostic). Give a rigorous worked analysis of Weak boundary conditions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Dirichlet conditions are built into H_0^1 ; Neumann data typically arise through boundary terms in integration by parts. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.17 (Galerkin approximation diagnostic). Give a rigorous worked analysis of Galerkin approximation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Finite-dimensional subspaces convert the weak problem into linear algebra and support constructive approximation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.18 (Uniqueness by energy diagnostic). Give a rigorous worked analysis of Uniqueness by energy in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Testing the difference of two solutions against itself often proves uniqueness from positivity or coercivity. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 23.19 (Lax–Milgram theorem proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If V is a Hilbert space, a is bounded and coercive, and $F \in V^*$, then there is a unique $u \in V$ with $a(u, v) = F(v)$ for all $v \in V$.

Solution. Use Riesz representation to write $a(u, v) = \langle Au, v \rangle$ for a bounded operator A . Coercivity makes A bounded below, hence injective with closed range; orthogonality to the range would contradict coercivity for the adjoint, so the range is all of V . This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 23.20 (Energy uniqueness proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If a is coercive and $a(u, v) = 0$ for all v , then $u = 0$.

Solution. Choose $v = u$. Then $0 = a(u, u) \geq c\|u\|_V^2$, hence $u = 0$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 23.21 (Galerkin stability proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For coercive a , the Galerkin solution u_h satisfies $\|u_h\|_V \leq c^{-1}\|F\|_{V^*}$.

Solution. Test the discrete equation with u_h and combine coercivity with the dual norm bound. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 23.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Weak boundary-value problems replace pointwise differential equations by integral identities tested against admissible functions. This is the natural formulation for rough coefficients, Sobolev solutions, and variational existence theory. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

23.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 23.23. State and apply the central principle behind Strong and weak formulations. Give one immediate consequence in the setting of this chapter.

Hint. Multiply the PDE by a test function, integrate over the domain, and integrate derivatives off the unknown to obtain the weak form. $\square$

Solution. A strong solution satisfies the PDE pointwise; a weak solution satisfies an integrated identity obtained by multiplying by test functions and integrating by parts. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.24. State and apply the central principle behind Energy spaces. Give one immediate consequence in the setting of this chapter.

Hint. Choose the trial space so the boundary condition is encoded structurally, for example by zero trace rather than a pointwise statement. $\square$

Solution. Sobolev spaces such as $H_0^1(\Omega)$ encode square-integrable weak gradients and homogeneous boundary conditions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.25. State and apply the central principle behind Bilinear forms. Give one immediate consequence in the setting of this chapter.

Hint. Bound the bilinear form using Hölder, Cauchy–Schwarz, or Sobolev inequalities until it is continuous in the chosen norm. $\square$

Solution. Second-order elliptic equations are represented by bilinear forms such as $a(u, v) = \int_{\Omega} A \nabla u \cdot \nabla v + cuv$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.26. State and apply the central principle behind Linear functionals. Give one immediate consequence in the setting of this chapter.

Hint. Coercivity requires a lower bound on $a(u, u)$; use Poincaré-type control when the gradient norm alone appears. $\square$

Solution. Forcing terms act as continuous linear functionals on the chosen energy space. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.27. State and apply the central principle behind Coercivity. Give one immediate consequence in the setting of this chapter.

Hint. Only after boundedness and coercivity are verified should you invoke Lax–Milgram for existence and uniqueness. $\square$

Solution. A coercive estimate $a(v, v) \geq c\|v\|_V^2$ supplies control of the solution norm. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.28. State and apply the central principle behind Weak boundary conditions. Give one immediate consequence in the setting of this chapter.

Hint. Interpret the right-hand side as a bounded linear functional on the test space and estimate its norm. $\square$

Solution. Dirichlet conditions are built into H_0^1 ; Neumann data typically arise through boundary terms in integration by parts. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.29. State and apply the central principle behind Galerkin approximation. Give one immediate consequence in the setting of this chapter.

Hint. To recover the strong PDE, integrate by parts backward only when the weak solution has enough additional regularity. $\square$

Solution. Finite-dimensional subspaces convert the weak problem into linear algebra and support constructive approximation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 23.30. State and apply the central principle behind Uniqueness by energy. Give one immediate consequence in the setting of this chapter.

Hint. Keep essential and natural boundary conditions distinct: they enter the variational formulation in different ways. $\square$

Solution. Testing the difference of two solutions against itself often proves uniqueness from positivity or coercivity. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

23.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 23.31 (Weak Poisson). What bilinear form arises for homogeneous Dirichlet Poisson?

Hint. Integrate by parts once. $\square$

Solution. $a(u,v) = \int \text{grad } u \cdot \text{grad } v$. $\square$

Exercise 23.32 (Energy space). What is the natural trial space for homogeneous Dirichlet Poisson?

Hint. Encode zero trace. $\square$

Solution. $H_0^1(\Omega)$. $\square$

Exercise 23.33 (Coercivity). What inequality is used to make gradient energy control full H^1 norm?

Hint. Use Poincare. □

Solution. $\|u\|_2 \leq C \|\text{grad } u\|_2$. □

Exercise 23.34 (Uniqueness test). What test function proves uniqueness for a homogeneous coercive problem?

Hint. Test with the solution difference. □

Solution. Use $v=u$. □

Exercise 23.35 (Galerkin matrix). What entries form the stiffness matrix?

Hint. Evaluate the bilinear form on basis pairs. □

Solution. $A_{ij} = a(\phi_j, \phi_i)$. □

Proofs

Exercise 23.36 (Weak derivation). Derive the weak Dirichlet Poisson formulation.

Hint. Multiply by a test and integrate by parts. □

Solution. Boundary terms vanish through the test/zero-trace structure, leaving gradient energy. □

Exercise 23.37 (Boundedness). Prove the Poisson bilinear form is bounded on H_0^1 .

Hint. Use Cauchy–Schwarz. □

Solution. $|a(u, v)| \leq \|\text{grad } u\|_2 \|\text{grad } v\|_2$. □

Exercise 23.38 (Energy uniqueness). Prove coercivity implies uniqueness.

Hint. Subtract two solutions and test with the difference. □

Solution. Coercivity forces its norm to zero. □

Exercise 23.39 (Galerkin stability). Derive the standard Galerkin norm bound.

Hint. Test the discrete equation with u_h . □

Solution. Coercivity and the dual norm give $\|u_h\| \leq c^{-1} \|F\|$. □

Counterexamples and hypothesis testing

Exercise 23.40 (No coercivity). Why can uniqueness fail if the bilinear form has a nontrivial nullspace?

Hint. Testing with the difference yields only zero energy, not zero vector. □

Solution. Neumann Laplacian constants are a standard example without normalization. □

Exercise 23.41 (Wrong trial space). Why can choosing H^1 instead of H_0^1 change a Dirichlet problem?

Hint. The homogeneous boundary condition is no longer built in. □

Solution. Extra boundary degrees of freedom alter the weak problem. □

Exercise 23.42 (Formal integration by parts). Why is pointwise C^2 regularity not required for a weak solution?

Hint. Derivatives are interpreted weakly and only first derivatives need be integrable. $\square$

Solution. The integrated identity is meaningful in Sobolev space. $\square$

Applications and investigations

Exercise 23.43 (Finite elements). Why is Galerkin the conceptual basis of finite-element methods?

Hint. Choose local finite-dimensional basis functions inside the energy space. $\square$

Solution. The PDE becomes a sparse linear system defined by the bilinear form. $\square$

Exercise 23.44 (Rough coefficients). Why are weak formulations useful for discontinuous elliptic coefficients?

Hint. The coefficient multiplies weak gradients inside an integral. $\square$

Solution. No classical derivative of the coefficient is needed in the divergence-form formulation. $\square$

Challenge problems

Exercise 23.45 (Lax–Milgram estimate). If a is coercive with constant c and F has dual norm M , derive the solution bound.

Hint. Test with u and use bounded action of F . $\square$

Solution. $c\|u\|^2 \leq a(u,u) = F(u) \leq M\|u\|$, hence $\|u\| \leq M/c$. $\square$

Exercise 23.46 (Cea preview). Why should a coercive Galerkin solution be close to the best approximation in V_h ?

Hint. Orthogonality of the error plus boundedness/coercivity compare true error with any competitor. $\square$

Solution. This yields Cea's lemma: error is bounded by a constant times the best approximation error. $\square$

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 24

Fundamental Solutions

A fundamental solution is a distributional inverse kernel for a differential operator. Once one is known, convolution converts a constant-coefficient PDE into an explicit representation formula.

Learning goals

After this chapter, the reader should be able to:

- verify that a candidate fundamental solution satisfies $L\Phi = \delta$ in the distributional sense;
- derive fundamental solutions from Fourier multipliers in constant-coefficient examples;
- compute solutions of forced equations by convolution with a fundamental solution when the operation is justified;
- track singularity, scaling, and dimensional dependence of standard fundamental solutions;
- distinguish a fundamental solution from a Green function adapted to boundary conditions;
- use distributional differentiation to check formulas that are singular at the source point;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

24.1 Definition

A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions.

24.2 Convolution solution

When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$.

24.3 Laplacian kernels

The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three.

Example 24.1 (Why the Newtonian power has the right scaling). For $n \geq 3$, the candidate Laplace fundamental solution has form

$$E(x) = c_n |x|^{2-n}.$$

Away from zero, direct radial differentiation gives $\Delta E = 0$. Its radial flux through a sphere of radius r is proportional to

$$r^{n-1} \partial_r (r^{2-n}),$$

which is independent of r . Choosing c_n so that this flux equals one normalizes the Dirac mass at the origin.

24.4 Heat kernel

The heat kernel is the spacetime propagation kernel for the heat operator with causal structure.

24.5 Wave kernels

Wave fundamental solutions reflect finite propagation and dimension-dependent support geometry.

24.6 Distributional verification

The singularity at the origin is detected by integrating by parts on punctured domains and passing to a boundary-flux limit.

Example 24.2 (Distributional verification by punctured integration). To verify $-\Delta E = \delta_0$, integrate on $\Omega_\varepsilon = \mathbb{R}^n \setminus B_\varepsilon(0)$. Twice applying integration by parts transfers derivatives from E to a test function. The volume term vanishes because E is harmonic away from zero; the outer boundary vanishes by compact support, while the inner sphere flux tends to $\phi(0)$. The singular source is exactly the limiting boundary flux.

24.7 Nonuniqueness

Adding any homogeneous solution to a fundamental solution yields another fundamental solution.

Example 24.3 (Fundamental solutions are not unique). If E satisfies $P(D)E = \delta_0$ and h satisfies the homogeneous equation $P(D)h = 0$, then

$$P(D)(E + h) = \delta_0.$$

Thus $E + h$ is another fundamental solution. Boundary conditions or growth requirements are what select a particular Green kernel or physically relevant representative from this affine family.

24.8 Regularity away from the pole

Fundamental solutions are smooth wherever the source point is absent.

24.9 Core structural results

Theorem 24.4 (Convolution inversion). *If $P(D)E = \delta_0$ and the relevant convolutions exist, then $P(D)(E * f) = f$.*

Proof. Constant-coefficient differentiation commutes with convolution, so $P(D)(E * f) = (P(D)E) * f = \delta_0 * f = f$. $\square$

Theorem 24.5 (Newtonian fundamental solution). *For $n \geq 3$, a constant multiple of $|x|^{2-n}$ is a fundamental solution of $-\Delta$.*

Proof. The function is harmonic away from zero. Integrate against a test function on the complement of a small ball, integrate by parts, and choose the constant so the shrinking boundary flux tends to the test value at zero. $\square$

Theorem 24.6 (Smoothness off the singularity). *If $P(D)E = \delta_0$, then $P(D)E = 0$ on every open set avoiding the origin.*

Proof. Test functions supported away from zero annihilate δ_0 , so the distribution solves the homogeneous equation there; standard regularity for the explicit classical kernels gives smoothness. $\square$

24.10 Worked examples

Example 24.7 (Definition diagnostic). Give a rigorous worked analysis of Definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 24.8 (Convolution solution diagnostic). Give a rigorous worked analysis of Convolution solution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 24.9 (Laplacian kernels diagnostic). Give a rigorous worked analysis of Laplacian kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 24.10 (Heat kernel diagnostic). Give a rigorous worked analysis of Heat kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The heat kernel is the spacetime propagation kernel for the heat operator with causal structure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

24.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 24.11 (Definition diagnostic). Give a rigorous worked analysis of Definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.12 (Convolution solution diagnostic). Give a rigorous worked analysis of Convolution solution in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.13 (Laplacian kernels diagnostic). Give a rigorous worked analysis of Laplacian kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.14 (Heat kernel diagnostic). Give a rigorous worked analysis of Heat kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The heat kernel is the spacetime propagation kernel for the heat operator with causal structure. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.15 (Wave kernels diagnostic). Give a rigorous worked analysis of Wave kernels in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Wave fundamental solutions reflect finite propagation and dimension-dependent support geometry. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.16 (Distributional verification diagnostic). Give a rigorous worked analysis of Distributional verification in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The singularity at the origin is detected by integrating by parts on punctured domains and passing to a boundary-flux limit. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.17 (Nonuniqueness diagnostic). Give a rigorous worked analysis of Nonuniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Adding any homogeneous solution to a fundamental solution yields another fundamental solution. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.18 (Regularity away from the pole diagnostic). Give a rigorous worked analysis of Regularity away from the pole in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Fundamental solutions are smooth wherever the source point is absent. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 24.19 (Convolution inversion proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $P(D)E = \delta_0$ and the relevant convolutions exist, then $P(D)(E * f) = f$.

Solution. Constant-coefficient differentiation commutes with convolution, so $P(D)(E * f) = (P(D)E) * f = \delta_0 * f = f$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 24.20 (Newtonian fundamental solution proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $n \geq 3$, a constant multiple of $|x|^{2-n}$ is a fundamental solution of $-\Delta$.

Solution. The function is harmonic away from zero. Integrate against a test function on the complement of a small ball, integrate by parts, and choose the constant so the shrinking boundary flux tends to the test value at zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 24.21 (Smoothness off the singularity proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $P(D)E = \delta_0$, then $P(D)E = 0$ on every open set avoiding the origin.

Solution. Test functions supported away from zero annihilate δ_0 , so the distribution solves the homogeneous equation there; standard regularity for the explicit classical kernels gives smoothness. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 24.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. A fundamental solution is a distributional inverse kernel for a differential operator. Once one is known, convolution converts a constant-coefficient PDE into an explicit representation formula. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

24.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 24.23. State and apply the central principle behind Definition. Give one immediate consequence in the setting of this chapter.

Hint. To verify $L\Phi = \delta$, pair both sides with a test function and move L onto the test function. $\square$

Solution. A fundamental solution E for $P(D)$ satisfies $P(D)E = \delta_0$ in distributions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.24. State and apply the central principle behind Convolution solution. Give one immediate consequence in the setting of this chapter.

Hint. Fourier transform converts a constant-coefficient operator into multiplication by its symbol; solve algebraically in frequency space. $\square$

Solution. When convolution is legitimate, $u = E * f$ satisfies $P(D)u = f$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.25. State and apply the central principle behind Laplacian kernels. Give one immediate consequence in the setting of this chapter.

Hint. After finding $\widehat{\Phi}$, identify the inverse transform and track singular behavior at the source. $\square$

Solution. The Laplacian has logarithmic fundamental solution in two dimensions and Newtonian power kernels in dimensions at least three. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.26. State and apply the central principle behind Heat kernel. Give one immediate consequence in the setting of this chapter.

Hint. A solution $u = \Phi * f$ is valid only when the convolution and differentiation operations are justified in the chosen function/distribution class. $\square$

Solution. The heat kernel is the spacetime propagation kernel for the heat operator with causal structure. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.27. State and apply the central principle behind Wave kernels. Give one immediate consequence in the setting of this chapter.

Hint. Use scaling of the operator to predict the homogeneity of the fundamental solution before computing constants. $\square$

Solution. Wave fundamental solutions reflect finite propagation and dimension-dependent support geometry. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.28. State and apply the central principle behind Distributional verification. Give one immediate consequence in the setting of this chapter.

Hint. For Laplace-type operators, separate the singular point from the smooth region and verify harmonicity away from the source. $\square$

Solution. The singularity at the origin is detected by integrating by parts on punctured domains and passing to a boundary-flux limit. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.29. State and apply the central principle behind Nonuniqueness. Give one immediate consequence in the setting of this chapter.

Hint. Boundary conditions are not part of a free-space fundamental solution; do not confuse it with a Green function. $\square$

Solution. Adding any homogeneous solution to a fundamental solution yields another fundamental solution. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 24.30. State and apply the central principle behind Regularity away from the pole. Give one immediate consequence in the setting of this chapter.

Hint. Check the normalization constant by integrating the flux over a small sphere around the singularity or by testing against a bump. $\square$

Solution. Fundamental solutions are smooth wherever the source point is absent. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

24.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 24.31 (Definition). What equation defines a fundamental solution E for $P(D)$?

Hint. Use the distributional inverse definition. $\square$

Solution. $P(D)E = \delta_0$. $\square$

Exercise 24.32 (Convolution). If $P(D)E = \delta$ and $u = E * f$, what is $P(D)u$?

Hint. Differentiate through convolution. $\square$

Solution. f , when convolution is justified. $\square$

Exercise 24.33 (Dimension 3). What radial power appears for the Laplacian in three dimensions?

Hint. Use $|x|^{-(2-n)}$. $\square$

Solution. $1/|x|$ up to normalization. $\square$

Exercise 24.34 (Dimension 2). What qualitative form replaces the power kernel in two dimensions?

Hint. Recall the critical radial case. $\square$

Solution. A logarithm. □

Exercise 24.35 (Off-source). What equation does a fundamental solution satisfy away from the pole?

Hint. Delta vanishes on tests away from zero. □

Solution. The homogeneous equation $P(D)E=0$. □

Proofs

Exercise 24.36 (Convolution inversion). Prove $E*f$ solves the forced equation.

Hint. Use constant-coefficient differentiation and delta identity. □

Solution. $P(D)(E*f)=(P(D)E)*f=\delta*f=f$. □

Exercise 24.37 (Radial harmonicity). Verify $|x|^{-(2-n)}$ is harmonic away from zero for $n \geq 3$.

Hint. Use the radial Laplacian formula. □

Solution. The radial second derivative plus $(n-1)/r$ times first derivative cancels. □

Exercise 24.38 (Flux normalization). Explain why the radial flux is independent of sphere radius.

Hint. Differentiate $r^{-(2-n)}$ and multiply by sphere area scale r^{n-1} . □

Solution. The powers cancel to r^0 . □

Exercise 24.39 (Smooth off pole). Why is the source-free equation valid on open sets avoiding zero?

Hint. Test functions there annihilate δ_0 . □

Solution. The distributional equation reduces to homogeneous there. □

Counterexamples and hypothesis testing

Exercise 24.40 (Fundamental versus Green). Why is a free-space fundamental solution not automatically a Green function on a bounded domain?

Hint. It does not satisfy the boundary conditions. □

Solution. A boundary-correction term is needed. □

Exercise 24.41 (Uniqueness failure). Why is a fundamental solution not unique?

Hint. Add any homogeneous solution. □

Solution. The operator kills the added term. □

Exercise 24.42 (Classical failure at pole). Why should one not demand the PDE hold classically at the source point?

Hint. The kernel is singular there. □

Solution. The correct equation is distributional and encodes the point source. □

Applications and investigations

Exercise 24.43 (Point-source field). Why does convolving a fundamental solution with f represent superposition of point responses?

Hint. The integral sums translated impulse responses weighted by f . □

Solution. Linearity turns the kernel into the response to a distributed source. □

Exercise 24.44 (Electrostatics). What is the PDE interpretation of the Newtonian kernel?

Hint. It is the free-space inverse of the Laplacian. □

Solution. It gives the potential generated by point or distributed charge, up to sign/normalization. □

Challenge problems

Exercise 24.45 (Scaling). Derive the homogeneity degree of the Laplacian fundamental solution from operator scaling.

Hint. The Laplacian scales like length^{-2} , while Δ scales like length^{-n} . □

Solution. The kernel must scale like length^{2-n} . □

Exercise 24.46 (Fourier derivation). Explain formally how the Laplacian fundamental solution arises from Fourier space.

Hint. Transform $-\Delta E = \delta$. □

Solution. The symbol $4\pi^2|\xi|^2$ gives $\hat{E}$ proportional to $|\xi|^{-2}$, interpreted distributionally at zero and inverse transformed. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 25

Green Functions

Green functions adapt fundamental solutions to domains and boundary conditions. They separate the singular response to a point source from a correction chosen to enforce the boundary data.

Learning goals

After this chapter, the reader should be able to:

- construct Green functions from fundamental solutions plus boundary-correction terms in elementary domains;
- verify the defining differential equation and boundary conditions separately;
- derive representation formulas for solutions using Green functions;
- interpret symmetry and positivity properties when the operator and domain permit them;
- distinguish free-space kernels from domain-dependent Green functions;
- check jump or derivative conditions that determine one-dimensional Green functions uniquely;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

25.1 Green function definition

For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior.

25.2 Singular plus regular decomposition

Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values.

25.3 Representation formula

Solutions are expressed by integrating the forcing against G , plus boundary terms when nonhomogeneous data are present.

25.4 Symmetry

Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$.

Example 25.1 (Symmetry reflects self-adjointness). For the preceding kernel,

$$G(x, \xi) = G(\xi, x)$$

is immediate from the piecewise formula. This is not accidental: the Dirichlet operator $-d^2/dx^2$ is self-adjoint. The symmetry of the inverse kernel is the integral-kernel version of

$$\langle L^{-1}f, g \rangle = \langle f, L^{-1}g \rangle.$$

25.5 Positivity

Maximum principles can imply nonnegativity of Dirichlet Green functions for positive elliptic operators.

25.6 One-dimensional construction

For second-order ODEs, Green kernels are assembled from two homogeneous solutions matched continuously with a derivative jump.

Example 25.2 (Dirichlet Green function for minus second derivative). For

$$-u'' = f \quad \text{on } (0, 1), \quad u(0) = u(1) = 0,$$

the Green kernel is

$$G(x, \xi) = \begin{cases} x(1 - \xi), & x \leq \xi, \\ \xi(1 - x), & x \geq \xi. \end{cases}$$

It is continuous at $x = \xi$, vanishes at both endpoints, and its derivative jump satisfies

$$-(G_x(\xi^+, \xi) - G_x(\xi^-, \xi)) = 1.$$

These three checks certify the point-source equation and boundary conditions.

25.7 Boundary dependence

Changing Dirichlet to Neumann conditions changes the correction and may require compatibility conditions.

25.8 Operator inverse viewpoint

The Green kernel is the integral kernel of the inverse or resolvent of the boundary-value operator.

Example 25.3 (Green kernel as inverse operator). Define

$$u(x) = \int_0^1 G(x, \xi) f(\xi) d\xi.$$

Differentiating piecewise in x and using the derivative jump recovers $-u'' = f$, while $G(0, \xi) = G(1, \xi) = 0$ gives the boundary conditions. Thus G is literally the integral kernel of L^{-1} for the Dirichlet boundary-value operator.

25.9 Core structural results

Theorem 25.4 (Green representation). *If G is a Dirichlet Green function for L and $Lu = f$, then formally $u(x) = \int_{\Omega} G(x, y) f(y) dy$ for homogeneous boundary data.*

Proof. Apply Green's identity to u and $G(x, \cdot)$, use the point-source equation for G , and use the boundary condition to eliminate the boundary term. $\square$

Theorem 25.5 (Symmetry for self-adjoint problems). *Under suitable self-adjoint boundary conditions, $G(x, y) = G(y, x)$.*

Proof. Apply Green's identity to the two Green functions with poles at x and y , excise small neighborhoods of the poles, and pass to the limit; boundary terms vanish by the shared boundary condition. $\square$

Theorem 25.6 (Derivative jump in one dimension). *For $-u'' = f$ on an interval, the Green kernel is continuous across $x = y$ and its first derivative has the jump required to create a Dirac mass.*

Proof. Integrating the equation $-G_{xx} = \delta_y$ across a small interval around y gives $-(G_x(y^+) - G_x(y^-)) = 1$, while absence of a delta-prime term forces continuity. $\square$

25.10 Worked examples

Example 25.7 (Green function definition diagnostic). Give a rigorous worked analysis of Green function definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 25.8 (Singular plus regular decomposition diagnostic). Give a rigorous worked analysis of Singular plus regular decomposition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 25.9 (Representation formula diagnostic). Give a rigorous worked analysis of Representation formula in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Solutions are expressed by integrating the forcing against G , plus boundary terms when nonhomogeneous data are present. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 25.10 (Symmetry diagnostic). Give a rigorous worked analysis of Symmetry in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

25.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 25.11 (Green function definition diagnostic). Give a rigorous worked analysis of Green function definition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.12 (Singular plus regular decomposition diagnostic). Give a rigorous worked analysis of Singular plus regular decomposition in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.13 (Representation formula diagnostic). Give a rigorous worked analysis of Representation formula in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Solutions are expressed by integrating the forcing against G , plus boundary terms when nonhomogeneous data are present. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.14 (Symmetry diagnostic). Give a rigorous worked analysis of Symmetry in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.15 (Positivity diagnostic). Give a rigorous worked analysis of Positivity in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Maximum principles can imply nonnegativity of Dirichlet Green functions for positive elliptic operators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.16 (One-dimensional construction diagnostic). Give a rigorous worked analysis of One-dimensional construction in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For second-order ODEs, Green kernels are assembled from two homogeneous solutions matched continuously with a derivative jump. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.17 (Boundary dependence diagnostic). Give a rigorous worked analysis of Boundary dependence in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Changing Dirichlet to Neumann conditions changes the correction and may require compatibility conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.18 (Operator inverse viewpoint diagnostic). Give a rigorous worked analysis of Operator inverse viewpoint in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Green kernel is the integral kernel of the inverse or resolvent of the boundary-value operator. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 25.19 (Green representation proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If G is a Dirichlet Green function for L and $Lu = f$, then formally $u(x) = \int_{\Omega} G(x, y)f(y)dy$ for homogeneous boundary data.

Solution. Apply Green's identity to u and $G(x, \cdot)$, use the point-source equation for G , and use the boundary condition to eliminate the boundary term. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 25.20 (Symmetry for self-adjoint problems proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Under suitable self-adjoint boundary conditions, $G(x, y) = G(y, x)$.

Solution. Apply Green's identity to the two Green functions with poles at x and y , excise small neighborhoods of the poles, and pass to the limit; boundary terms vanish by the shared boundary condition. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 25.21 (Derivative jump in one dimension proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $-u'' = f$ on an interval, the Green kernel is continuous across $x = y$ and its first derivative has the jump required to create a Dirac mass.

Solution. Integrating the equation $-G_{xx} = \delta_y$ across a small interval around y gives $-(G_x(y^+) - G_x(y^-)) = 1$, while absence of a delta-prime term forces continuity. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 25.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Green functions adapt fundamental solutions to domains and boundary conditions. They separate the singular response to a point source from a correction chosen to enforce the boundary data. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

25.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 25.23. State and apply the central principle behind Green function definition. Give one immediate consequence in the setting of this chapter.

Hint. Write the Green function as a free-space singular kernel plus a correction chosen to enforce the boundary condition. $\square$

Solution. For a domain operator, $G(x, y)$ solves the differential equation in x with a point source at y and prescribed boundary behavior. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.24. State and apply the central principle behind Singular plus regular decomposition. Give one immediate consequence in the setting of this chapter.

Hint. Verify the differential equation away from the pole and then verify the delta source distributionally at the pole. $\square$

Solution. Typically $G(x, y) = E(x - y) - H(x, y)$, where E is a whole-space fundamental solution and H corrects the boundary values. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.25. State and apply the central principle behind Representation formula. Give one immediate consequence in the setting of this chapter.

Hint. Check the boundary condition in the spatial variable separately from the source-variable dependence. $\square$

Solution. Solutions are expressed by integrating the forcing against G , plus boundary terms when nonhomogeneous data are present. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.26. State and apply the central principle behind Symmetry. Give one immediate consequence in the setting of this chapter.

Hint. Use Green's identity to derive the representation formula rather than guessing the integral kernel. $\square$

Solution. Self-adjoint elliptic problems often yield $G(x, y) = G(y, x)$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.27. State and apply the central principle behind Positivity. Give one immediate consequence in the setting of this chapter.

Hint. Symmetry $G(x, y) = G(y, x)$ follows from self-adjoint structure when the hypotheses hold; identify that structure first. $\square$

Solution. Maximum principles can imply nonnegativity of Dirichlet Green functions for positive elliptic operators. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.28. State and apply the central principle behind One-dimensional construction. Give one immediate consequence in the setting of this chapter.

Hint. Positivity requires a maximum-principle or energy argument; it is not automatic for every operator. $\square$

Solution. For second-order ODEs, Green kernels are assembled from two homogeneous solutions matched continuously with a derivative jump. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.29. State and apply the central principle behind Boundary dependence. Give one immediate consequence in the setting of this chapter.

Hint. In one dimension, solve the homogeneous ODE on each side of the source and match continuity plus derivative jump. $\square$

Solution. Changing Dirichlet to Neumann conditions changes the correction and may require compatibility conditions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 25.30. State and apply the central principle behind Operator inverse viewpoint. Give one immediate consequence in the setting of this chapter.

Hint. Distinguish Dirichlet, Neumann, and periodic Green functions by the boundary/compatibility conditions they satisfy. $\square$

Solution. The Green kernel is the integral kernel of the inverse or resolvent of the boundary-value operator. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

25.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 25.31 (Boundary check). What must a Dirichlet Green kernel do at the boundary?

Hint. Enforce the homogeneous data in the x variable. $\square$

Solution. It vanishes at the boundary. $\square$

Exercise 25.32 (Jump). For $-G_{xx} = \delta$, what derivative jump is required?

Hint. Integrate across the source. □

Solution. $G_x(\text{right}) - G_x(\text{left}) = -1$. □

Exercise 25.33 (Continuity). Why is G continuous across a second-order delta source?

Hint. A jump in G would create a delta-prime term. □

Solution. Continuity removes that unwanted singularity. □

Exercise 25.34 (Symmetry). What property often gives $G(x,y) = G(y,x)$?

Hint. Self-adjointness. □

Solution. A self-adjoint inverse has a symmetric kernel. □

Exercise 25.35 (Representation). How is the homogeneous Dirichlet solution represented?

Hint. Integrate forcing against G . □

Solution. $u(x) = \int G(x,y)f(y)dy$. □

Proofs

Exercise 25.36 (One-dimensional construction). Derive the piecewise Green kernel for $-u''$ on $(0,1)$ with zero endpoints.

Hint. Use a linear homogeneous solution on each side, continuity, boundary values, and derivative jump. □

Solution. Solving those four conditions gives $x(1-y)$ for $x < y$ and $y(1-x)$ for $x > y$. □

Exercise 25.37 (Symmetry proof). Explain Green symmetry from self-adjointness.

Hint. Pair the inverse with two forcing functions. □

Solution. Equality of the two inner products implies symmetry of the kernel in the standard setting. □

Exercise 25.38 (Representation proof). Verify $u = G * f$ solves the boundary problem.

Hint. Apply the differential operator in x under the integral. □

Solution. The delta equation recovers f and boundary values follow from G . □

Exercise 25.39 (Jump derivation). Derive the derivative jump from $-G_{xx} = \delta_y$.

Hint. Integrate over $(y-\epsilon, y+\epsilon)$. □

Solution. The integral of $-G_{xx}$ tends to $-(G_x^+ - G_x^-) = 1$. □

Counterexamples and hypothesis testing

Exercise 25.40 (Free-space kernel). Why does a fundamental solution generally violate domain boundary conditions?

Hint. It knows only the operator and source, not the boundary. $\square$

Solution. A correction term is needed to form G . $\square$

Exercise 25.41 (Neumann compatibility). Why can a pure Neumann Green problem require a zero-mean forcing condition?

Hint. Constants lie in the kernel of the Neumann Laplacian. $\square$

Solution. Solvability requires forcing orthogonal to the nullspace. $\square$

Exercise 25.42 (Symmetry can fail). Why can Green symmetry fail for non-self-adjoint problems?

Hint. The inverse operator need not be self-adjoint. $\square$

Solution. Its kernel has no reason to be symmetric. $\square$

Applications and investigations

Exercise 25.43 (Boundary response). Why are Green functions useful for domain-specific physical models?

Hint. They build both propagation and boundary constraints into one kernel. $\square$

Solution. A distributed forcing is solved by a single integral against the domain-adapted response. $\square$

Exercise 25.44 (Inverse matrix analogy). How is a Green kernel analogous to an inverse matrix?

Hint. The source coordinate is like a column index. $\square$

Solution. Integrating G against f is the continuous analogue of multiplying an inverse matrix by a vector. $\square$

Challenge problems

Exercise 25.45 (Positivity). For the one-dimensional Dirichlet kernel above, verify $G \geq 0$.

Hint. Inspect each product factor on $0 \leq x, y \leq 1$. $\square$

Solution. All factors are nonnegative, consistent with maximum-principle positivity. $\square$

Exercise 25.46 (Eigenfunction expansion). Explain formally why a self-adjoint Green kernel can be expanded as a sum $\phi_k(x)\phi_k(y)/\lambda_k$.

Hint. Expand f in eigenfunctions and invert each scalar eigenvalue. $\square$

Solution. Reading the coefficient formula as an integral operator yields the kernel expansion. $\square$

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 26

Sturm–Liouville Green Kernels

Sturm–Liouville Green kernels combine boundary-value ODE theory, self-adjointness, Wronskians, and spectral expansions. They are the one-dimensional prototype of resolvent kernels for elliptic operators.

Learning goals

After this chapter, the reader should be able to:

- construct Green kernels for second-order Sturm–Liouville boundary-value problems from two homogeneous solutions;
- compute the Wronskian and use it to normalize the kernel;
- verify continuity of the kernel and the prescribed derivative jump across the diagonal;
- check boundary conditions in each variable before accepting a proposed kernel;
- derive solution representations by integrating the forcing term against the kernel;
- distinguish regular, periodic, Dirichlet, and Neumann kernel constructions by their boundary conditions;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

26.1 Sturm–Liouville operator

Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight.

26.2 Homogeneous fundamental solutions

Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition.

26.3 Wronskian structure

A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel.

Example 26.1 (Wronskian normalization for the Dirichlet Laplacian). For $-y''$ on $(0, 1)$ with Dirichlet conditions, choose $u(x) = x$ satisfying the left condition and $v(x) = 1 - x$ satisfying the right condition. Their Wronskian is

$$W = uv' - u'v = x(-1) - 1(1 - x) = -1,$$

constant as expected. The Green kernel is proportional to $u(x_{<})v(x_{>})$, and the constant Wronskian supplies exactly the normalization needed for the derivative jump.

26.4 Piecewise kernel

Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$.

26.5 Matching conditions

Continuity at the source and a derivative jump enforce the Dirac equation.

Example 26.2 (Matching conditions determine the kernel). A second-order Green kernel is built from one homogeneous branch on each side of $x = \xi$. Two conditions meet at the source: continuity of G , which prevents a delta-prime term, and the flux jump

$$-p(\xi)(G_x(\xi^+, \xi) - G_x(\xi^-, \xi)) = 1,$$

which creates the desired Dirac mass. Boundary conditions determine the outer normalizations. Together these conditions uniquely fix the nonresonant kernel.

26.6 Self-adjoint symmetry

Self-adjoint boundary conditions make the Green kernel symmetric.

26.7 Eigenfunction expansion

Away from eigenvalues, the resolvent kernel can be expanded in normalized eigenfunctions with spectral denominators.

26.8 Resonance

At an eigenvalue the inverse fails and solvability requires orthogonality compatibility with the nullspace.

Example 26.3 (Resonance blocks inversion). Suppose $L\phi = \lambda_0\phi$ and one tries to invert $L - \lambda_0$. The eigenfunction coefficient equation becomes

$$(\lambda_0 - \lambda_0)c = \langle f, \phi \rangle.$$

A solution can exist only if $\langle f, \phi \rangle = 0$. Even then the solution is nonunique because any multiple of ϕ may be added. Thus the pole of the Green resolvent at an eigenvalue is the analytic signature of resonance.

26.9 Core structural results

Theorem 26.4 (Green-kernel construction). *Let u satisfy the left boundary condition and v the right one. Away from eigenvalues, $G(x, \xi)$ is proportional to $u(x_{<})v(x_{>})$ divided by the weighted Wronskian.*

Proof. Each branch solves the homogeneous equation and the boundary condition on its side. Continuity and the prescribed flux jump at $x = \xi$ determine the common normalization by the Wronskian. $\square$

Theorem 26.5 (Symmetry). *For a regular self-adjoint Sturm–Liouville problem, $G(x, \xi) = G(\xi, x)$.*

Proof. The product formula is symmetric after swapping x and ξ , or equivalently follows from self-adjointness of the inverse. $\square$

Theorem 26.6 (Spectral resolvent expansion). *If $\{\phi_n\}$ is an orthonormal eigenbasis and λ avoids the spectrum, then the Green kernel has formal expansion $G_\lambda(x, \xi) = \sum_n \phi_n(x)\phi_n(\xi)/(\lambda_n - \lambda)$.*

Proof. Expand the forcing in the eigenbasis, solve each scalar coefficient equation, and read off the integral kernel of the resulting inverse. $\square$

26.10 Worked examples

Example 26.7 (Sturm–Liouville operator diagnostic). Give a rigorous worked analysis of Sturm–Liouville operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 26.8 (Homogeneous fundamental solutions diagnostic). Give a rigorous worked analysis of Homogeneous fundamental solutions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 26.9 (Wronskian structure diagnostic). Give a rigorous worked analysis of Wronskian structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 26.10 (Piecewise kernel diagnostic). Give a rigorous worked analysis of Piecewise kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

26.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 26.11 (Sturm–Liouville operator diagnostic). Give a rigorous worked analysis of Sturm–Liouville operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.12 (Homogeneous fundamental solutions diagnostic). Give a rigorous worked analysis of Homogeneous fundamental solutions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.13 (Wronskian structure diagnostic). Give a rigorous worked analysis of Wronskian structure in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.14 (Piecewise kernel diagnostic). Give a rigorous worked analysis of Piecewise kernel in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.15 (Matching conditions diagnostic). Give a rigorous worked analysis of Matching conditions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Continuity at the source and a derivative jump enforce the Dirac equation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.16 (Self-adjoint symmetry diagnostic). Give a rigorous worked analysis of Self-adjoint symmetry in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Self-adjoint boundary conditions make the Green kernel symmetric. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.17 (Eigenfunction expansion diagnostic). Give a rigorous worked analysis of Eigenfunction expansion in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Away from eigenvalues, the resolvent kernel can be expanded in normalized eigenfunctions with spectral denominators. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.18 (Resonance diagnostic). Give a rigorous worked analysis of Resonance in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. At an eigenvalue the inverse fails and solvability requires orthogonality compatibility with the nullspace. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 26.19 (Green-kernel construction proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: Let u satisfy the left boundary condition and v the right one. Away from eigenvalues, $G(x, \xi)$ is proportional to $u(x_{<})v(x_{>})$ divided by the weighted Wronskian.

Solution. Each branch solves the homogeneous equation and the boundary condition on its side. Continuity and the prescribed flux jump at $x = \xi$ determine the common normalization by the Wronskian. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 26.20 (Symmetry proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For a regular self-adjoint Sturm–Liouville problem, $G(x, \xi) = G(\xi, x)$.

Solution. The product formula is symmetric after swapping x and ξ , or equivalently follows from self-adjointness of the inverse. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 26.21 (Spectral resolvent expansion proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $\{\phi_n\}$ is an orthonormal eigenbasis and λ avoids the spectrum, then the Green kernel has formal expansion $G_\lambda(x, \xi) = \sum_n \phi_n(x)\phi_n(\xi)/(\lambda_n - \lambda)$.

Solution. Expand the forcing in the eigenbasis, solve each scalar coefficient equation, and read off the integral kernel of the resulting inverse. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 26.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Sturm–Liouville Green kernels combine boundary-value ODE theory, self-adjointness, Wronskians, and spectral expansions. They are the one-dimensional prototype of resolvent kernels for elliptic operators. The recurring strategy is to encode the object in the chapter’s natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

26.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 26.23. State and apply the central principle behind Sturm–Liouville operator. Give one immediate consequence in the setting of this chapter.

Hint. Choose one homogeneous solution satisfying the left boundary condition and another satisfying the right boundary condition. $\square$

Solution. Consider $Ly = -(py')' + qy$ with separated boundary conditions and positive weight. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.24. State and apply the central principle behind Homogeneous fundamental solutions. Give one immediate consequence in the setting of this chapter.

Hint. Compute the Wronskian; its nonvanishing is what lets the two homogeneous solutions normalize the kernel. $\square$

Solution. Choose one solution satisfying the left boundary condition and one satisfying the right boundary condition. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.25. State and apply the central principle behind Wronskian structure. Give one immediate consequence in the setting of this chapter.

Hint. Define the kernel piecewise on $x < \xi$ and $x > \xi$ using the two boundary-adapted solutions. $\square$

Solution. A weighted Wronskian is constant for homogeneous solutions and controls the denominator of the Green kernel. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.26. State and apply the central principle behind Piecewise kernel. Give one immediate consequence in the setting of this chapter.

Hint. Enforce continuity of the kernel at $x = \xi$ before imposing the derivative jump. $\square$

Solution. Use one product of homogeneous solutions for $x < \xi$ and the reversed product for $x > \xi$. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.27. State and apply the central principle behind Matching conditions. Give one immediate consequence in the setting of this chapter.

Hint. Integrate the differential equation across a small interval around ξ to derive the exact jump in the first derivative. $\square$

Solution. Continuity at the source and a derivative jump enforce the Dirac equation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.28. State and apply the central principle behind Self-adjoint symmetry. Give one immediate consequence in the setting of this chapter.

Hint. Check the boundary conditions on the kernel before inserting it into the solution integral. $\square$

Solution. Self-adjoint boundary conditions make the Green kernel symmetric. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.29. State and apply the central principle behind Eigenfunction expansion. Give one immediate consequence in the setting of this chapter.

Hint. For self-adjoint Sturm–Liouville problems, test whether the kernel should be symmetric after including the weight correctly. $\square$

Solution. Away from eigenvalues, the resolvent kernel can be expanded in normalized eigenfunctions with spectral denominators. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 26.30. State and apply the central principle behind Resonance. Give one immediate consequence in the setting of this chapter.

Hint. Periodic or Neumann problems may require compatibility conditions or removal of a zero mode; inspect the homogeneous kernel first. $\square$

Solution. At an eigenvalue the inverse fails and solvability requires orthogonality compatibility with the nullspace. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

26.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 26.31 (Wronskian). Compute the Wronskian of $u=x$ and $v=1-x$.

Hint. Use $u'v - u v'$. $\square$

Solution. It is -1 . $\square$

Exercise 26.32 (Piecewise product). What product form is used for $x < x_i$?

Hint. Use left solution at x and right solution at x_i . $\square$

Solution. $u(x)v(x_i)$ up to Wronskian normalization. $\square$

Exercise 26.33 (Continuity). What matching condition prevents a delta-prime?

Hint. Keep the kernel value continuous. $\square$

Solution. $G(x_i^-, x_i) = G(x_i^+, x_i)$. $\square$

Exercise 26.34 (Flux jump). What creates the delta source?

Hint. A prescribed derivative/flux jump. □

Solution. The weighted derivative jump equals the inverse leading coefficient normalization. □

Exercise 26.35 (Resonance). What happens to the resolvent at an eigenvalue?

Hint. A spectral denominator vanishes. □

Solution. The inverse fails unless forcing satisfies compatibility, and solutions are nonunique. □

Proofs

Exercise 26.36 (Wronskian constancy). For a self-adjoint Sturm–Liouville equation, explain why the weighted Wronskian is constant for homogeneous solutions.

Hint. Differentiate $p(u'v - uv')$ and use both equations. □

Solution. The derivative cancels to zero. □

Exercise 26.37 (Kernel construction). Prove the piecewise product solves the homogeneous equation away from the source and boundary conditions.

Hint. Each factor is the selected homogeneous solution. □

Solution. On each side only a scalar multiple of the correct homogeneous solution appears. □

Exercise 26.38 (Jump normalization). Derive the weighted derivative jump.

Hint. Integrate the differential equation across the source. □

Solution. The qG term vanishes in the shrinking interval and the pG' flux contributes the unit mass. □

Exercise 26.39 (Symmetry). Prove the product Green formula is symmetric in x and x_i for the self-adjoint case.

Hint. Exchange min and max variables. □

Solution. $u(x_{<})v(x_{>})$ is unchanged by swapping the arguments. □

Counterexamples and hypothesis testing

Exercise 26.40 (Eigenvalue denominator). Why is the spectral Green expansion invalid at an eigenvalue?

Hint. A denominator $\lambda_k - \lambda$ becomes zero. □

Solution. The resolvent does not exist as a bounded inverse there. □

Exercise 26.41 (Nonself-adjoint symmetry). Why should one not expect a symmetric Green kernel for a nonself-adjoint problem?

Hint. The inverse operator is not self-adjoint. □

Solution. Kernel symmetry encodes self-adjointness. □

Exercise 26.42 (Wrong boundary solution). What fails if both homogeneous factors satisfy the same endpoint condition?

Hint. The other boundary condition is not enforced. □

Solution. The piecewise kernel will generally violate the required right or left boundary data. □

Applications and investigations

Exercise 26.43 (Mode response). How does a Green resolvent amplify a forcing component near an eigenvalue?

Hint. Inspect $1/(\lambda_k - \lambda)$. □

Solution. The corresponding modal response becomes large near resonance. □

Exercise 26.44 (Vibration model). Why is Sturm–Liouville theory natural for strings and diffusion in one dimension?

Hint. Separation of variables yields self-adjoint second-order eigenproblems. □

Solution. Green kernels and eigenmodes describe forced and transient responses. □

Challenge problems

Exercise 26.45 (Compatibility). At resonance, derive the orthogonality condition on f .

Hint. Take inner product of $(L - \lambda_0)y = f$ with the null eigenfunction. □

Solution. Self-adjointness makes the left side zero, so f must be orthogonal to the eigenfunction. □

Exercise 26.46 (Resolvent pole). Explain why the spectral term $1/(\lambda_k - \lambda)$ predicts a simple pole of the resolvent.

Hint. As λ approaches an isolated simple eigenvalue, one denominator dominates. □

Solution. The corresponding rank-one eigenprojection is multiplied by a factor diverging like the reciprocal distance to the eigenvalue. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 27

Elliptic Operators and Maximum Principles

Maximum principles turn the sign of an elliptic operator into strong qualitative control. They yield uniqueness, comparison, boundary estimates, and positivity of Green functions without explicitly solving the PDE.

Learning goals

After this chapter, the reader should be able to:

- classify elementary second-order operators as elliptic by inspecting the principal symbol or coefficient matrix;
- apply weak or classical maximum principles only after checking ellipticity and sign hypotheses;
- use maximum principles to prove uniqueness of boundary-value problems;
- derive comparison principles from the maximum principle;
- construct examples showing how sign changes or degeneracy can invalidate the conclusion;
- connect elliptic estimates with qualitative control of interior extrema and boundary data;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

27.1 Elliptic operators

Second-order ellipticity means the principal coefficient matrix is uniformly positive definite.

Example 27.1 (Checking uniform ellipticity from a matrix). Consider

$$Lu = 3u_{xx} + 2u_{xy} + 2u_{yy}.$$

The symmetric principal matrix is

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its eigenvalues are positive because $\operatorname{tr} A = 5 > 0$ and $\det A = 5 > 0$. Hence

$$\xi^T A \xi \geq \lambda_{\min} |\xi|^2,$$

so the operator is uniformly elliptic with ellipticity constant $\lambda_{\min} > 0$.

27.2 Weak maximum principle

For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum.

27.3 Strong maximum principle

A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses.

27.4 Comparison principle

If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows.

Example 27.2 (Comparison gives uniqueness in one line). Suppose $Lu = Lv$ in Ω and $u = v$ on the boundary. Apply the comparison principle first to u, v and then to v, u . It gives $u \leq v$ and $v \leq u$, hence $u = v$. Equivalently the maximum principle applied to $w = u - v$ and $-w$ shows both extrema of w are zero.

27.5 Uniqueness

The maximum principle applied to the difference of two solutions yields uniqueness for Dirichlet problems.

27.6 Hopf boundary lemma

At a nonconstant boundary extremum, the outward normal derivative has a strict sign under suitable smoothness.

27.7 Positivity preservation

Nonnegative data often produce nonnegative solutions.

27.8 Barriers

Explicit super- and subsolutions transfer geometric boundary information into quantitative estimates.

Example 27.3 (A quadratic barrier converts geometry into a bound). In a ball B_R , the function

$$b(x) = R^2 - |x|^2$$

is nonnegative inside, zero on the boundary, and satisfies

$$-\Delta b = 2n.$$

Multiples of b can therefore dominate solutions of Poisson inequalities with bounded forcing. Comparing u to Cb transfers a source bound into an interior pointwise estimate without solving the PDE explicitly.

27.9 Core structural results

Theorem 27.4 (Weak maximum principle). *If $Lu \geq 0$ for a uniformly elliptic operator with appropriate lower-order signs, then the maximum of u is controlled by the boundary maximum.*

Proof. At a hypothetical strict interior maximum, the gradient vanishes and the Hessian is negative semidefinite; ellipticity makes the principal second-order contribution have the wrong sign, contradicting the differential inequality after the standard perturbation argument. $\square$

Theorem 27.5 (Comparison principle). *If $Lu \geq Lv$ in Ω and $u \leq v$ on the boundary, then $u \leq v$ in Ω under the maximum-principle hypotheses.*

Proof. Apply the weak maximum principle to $u - v$. $\square$

Theorem 27.6 (Dirichlet uniqueness). *A homogeneous Dirichlet solution of $Lu = 0$ is identically zero under the maximum-principle hypotheses.*

Proof. Apply the maximum principle to u and to $-u$ to show both maximum and minimum are zero. $\square$

27.10 Worked examples

Example 27.7 (Elliptic operators diagnostic). Give a rigorous worked analysis of Elliptic operators in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Second-order ellipticity means the principal coefficient matrix is uniformly positive definite. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 27.8 (Weak maximum principle diagnostic). Give a rigorous worked analysis of Weak maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 27.9 (Strong maximum principle diagnostic). Give a rigorous worked analysis of Strong maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 27.10 (Comparison principle diagnostic). Give a rigorous worked analysis of Comparison principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

27.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 27.11 (Elliptic operators diagnostic). Give a rigorous worked analysis of Elliptic operators in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Second-order ellipticity means the principal coefficient matrix is uniformly positive definite. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.12 (Weak maximum principle diagnostic). Give a rigorous worked analysis of Weak maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.13 (Strong maximum principle diagnostic). Give a rigorous worked analysis of Strong maximum principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.14 (Comparison principle diagnostic). Give a rigorous worked analysis of Comparison principle in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.15 (Uniqueness diagnostic). Give a rigorous worked analysis of Uniqueness in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The maximum principle applied to the difference of two solutions yields uniqueness for Dirichlet problems. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.16 (Hopf boundary lemma diagnostic). Give a rigorous worked analysis of Hopf boundary lemma in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. At a nonconstant boundary extremum, the outward normal derivative has a strict sign under suitable smoothness. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.17 (Positivity preservation diagnostic). Give a rigorous worked analysis of Positivity preservation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Nonnegative data often produce nonnegative solutions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.18 (Barriers diagnostic). Give a rigorous worked analysis of Barriers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Explicit super- and subsolutions transfer geometric boundary information into quantitative estimates. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 27.19 (Weak maximum principle proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $Lu \geq 0$ for a uniformly elliptic operator with appropriate lower-order signs, then the maximum of u is controlled by the boundary maximum.

Solution. At a hypothetical strict interior maximum, the gradient vanishes and the Hessian is negative semidefinite; ellipticity makes the principal second-order contribution have the wrong sign, contradicting the differential inequality after the standard perturbation argument. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 27.20 (Comparison principle proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $Lu \geq Lv$ in Ω and $u \leq v$ on the boundary, then $u \leq v$ in Ω under the maximum-principle hypotheses.

Solution. Apply the weak maximum principle to $u - v$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 27.21 (Dirichlet uniqueness proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: A homogeneous Dirichlet solution of $Lu = 0$ is identically zero under the maximum-principle hypotheses.

Solution. Apply the maximum principle to u and to $-u$ to show both maximum and minimum are zero. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 27.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Maximum principles turn the sign of an elliptic operator into strong qualitative control. They yield uniqueness, comparison, boundary estimates, and positivity of Green functions without explicitly solving the PDE. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

27.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 27.23. State and apply the central principle behind Elliptic operators. Give one immediate consequence in the setting of this chapter.

Hint. Inspect the principal coefficient matrix or principal symbol; positive definiteness is the basic ellipticity test in the scalar second-order case. $\square$

Solution. Second-order ellipticity means the principal coefficient matrix is uniformly positive definite. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.24. State and apply the central principle behind Weak maximum principle. Give one immediate consequence in the setting of this chapter.

Hint. For a maximum principle, assume an interior positive maximum and evaluate the first and second derivative information there. $\square$

Solution. For suitable operators, a subsolution cannot achieve a strict positive interior maximum beyond its boundary maximum. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.25. State and apply the central principle behind Strong maximum principle. Give one immediate consequence in the setting of this chapter.

Hint. Use the sign of the zeroth-order term exactly where the theorem requires it; changing that sign can destroy the conclusion. $\square$

Solution. A nonconstant elliptic solution cannot attain an interior extremum under the standard sign hypotheses. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.26. State and apply the central principle behind Comparison principle. Give one immediate consequence in the setting of this chapter.

Hint. Uniqueness follows by applying the maximum principle to the difference of two solutions with zero boundary data. $\square$

Solution. If $Lu \leq Lv$ and the boundary ordering agrees, then the interior ordering follows. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.27. State and apply the central principle behind Uniqueness. Give one immediate consequence in the setting of this chapter.

Hint. Comparison follows by applying the maximum principle to the difference between the candidate sub- and supersolutions. $\square$

Solution. The maximum principle applied to the difference of two solutions yields uniqueness for Dirichlet problems. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.28. State and apply the central principle behind Hopf boundary lemma. Give one immediate consequence in the setting of this chapter.

Hint. For a counterexample, alter ellipticity or the sign condition and look for an interior extremum that violates the expected bound. $\square$

Solution. At a nonconstant boundary extremum, the outward normal derivative has a strict sign under suitable smoothness. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.29. State and apply the central principle behind Positivity preservation. Give one immediate consequence in the setting of this chapter.

Hint. In weak settings, translate the maximum principle into an energy inequality for the positive part when that method is available. $\square$

Solution. Nonnegative data often produce nonnegative solutions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 27.30. State and apply the central principle behind Barriers. Give one immediate consequence in the setting of this chapter.

Hint. Keep boundary regularity and domain assumptions visible; the qualitative theorem is not independent of the setting. $\square$

Solution. Explicit super- and subsolutions transfer geometric boundary information into quantitative estimates. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

27.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 27.31 (Elliptic matrix). Is $\text{diag}(2,5)$ uniformly elliptic?

Hint. Check eigenvalues. $\square$

Solution. Yes; the smallest eigenvalue is 2. $\square$

Exercise 27.32 (Degenerate matrix). Is $\text{diag}(1,0)$ uniformly elliptic?

Hint. Check positive definiteness. $\square$

Solution. No; one direction has zero quadratic form. $\square$

Exercise 27.33 (Uniqueness). What theorem proves uniqueness for homogeneous Dirichlet harmonic functions?

Hint. Apply maximum principle to u and $-u$. $\square$

Solution. The Dirichlet uniqueness consequence of the maximum principle. $\square$

Exercise 27.34 (Comparison). If $Lu \geq Lv$ and $u \leq v$ on boundary, what follows under the chapter hypotheses?

Hint. Apply comparison. □

Solution. $u \leq v$ in the domain. □

Exercise 27.35 (Barrier Laplacian). Compute $-\Delta(R^2 - |x|^2)$ in n dimensions.

Hint. Each second derivative of $|x|^2$ is 2. □

Solution. The value is $2n$. □

Proofs

Exercise 27.36 (Maximum uniqueness). Prove homogeneous Dirichlet uniqueness.

Hint. Apply maximum principle to u and $-u$. □

Solution. Both maximum and minimum are zero. □

Exercise 27.37 (Comparison proof). Derive comparison from maximum principle.

Hint. Apply it to $u - v$ with the appropriate sign. □

Solution. Boundary order and operator inequality force the interior order. □

Exercise 27.38 (Ellipticity check). Show a symmetric coefficient matrix with eigenvalues between λ and Λ gives uniform ellipticity.

Hint. Diagonalize orthogonally. □

Solution. The quadratic form is bounded between $\lambda|x|^2$ and $\Lambda|x|^2$. □

Exercise 27.39 (Barrier method). Explain how a supersolution bounds a subsolution.

Hint. Apply comparison to their difference. □

Solution. Boundary and operator inequalities force the desired pointwise order. □

Counterexamples and hypothesis testing

Exercise 27.40 (Wrong lower-order sign). Why can a bad zeroth-order sign invalidate a maximum principle?

Hint. At an interior maximum, the lower-order term may reinforce rather than contradict the maximum. □

Solution. The standard sign hypothesis is needed in the differential inequality. □

Exercise 27.41 (Degenerate ellipticity). Why can maximum-principle conclusions weaken for degenerate operators?

Hint. The principal quadratic form does not control all directions. □

Solution. The interior Hessian sign may not force a contradiction. □

Exercise 27.42 (Boundary condition absent). Why does a maximum principle not by itself determine a unique solution without boundary data?

Hint. Many harmonic functions exist. □

Solution. The boundary values are what pin down the solution. □

Applications and investigations

Exercise 27.43 (Positivity). Why do nonnegative forcing/boundary data often imply a sign for elliptic solutions?

Hint. Apply comparison with zero or a suitable barrier. □

Solution. The maximum principle transfers boundary/source sign information to the interior. □

Exercise 27.44 (A priori bounds). Why are maximum principles valuable numerically?

Hint. They provide solution bounds independent of mesh resolution. □

Solution. A discrete solution violating known bounds can signal instability or implementation error. □

Challenge problems

Exercise 27.45 (Hopf intuition). Why should a nonconstant solution attaining a strict boundary maximum have a nonzero outward normal derivative under Hopf hypotheses?

Hint. A tangent-flat contact would contradict the strong interior comparison with a barrier. □

Solution. The barrier enforces strict first-order escape from the extremum. □

Exercise 27.46 (Green positivity). Explain why a Dirichlet Green kernel for $-\Delta$ is nonnegative in suitable domains.

Hint. Approximate a positive point source and apply positivity preservation. □

Solution. The response to nonnegative forcing cannot become negative under the maximum principle. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.

Chapter 28

Spectral and Transform Methods for PDE

Spectral and transform methods solve linear PDEs by diagonalizing the operator. Fourier transforms diagonalize translation-invariant operators, while eigenfunction expansions diagonalize boundary-value operators on bounded domains.

Learning goals

After this chapter, the reader should be able to:

- diagonalize translation-invariant or separable PDE operators by Fourier or eigenfunction transforms;
- derive transformed algebraic or ordinary differential equations from PDEs;
- solve model heat, wave, or elliptic equations by inverse transform under the stated hypotheses;
- interpret spectral multipliers as frequency-dependent propagation, damping, or resolvent factors;
- compare Fourier-transform, Fourier-series, and eigenfunction-expansion methods according to the domain and boundary conditions;
- check transformed solutions by substituting back into the original PDE and verifying initial or boundary data;

Conceptual roadmap

structure $\longrightarrow$ estimate $\longrightarrow$ limit or representation $\longrightarrow$ application.

28.1 Symbol of an operator

A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform.

28.2 Elliptic multipliers

Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful.

28.3 Heat equation

The heat transform multiplier $e^{-4\pi^2 t |\xi|^2}$ gives smoothing and decay.

Example 28.1 (Heat equation diagonalizes frequency by frequency). For $u_t = \Delta u$ on $\mathbb{R}^n$, Fourier transformation gives

$$\partial_t \hat{u}(t, \xi) = -4\pi^2 |\xi|^2 \hat{u}(t, \xi).$$

Solving this scalar ODE yields

$$\hat{u}(t, \xi) = e^{-4\pi^2 t |\xi|^2} \hat{u}_0(\xi).$$

High frequencies decay faster, which simultaneously explains smoothing, energy decay, and the Gaussian heat kernel after inverse transformation.

28.4 Wave equation

Frequency modes oscillate with phase determined by the dispersion relation.

Example 28.2 (Wave modes oscillate instead of decay). For $u_{tt} = \Delta u$, Fourier transformation gives

$$\partial_{tt} \hat{u} + 4\pi^2 |\xi|^2 \hat{u} = 0.$$

Each frequency is a harmonic oscillator:

$$\hat{u}(t, \xi) = \hat{u}_0(\xi) \cos(2\pi |\xi| t) + \frac{\hat{u}_1(\xi)}{2\pi |\xi|} \sin(2\pi |\xi| t),$$

with the zero-frequency limit interpreted continuously. Unlike heat flow, the multiplier has unit-size oscillation rather than exponential damping.

28.5 Poisson equation

The Fourier multiplier $|\xi|^{-2}$ represents the Newtonian potential, interpreted distributionally at low frequency.

28.6 Eigenfunction expansions

On bounded domains, self-adjoint elliptic operators admit orthogonal eigenfunctions that replace Fourier exponentials.

Example 28.3 (Eigenfunction expansion solves the bounded-domain heat problem). If $L\phi_k = \lambda_k \phi_k$ with an orthonormal eigenbasis and

$$u_0 = \sum_k c_k \phi_k,$$

then the heat-type problem $u_t + Lu = 0$ has formal solution

$$u(t) = \sum_k e^{-t\lambda_k} c_k \phi_k.$$

Each PDE mode reduces to the scalar ODE $c'_k(t) + \lambda_k c_k(t) = 0$. Boundary conditions are already encoded in the eigenfunctions, making spectral expansion the bounded-domain analogue of the Fourier transform.

28.7 Semigroups

Spectral calculus defines e^{-tL} and links PDE evolution to functional calculus.

28.8 Boundary and whole-space synthesis

Transform methods exploit translation symmetry; spectral methods adapt the same diagonalization idea to domains and boundary conditions.

28.9 Core structural results

Theorem 28.4 (Heat transform solution). *For $u_t = \Delta u$ on $\mathbb{R}^n$ with initial data u_0 , $\hat{u}(t, \xi) = e^{-4\pi^2 t |\xi|^2} \hat{u}_0(\xi)$.*

Proof. Fourier transform converts the PDE into the scalar ODE $\partial_t \hat{u} = -4\pi^2 |\xi|^2 \hat{u}$ for each frequency; solve the ODE with the transformed initial data. $\square$

Theorem 28.5 (Spectral heat expansion). *If $L\phi_k = \lambda_k \phi_k$ on a bounded self-adjoint problem, then $u(t) = \sum_k e^{-t\lambda_k} \langle u_0, \phi_k \rangle \phi_k$ solves $u_t + Lu = 0$.*

Proof. Expand the initial data in the eigenbasis; each coefficient satisfies the scalar ODE $c'_k + \lambda_k c_k = 0$. $\square$

Theorem 28.6 (Resolvent diagonalization). *For λ off the spectrum, $(L - \lambda)^{-1}$ multiplies each eigenfunction coefficient by $(\lambda_k - \lambda)^{-1}$.*

Proof. Apply $L - \lambda$ to the eigenfunction expansion and solve the scalar coefficient equations independently. $\square$

28.10 Worked examples

Example 28.7 (Symbol of an operator diagnostic). Give a rigorous worked analysis of Symbol of an operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 28.8 (Elliptic multipliers diagnostic). Give a rigorous worked analysis of Elliptic multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 28.9 (Heat equation diagnostic). Give a rigorous worked analysis of Heat equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. The heat transform multiplier $e^{-4\pi^2 t|\xi|^2}$ gives smoothing and decay. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

Example 28.10 (Wave equation diagnostic). Give a rigorous worked analysis of Wave equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion. Frequency modes oscillate with phase determined by the dispersion relation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone.

28.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 28.11 (Symbol of an operator diagnostic). Give a rigorous worked analysis of Symbol of an operator in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.12 (Elliptic multipliers diagnostic). Give a rigorous worked analysis of Elliptic multipliers in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.13 (Heat equation diagnostic). Give a rigorous worked analysis of Heat equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The heat transform multiplier $e^{-4\pi^2 t|\xi|^2}$ gives smoothing and decay. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.14 (Wave equation diagnostic). Give a rigorous worked analysis of Wave equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Frequency modes oscillate with phase determined by the dispersion relation. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.15 (Poisson equation diagnostic). Give a rigorous worked analysis of Poisson equation in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. The Fourier multiplier $|\xi|^{-2}$ represents the Newtonian potential, interpreted distributionally at low frequency. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.16 (Eigenfunction expansions diagnostic). Give a rigorous worked analysis of Eigenfunction expansions in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. On bounded domains, self-adjoint elliptic operators admit orthogonal eigenfunctions that replace Fourier exponentials. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.17 (Semigroups diagnostic). Give a rigorous worked analysis of Semigroups in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Spectral calculus defines e^{-tL} and links PDE evolution to functional calculus. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.18 (Boundary and whole-space synthesis diagnostic). Give a rigorous worked analysis of Boundary and whole-space synthesis in the setting of this chapter. State the decisive definition or hypothesis and derive the central conclusion.

Solution. Transform methods exploit translation symmetry; spectral methods adapt the same diagonalization idea to domains and boundary conditions. The decisive point is to use the chapter definition at the correct countable, norm, transform, or distributional level rather than rely on pointwise intuition alone. $\square$

Problem 28.19 (Heat transform solution proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For $u_t = \Delta u$ on $\mathbb{R}^n$ with initial data u_0 , $\hat{u}(t, \xi) = e^{-4\pi^2 t |\xi|^2} \hat{u}_0(\xi)$.

Solution. Fourier transform converts the PDE into the scalar ODE $\partial_t \hat{u} = -4\pi^2 |\xi|^2 \hat{u}$ for each frequency; solve the ODE with the transformed initial data. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 28.20 (Spectral heat expansion proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: If $L\phi_k = \lambda_k \phi_k$ on a bounded self-adjoint problem, then $u(t) = \sum_k e^{-t\lambda_k} \langle u_0, \phi_k \rangle \phi_k$ solves $u_t + Lu = 0$.

Solution. Expand the initial data in the eigenbasis; each coefficient satisfies the scalar ODE $c'_k + \lambda_k c_k = 0$. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 28.21 (Resolvent diagonalization proof dossier). Prove the following structural statement and identify the step where its main hypothesis is used: For λ off the spectrum, $(L - \lambda)^{-1}$ multiplies each eigenfunction coefficient by $(\lambda_k - \lambda)^{-1}$.

Solution. Apply $L - \lambda$ to the eigenfunction expansion and solve the scalar coefficient equations independently. This proof also explains why the stated hypothesis is structural rather than cosmetic. $\square$

Problem 28.22 (Chapter synthesis). Connect the main definitions and structural theorems of this chapter into one proof strategy. Explain what object is controlled, what estimate or closure principle is used, and what conclusion becomes available.

Solution. Spectral and transform methods solve linear PDEs by diagonalizing the operator. Fourier transforms diagonalize translation-invariant operators, while eigenfunction expansions diagonalize boundary-value operators on bounded domains. The recurring strategy is to encode the object in the chapter's natural measurable, normed, Fourier, or distributional structure, establish the controlling estimate, and then pass to the desired limit or representation. $\square$

28.12 Exercises with complete solutions

The shorter exercise layer remains separate from the solved dossiers.

Exercise 28.23. State and apply the central principle behind Symbol of an operator. Give one immediate consequence in the setting of this chapter.

Hint. Transform in the variables where the operator is translation invariant; leave the remaining variables untouched if separation is more natural. $\square$

Solution. A constant-coefficient differential operator becomes multiplication by its polynomial symbol under the Fourier transform. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.24. State and apply the central principle behind Elliptic multipliers. Give one immediate consequence in the setting of this chapter.

Hint. The Fourier transform turns spatial derivatives into multiplication by frequency, reducing the PDE to algebra or an ODE in time. $\square$

Solution. Solving $P(D)u = f$ in frequency space amounts to dividing by the symbol where the division is meaningful. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.25. State and apply the central principle behind Heat equation. Give one immediate consequence in the setting of this chapter.

Hint. For the heat equation, solve $\partial_t \hat{u} = -|\xi|^2 \hat{u}$ and then invert the transform. $\square$

Solution. The heat transform multiplier $e^{-4\pi^2 t |\xi|^2}$ gives smoothing and decay. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.26. State and apply the central principle behind Wave equation. Give one immediate consequence in the setting of this chapter.

Hint. For the wave equation, solve the frequency-space oscillator and impose both initial displacement and velocity data. $\square$

Solution. Frequency modes oscillate with phase determined by the dispersion relation. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.27. State and apply the central principle behind Poisson equation. Give one immediate consequence in the setting of this chapter.

Hint. For elliptic resolvents, divide by the symbol only after checking zeros and the distribution/function space in which the multiplier acts. $\square$

Solution. The Fourier multiplier $|\xi|^{-2}$ represents the Newtonian potential, interpreted distributionally at low frequency. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.28. State and apply the central principle behind Eigenfunction expansions. Give one immediate consequence in the setting of this chapter.

Hint. On bounded or periodic domains, replace the continuous Fourier transform by Fourier series or an eigenfunction expansion. $\square$

Solution. On bounded domains, self-adjoint elliptic operators admit orthogonal eigenfunctions that replace Fourier exponentials. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.29. State and apply the central principle behind Semigroups. Give one immediate consequence in the setting of this chapter.

Hint. Interpret large-frequency behavior of the multiplier as smoothing, damping, or amplification before estimating the inverse transform. $\square$

Solution. Spectral calculus defines e^{-tL} and links PDE evolution to functional calculus. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

Exercise 28.30. State and apply the central principle behind Boundary and whole-space synthesis. Give one immediate consequence in the setting of this chapter.

Hint. Verify the final formula by checking the transformed equation and then the original initial or boundary conditions. $\square$

Solution. Transform methods exploit translation symmetry; spectral methods adapt the same diagonalization idea to domains and boundary conditions. As an immediate consequence, the same principle can be inserted into later convergence, approximation, transform, or weak-form arguments whenever its hypotheses are verified. $\square$

28.13 Graded supplementary exercises

These exercises extend the preserved foundational set and are graded by mathematical role.

Standard computations

Exercise 28.31 (Heat multiplier). What is the heat multiplier at frequency ξ ?

Hint. Transform the Laplacian. $\square$

Solution. $\exp(-4 \pi^2 t |\xi|^2)$. $\square$

Exercise 28.32 (Wave frequency). What ODE does each wave Fourier mode satisfy?

Hint. Transform the Laplacian. □

Solution. $y'' + 4\pi^2 |x|^2 y = 0$. □

Exercise 28.33 (Poisson symbol). What multiplier formally inverts $-\Delta$?

Hint. Divide by $4\pi^2 |x|^2$. □

Solution. $1/(4\pi^2 |x|^2)$, interpreted distributionally at zero. □

Exercise 28.34 (Spectral heat). What happens to an eigenmode with eigenvalue λ under heat flow?

Hint. Solve $c' + \lambda c = 0$. □

Solution. It is multiplied by $\exp(-\lambda t)$. □

Exercise 28.35 (Resolvent). How does $(L - \lambda)^{-1}$ act on eigenmode ϕ_k ?

Hint. Solve the scalar coefficient equation. □

Solution. Multiply by $1/(\lambda_k - \lambda)$. □

Proofs

Exercise 28.36 (Heat transform). Derive the transformed heat solution.

Hint. Solve the scalar ODE for each x . □

Solution. The multiplier is $\exp(-4\pi^2 t |x|^2)$. □

Exercise 28.37 (Wave transform). Derive the transformed wave solution from initial displacement and velocity.

Hint. Solve the second-order oscillator ODE. □

Solution. Cosine and sine multipliers appear with frequency $2\pi|x|$. □

Exercise 28.38 (Spectral heat). Prove the eigenfunction heat expansion.

Hint. Expand initial data and equate each eigenmode coefficient. □

Solution. Each coefficient solves $c_k' + \lambda_k c_k = 0$. □

Exercise 28.39 (Resolvent diagonalization). Prove the resolvent coefficient formula.

Hint. Apply $L - \lambda$ to an eigenfunction expansion. □

Solution. Each coefficient is divided by $\lambda_k - \lambda$. □

Counterexamples and hypothesis testing

Exercise 28.40 (Symbol zero). Why is Poisson inversion delicate at $\xi=0$?

Hint. The symbol vanishes. □

Solution. The reciprocal is singular and must be interpreted with compatibility/distributional information. □

Exercise 28.41 (Wrong basis). Why can ordinary Fourier exponentials be inappropriate on a bounded domain with nonperiodic boundary conditions?

Hint. They need not satisfy the boundary data. □

Solution. Use operator-adapted eigenfunctions instead. □

Exercise 28.42 (Heat reversibility). Why is backward heat evolution severely unstable?

Hint. It multiplies frequencies by $\exp(+4 \pi^2 t |\xi|^2)$. □

Solution. Tiny high-frequency errors are exponentially amplified. □

Applications and investigations

Exercise 28.43 (Diffusion smoothing). Why does heat flow smooth initial data?

Hint. High-frequency spectral components are strongly damped. □

Solution. The multiplier decays faster for larger $|\xi|$. □

Exercise 28.44 (Modal dynamics). Why are eigenfunction methods useful in vibrating or diffusing bounded systems?

Hint. They decouple the PDE into scalar modal ODEs. □

Solution. Each mode evolves independently with a frequency or decay rate set by its eigenvalue. □

Challenge problems

Exercise 28.45 (Energy decay). For a positive self-adjoint L , derive monotone L^2 decay of heat flow from spectral coefficients.

Hint. Differentiate the sum of $|c_k|^2 \exp(-2 \lambda_k t)$. □

Solution. Each term is nonincreasing, and strictly decays for positive eigenvalues. □

Exercise 28.46 (Semigroup property). Show spectral heat operators satisfy $\exp(-tL)\exp(-sL)=\exp(-(t+s)L)$.

Hint. Check every eigenmode. □

Solution. The scalar multipliers multiply: $\exp(-t \lambda_k)\exp(-s \lambda_k)=\exp(-(t+s)\lambda_k)$. □

Chapter summary

The chapter now supplies a canonical theorem-and-dossier layer for subsequent measure, Fourier, distributional, Sobolev, and PDE arguments.